# Copernicus Global Land Operations
# "Vegetation and Energy"
## "CGLOPS-1"

**Framework Service Contract N° 945120-IPR-2023**

## VALIDATION REPORT

## LAND SURFACE PHENOLOGY FROM SENTINEL-3/OLCI

## VERSION 1.0

## Issue 1.10

Organization name of lead contractor for this deliverable: EOLAB

| | |
|---|---|
| Book Captain: | Jorge Sánchez-Zapero |
| Contributing Authors: | Enrique Martínez-Sánchez |
| | Fernando Camacho |

| Dissemination Level | | |
|---|---|---|
| PU | Public | X |
| PP | Restricted to other programme participants (including the Commission Services) | |
| RE | Restricted to a group specified by the consortium (including the Commission Services) | |
| CO | Confidential, only for members of the consortium (including the Commission Services) | |

# Document Release Sheet

| | | | |
|---|---|---|---|
| Book captain: | Jorge Sánchez-Zapero (EOLAB) | Sign Jorge Sanchez-Zapero | Date 09.07.2025 |
| Approval: | Roselyne Lacaze (HYGEOS) | Sign Roselyne Lacaze | Date 17.07.2025 |
| Endorsement: | Nadine Gobron (JRC) | Sign | Date |
| Distribution: | Public | | |


© C-GLOPS Lot1 consortium

# Change Record

| Date | Page(s) | Description of Change | Release |
|---|---|---|---|
| 01.07.2024 | All | First issue | I1.00 |
| 09.07.2025 | 3.2.2.1, 3.2.2.2, 4.2, 4.3.4, 4.3.5, 4.3.6, 4.4.5, Annex IV | Update product intercomparison with MODIS V6.1 and VIIRS V2 phenology products.<br>Add direct validation for the 2024 year. | I1.10 |
| | | | |
| | | | |


© C-GLOPS Lot1 consortium

# TABLE OF CONTENTS

*Executive summary* ..... *22*

*1 Background of the document* ..... *24*

**1.1 Scope and Objectives** ..... **24**

**1.2 Content of the document** ..... **24**

**1.3 Related documents** ..... **24**

1.3.1 Applicable documents ..... 24
1.3.2 Input documents ..... 24
1.3.3 Output documents ..... 25
1.3.4 External documents ..... 25

*2 Requirements* ..... *27*

**2.1 specific technical details and requirements** ..... **27**

**2.2 Further requirements** ..... **27**

2.2.1 Output product composition ..... 27
2.2.2 Data Structure ..... 28
2.2.3 Data format ..... 28
2.2.4 Uncertainties and Validation ..... 28
2.2.5 Input data ..... 28
2.2.6 Product delivery ..... 29

*3 Quality Assessment Method* ..... *30*

**3.1 Overvall procedure** ..... **30**

3.1.1 Product completeness ..... 32
3.1.2 Spatial consistency ..... 32
3.1.3 Error evaluation ..... 35
3.1.4 Summary of the Quality Assessment Procedure ..... 36

**3.2 Satellite Products** ..... **37**

3.2.1 Evaluated dataset: CLMS OLCI LSP Collection 300m V1 (OLCI LSP) ..... 39
3.2.2 Reference phenology global products ..... 41
3.2.3 Reference global productivity products ..... 43
3.2.4 Pan-European High-Resolution Vegetation Phenology and Productivity (HR-VPP) ..... 45
3.2.5 Pan-European Medium-Resolution Vegetation Phenology and Productivity (MR-VPP) Version 4 ..... 46

**3.3 In-situ Reference Products** ..... **47**

3.3.1 Pan-European Phenology Project (PEP725) ..... 47
3.3.2 TEMPO ..... 50
3.3.3 National Science Foundation's National Ecological Observatory Network (NEON) ..... 51
3.3.4 Ecosystem Phenology Camera (PhenoCam V3) ..... 52

*4 Results ..... 55*

**4.1 Product Content & Visual inspection of global maps ..... 55**

**4.2 Product completeness ..... 61**

**4.3 Spatial consistency ..... 64**

4.3.1 Visual inspection over subcontinental areas ..... 64
4.3.2 Maps of residuals with MR-VPP ..... 69
4.3.3 Spatial consistency between CLMS OLCI LSP vs. HR-VPP over AOIs ..... 72
4.3.4 Maps of differences with global LSP products ..... 76
4.3.5 Latitudinal transect ..... 79
4.3.6 Spatial variability (precision) ..... 81

**4.4 Error evaluation ..... 83**

4.4.1 CLMS OLCI LSP versus ground references ..... 83
4.4.2 CLMS OLCI LSP versus near-ground-based PhenoCam V3 dataset ..... 95
4.4.3 Overall consistency over Europe with MR-VPP ..... 97
4.4.4 Overall consistency over Europe with HR-VPP ..... 99
4.4.5 Overall consistency with global phenology products ..... 102
4.4.6 Overall consistency with global productivity products ..... 112

*5 Summary and Conclusions ..... 120*

*6 References ..... 125*

*Annex I: Difference maps between CLMS OLCI LSP and MR-VPP ..... 128*

*Annex II: Spatial consistency between CLMS OLCI LSP and HR-VPP over AOIs ..... 130*

*Annex III: Maps and spatial precision of CLMS OLCI LSP over selected ROIs ..... 138*

*Annex IV: Direct validation results for 2024 year ..... 144*

PEP725 ..... 144
TEMPO ..... 146
NEON ..... 148
PhenoCam V3 ..... 150

## List of Figures

Figure 1: Sampling strategy for product intercomparison with reference global (LANDVAL V2, left side) and European (EVAL right side) products. ..... 31

Figure 2: Location of the subcontinental areas (red boxes), AOIs (dark blue crosses), ROIs (light blue crosses) and latitudinal transect (pink vertical line) used for spatial consistency. ..... 33

Figure 3. Examples of TIMESAT-derived phenology and productivity parameters: (a) start of season, (b) end of season, (c) length of season, (d) minimum value, (e) time of maximum, (f) maximum value, (g) amplitude, (h) seasonal productivity, (h+i) total productivity. Source: https://web.nateko.lu.se/timesat/timesat.asp. ..... 40

Figure 4: Distribution of PEP725, TEMPO, NEON and PhenoCam locations used in direct validation for 2023. ..... 47

Figure 5: Vegetation group strategy. Top: Google Earth view of 4 km x 4 km around PEP site ID=110. Bottom: CLC around the site at 100m of spatial resolution (left), aggregated CLC around the site at 100 m resolution (center) and aggregated CLC around the site resample at 300 m resolution (right). ..... 49

Figure 6: Example of two PhenoCam V3 ROIs (left side) and Google Earth views at 300 m x 300 m spatial extension. ..... 54

Figure 7: Global maps of CLMS OLCI LSP following layers for 2023: SOSD, EOSD, MAXD and LENGTH (from top to bottom). Season 1 and season 2 are displayed at left and right sides, respectively. ..... 56

Figure 8: Global maps of CLMS OLCI LSP following layers for 2023 season 1: SOSV, EOSV, MINV, MAXV and AMPL (from left to right and top to bottom). Unit: $m^2 \cdot m^{-2}$. ..... 57

Figure 9: Global maps of CLMS LSP OLCI LSLOPE (left) and RSLOPE (right) for 2023 season 1. Unit: $m^2 \cdot m^{-2} \cdot day^{-1}$. ..... 58

Figure 10: Global maps of CLMS OLCI LSP TPROD (left) and SPROD (right) for 2023 season 1. Unit: $m^2 \cdot m^{-2} \cdot day$. ..... 59

Figure 11: Global maps of CLMS OLCI LSP QA layers for 2023 season 1: SOS quality, Peak quality, EOSD quality and Overall quality (from left to right and top to bottom). ..... 61

Figure 12: Global maps of valid (blue) and non-valid (yellow) observations of start of season layer of CLMS OLCI LSP (1st row), MidGreenup layer of MODIS LCD V6.1 (2nd row) and Date_Mid_Greenup_Phase layer of VIIRS LCD V2 (3rd row) products for season 1 (2023). Maps considering all pixels are displayed at left side and considering 'best quality' pixels using quality

flags are displayed at right side. Grey stripes at northern and southern latitudes correspond to the areas where CLMS products are not provided. Percentage of valid pixels ('all' and 'best quality (BQ)' cases) is displayed at the bottom side. ..... 62

Figure 13: Same as in Figure 12 for season 2. ..... 63

Figure 14: Maps of CLMS OLCI LSP SOSD for 2023 season 1 over six sub-continental areas (10º latitude x 15º longitude). ..... 65

Figure 15: Same as in Figure 14 but for EOSD. ..... 66

Figure 16: Same as in Figure 14 but for SPROD. Unit: $m^2 \cdot m^{-2} \cdot day$. ..... 67

Figure 17: Same as in Figure 14 but for TPROD. Unit: $m^2 \cdot m^{-2} \cdot day$. ..... 68

Figure 18: Top: Maps and histograms of residuals between CLMS OLCI LSP SOSD and MR-VPP DSOS for 2023 (season 1). Bottom: percentage of global cases within goal (blue), threshold (yellow) user requirements and non-conform (purple). Unit: days. ..... 69

Figure 19: Top: Maps and histograms of residuals between CLMS OLCI LSP EOSD and MR-VPP DEOS for 2023 (season 1). Bottom: percentage of global cases within goal (blue), threshold (yellow) user requirements and non-conform (purple). Unit: days. ..... 70

Figure 20: Maps and histograms of residuals between CLMS OLCI LSP SPROD and MR-VPP SINT for 2023 (season 1). Unit: $m^2.m^{-2}.day$. ..... 71

Figure 21: Maps and histograms of residuals between CLMS OLCI LSP TPROD and MR-VPP LINT for 2023 (season 1). Unit: $m^2.m^{-2}.day$. ..... 71

Figure 22: Spatial consistency over Barrax cultivated site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SOSD (left) and EOSD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SOSD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for EOSD. ..... 73

Figure 23: Spatial consistency over Barrax cultivated site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SPROD (left) and TPROD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SPROD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for TPROD. ... 74

Figure 24: Same as in Figure 22, but for Bialowieza National Park DBF site. ..... 75

Figure 25: Same as in Figure 23, but for Bialowieza National Park DBF site. ..... 76

Figure 26: Global map (left) and histogram (right) of differences between CLMS OLCI LSP SOSD and MODIS V6.1 MidGreenup for 2023 (season 1) considering best-quality pixels. ..... 77

Figure 27: Global map (left) and histogram (right) of differences between CLMS OLCI LSP EOSD and MODIS V6.1 MidGreendown for 2023 (season 1) considering best-quality pixels. ..... 77

Figure 28: Global map (left) and histogram (right) of differences between CLMS OLCI LSP SOSD and VIIRS V2 Date_Mid_Greenup_Phase (all pixels) for 2023 (season 1). ..... 78

Figure 29: Global map (left) and histogram (right) of differences between CLMS OLCI LSP EOSD and VIIRS V2 Date_Mid_Senescence_Phase (all pixels) for 2023 (season 1). ..... 79

Figure 30: Latitudinal transect of CLMS OLCI LSP (magenta points) SOSD (top) and EOSD (bottom) during 2023 (season 1) and MODIS LCD V6.1 (blue points) and VIIRS LCD V2 (green points) similar phenometrics from Norway (70.315ºN, 20.119ºE) to South Africa (34.872ºS, 20.119ºE). ..... 80

Figure 31: CLMS OLCI LSP (top) and MR-VPP (bottom) SOSD (good quality pixels) versus PEP725 phases 10 (‘first leaves separated’, left), 11 (‘first leaves unfolded’, middle) and 223 (‘leaf unfolding ≥ 50%’, right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 84

Figure 32: CLMS OLCI LSP (top) and MR-VPP (bottom) EOSD (good quality pixels) versus PEP725 phases 95 (‘50% of leaves fallen’, left), and 205 (‘autumnal colouration ≥ 50%’, right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 85

Figure 33: CLMS OLCI LSP (top) and MR-VPP (bottom) SOSD (good quality pixels) versus TEMPO phases 0 (‘germination/sprouting/bud development’, left) and 1 (‘leaf development/main shoot’, right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 88

Figure 34: CLMS OLCI LSP (left) and MR-VPP (right) EOSD (good quality pixels) versus TEMPO phase 9 (‘beginning of dormancy/die off of leaves’) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 88

Figure 35: CLMS OLCI LSP SOSD (good quality pixels) versus NEON phases breaking leaves/buds (left) and emerging leaves/buds (right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 90

Figure 36: CLMS OLCI LSP EOSD (good quality pixels) versus NEON phases colored leaves/buds (left), and falling leaves/buds (right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 91

Figure 37: Diagram bars of Bias (left) and RMSD (right) of the comparison of CLMS OLCI LSP SOSD (top) and EOSD (bottom) versus different phenophases of PEP725, TEMPO and NEON ground databases. Analysis for 2023 (season 1). N indicates the number of samples for each comparison. ..... 93

Figure 38: Diagram bars of percentage of cases within goal (left) and threshold (right) user requirements of the comparison of CLMS OLCI LSP SOSD (top) and EOSD (bottom) versus different phenophases of PEP725, TEMPO and NEON ground databases. Analysis for 2023 (season 1). ..... 94

Figure 39: CLMS OLCI LSP SOSD (good quality pixels) versus PhenoCam V3 start of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2023 (season 1). ..... 95

Figure 40: CLMS OLCI LSP EOSD (good quality pixels) versus PhenoCam V3 end of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2023 (season 1). ..... 96

Figure 41: CLMS OLCI LSP SOSD and EOSD versus MR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels..... 97

Figure 42: CLMS OLCI LSP TPROD and SPROD versus MR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels. ..... 99

Figure 43: CLMS OLCI LSP SOSD and EOSD versus HR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels..... 100

Figure 44: CLMS OLCI LSP TPROD and SPROD versus HR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels. ..... 101

Figure 45: CLMS OLCI LSP SOSD versus MODIS V6.1 Greenup (left), MidGreenup (middle) and Maturity (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels..... 102

Figure 46: CLMS OLCI LSP EOSD versus MODIS V6.1 Senescence (left), MidGreendown (middle) and Dormancy (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels. ..... 103

Figure 47: CLMS OLCI LSP SOSD versus MODIS V6.1 MidGreenup products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels..... 104

Figure 48: CLMS OLCI LSP EOSD versus MODIS V6.1 MidGreendown products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels..... 106

Figure 49: CLMS OLCI LSP SOSD versus VIIRS V2 Onset_Greenness_Increase (left), Date_Mid_Greenup_Phase (middle) and Onset_Greenness_Maximum (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels..... 107

Figure 50: CLMS OLCI LSP EOSD versus VIIRS V2 Onset_Greeness_Decrease (left), Date_Mid_Senescence_Phase (middle) and Onset_Greeness_Minimum (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels. 108

Figure 51: CLMS OLCI LSP SOSD versus VIIRS V2 Date_Mid_Greenup_Phase products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). 109

Figure 52: CLMS OLCI LSP V1 EOSD versus VIIRS V2 Date_Senescence_Greenup_Phase products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). 111

Figure 53: Scatter-plots between CLMS OLCI LSP SPROD (left) and TPROD (right) ($m^2 \cdot m^{-2} \cdot day$) and the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1). 113

Figure 54: Scatter-plots CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1). 114

Figure 55: Scatter-plots CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1). 115

Figure 56: Scatter-plots between CLMS OLCI LSP SPROD and TPROD ($m^2 \cdot m^{-2} \cdot day$) and the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1). 116

Figure 57: Scatter-plots CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD and EOS. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1). 117

Figure 58: Scatter-plots CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1). 118

Figure 59: Top: Maps and histograms of differences between CLMS OLCI LSP SOSD and MR-VPP DSOS for 2023 (season 1). Unit: days. 128

Figure 60: Top: Maps and histograms of differences between CLMS OLCI LSP EOSD and MR-VPP DEOS for 2023 (season 1). Unit: days. 128

Figure 61: Top: Maps and histograms of differences between CLMS OLCI LSP SPROD and MR-VPP SINT for 2023 (season 1). Unit: $m^2 \cdot m^{-2} \cdot day$. 128

Figure 62: Top: Maps and histograms of differences between CLMS OLCI LSP TPROD and MR-VPP LINT for 2023 (season 1). Unit: $m^2 \cdot m^{-2} \cdot day$. ..... 129

Figure 63: Top: Maps and histograms of differences between CLMS OLCI LSP MAXV and MR-VPP 'largest values' for 2023 (season 1). Unit: $m^2 \cdot m^{-2}$. ..... 129

Figure 64: Spatial consistency over LUCAS 2777 DBF site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SOSD (left) and EOSD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SOSD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for EOSD. ..... 130

Figure 65: Spatial consistency over LUCAS 2777 DBF site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SPROD (left) and TPROD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SPROD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for TPROD. ..... 131

Figure 66: Same as in Figure 64, but for LUCAS 1208 NLF site. ..... 132

Figure 67: Same as in Figure 65, but for LUCAS 1208 NLF site. ..... 133

Figure 68: Same as in Figure 64, but for EOLAB 53 cultivated site. ..... 134

Figure 69: Same as in Figure 65, but for EOLAB 53 cultivated site. ..... 135

Figure 70: Same as in Figure 64, but for SPAR 113 shrublands site. ..... 136

Figure 71: Same as in Figure 65, but for SPAR 113 shrublands site. ..... 137

Figure 72: Spatial precision over 5 km x 5 km area of Evare EBF site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). In the temporal profiles, unfilled inverted triangles of EOSD represent the value corresponding to next year (EOSD > 365). ..... 138

Figure 73: Spatial precision over 5 km x 5 km area of Mountain Lake DBF site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). ..... 139

Figure 74: Spatial precision over 5 km x 5 km area of Oskoba NLF site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles

for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). In the temporal profiles, unfilled inverted triangles of EOSD represent the value corresponding to next year (EOSD > 365). ..... 140

Figure 75: Spatial precision over 5 km x 5 km area of Albufera CUL site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). ..... 141

Figure 76: Spatial precision over 5 km x 5 km area of Cimarron National Grassland Park HER site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). ..... 142

Figure 77: Spatial precision over 5 km x 5 km area of Jornada SHR site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). ..... 143

Figure 78: CLMS OLCI LSP SOSD (good quality pixels) versus PEP725 phases 10 (‘first leaves separated’, left), 11 (‘first leaves unfolded’, middle) and 223 (‘leaf unfolding ≥ 50%’, right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 144

Figure 79: CLMS OLCI LSP EOSD (good quality pixels) versus PEP725 phases 95 (‘50% of leaves fallen’, left), and 205 (‘autumnal colouration ≥ 50%’, right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 144

Figure 80: CLMS OLCI LSP SOSD (good quality pixels) versus TEMPO phases 0 (‘germination/sprouting/bud development’, left) and 1 (‘leaf development/main shoot’, right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 146

Figure 81: CLMS OLCI LSP EOSD (good quality pixels) versus TEMPO phase 9 (‘beginning of dormancy/die off of leaves’) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 146

Figure 82: CLMS OLCI LSP SOSD (good quality pixels) versus NEON phases breaking leaves/buds (left) and emerging leaves/buds (right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 148

Figure 83: CLMS OLCI LSP EOSD (good quality pixels) versus NEON phases colored leaves/buds (left), and falling leaves/buds (right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements. ..... 148

Figure 84: CLMS OLCI LSP SOSD (good quality pixels) versus PhenoCam V3 start of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2024 (season 1). ... 150

Figure 85: CLMS OLCI LSP EOSD (good quality pixels) versus PhenoCam V3 end of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2024 (season 1). ..... 150


© C-GLOPS Lot1 consortium

## List of Tables

Table 1: Summary of central coordinates of selected AOIs (20km x 20km) for spatial consistency. ..... 33

Table 2: Relevant information and central coordinates of selected ROIs (5 km x 5 km). ..... 34

Table 3: Validation metrics. 'x' and 'y' refer to reference and evaluated products. ..... 36

Table 4: Summary of the QA procedure for CLMS OLCI LSP validation. ..... 37

Table 5: Characteristics of the remote sensing phenology products used in this exercise. ..... 38

Table 6: Characteristics of the remote sensing productivity products used in this exercise. ..... 38

Table 7: Quality Flag information used to select good quality pixels. ..... 39

Table 8: Grouping strategy for linking OLCI LSP with PEP725 phenology observations. ..... 48

Table 9: Phases used for linking CLMS OLCI LSP SOSD and EOSD with PEP725 phenology observations. ..... 50

Table 10: Grouping strategy for linking OLCI LSP SOSD and EOSD with TEMPO phenology observations. ..... 51

Table 11: Phases used for linking CLMS OLCI LSP SOSD and EOSD with TEMPO phenology observations. ..... 51

Table 12: Growth Forms aggregation for NEON data. ..... 52

Table 13: Strategy for choosing PhenoCam ROIs vegetation types according to CLMS LC300 V3. ..... 53

Table 14: Vegetation Phenology and Productivity parameters generated in CLMS LSP products. ..... 55

Table 15: CLMS LSP QA flags and descriptions of phenology quality. ..... 60

Table 16: Summary of spatial variability STD and MAD (i.e., indicators of precision) of CLMS OLCI LSP SOSD and EOSD (season1) evaluated over 5 km x 5 km ROIs during 2023. ..... 81

Table 17: Summary of spatial variability STD (precision) and MAD indicators of CLMS OLCI LSP SPROD and TPROD evaluated over 5 km x 5 km ROIs during 2023. Relative values in percentage are computed respect to the mean values of the whole ROI. ..... 82

Table 18: Performance statistics between CLMS OLCI LSP and MR-VPP versus PEP725 database: SOSD versus PEP725 Phases 10, 11 and 223; and EOSD versus PEP725 Phases 95 and 205. Analysis for 2023 (season 1). ..... 86

Table 19: Performance statistics between CLMS OLCI LSP and MR-VPP versus TEMPO database: SOSD versus TEMPO Phases 0 and 1; and EOSD versus TEMPO Phase 9. Analysis for 2023 (season 1). ..... 89

Table 20: Performance statistics between CLMS OLCI LSP SOSD and EOSD with NEON breaking&emerging, colored and falling leaves/buds. ..... 91

Table 21: Performance statistics between CLMS OLCI LSP SOSD and EOSD with PhenoCam V3 equivalent phases defined at 10%, 25% and 50% for 2023 (season 1). ..... 96

Table 22: Performance statistics between CLMS OLCI LSP SOSD and EOSD versus MR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1). ..... 98

Table 23: Performance statistics between CLMS OLCI LSP TPROD and SPROD versus MR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1). ..... 99

Table 24: Performance statistics between CLMS OLCI LSP SOSD and EOSD versus HR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1). ..... 100

Table 25: Performance statistics between CLMS OLCI LSP TPROD and SPROD versus HR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1). ..... 101

Table 26: Performance statistics between CLMS OLCI LSP SOSD and EOSD with MODIS V6.1 Greenup/MidGreenup/Maturity and Dormancy/MidGreendown/Senescence phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). ..... 103

Table 27: Performance statistics between CLMS OLCI LSP SOSD with MODIS V6.1 Greenup and MidGreenup phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). ..... 105

Table 28: Performance statistics between CLMS OLCI LSP EOSD with MODIS V6.1 and Dormancy and MidGreendown phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). ..... 106

Table 29: Performance statistics between CLMS OLCI LSP SOSD and EOSD with VIIRS V2 Onset_Greenness_Increase/Date_Mid_Greenup_Phase/Onset_Greenness_Maximum and Onset_Greeness_Decrease/Date_Mid_Senescence_Phase/Onset_Greeness_Minimum phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). ..... 108

Table 30: Performance statistics between CLMS OLCI LSP SOSD with VIIRS V2 Onset_Greenness_Increase and Date_Mid_Greenup_Phase. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). ..... 110

Table 31: Performance statistics between CLMS OLCI LSP EOSD with VIIRS V2 Date_Mid_ Senescence _Phase (left) and Onset_Greeness_Minimum (right). Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1). ..... 111

Table 32: Performance statistics between CLMS OLCI LSP SPROD and TPROD ($m^2 \cdot m^{-2} \cdot day$) versus CLMS OLCI GPP aggregated values ($kgC \cdot m^{-2} \cdot day$) for the same season length. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1). ..... 113

Table 33: Performance statistics between CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1). ..... 114

Table 34: Performance statistics between CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1). ..... 115

Table 35: Performance statistics between CLMS OLCI LSP SPROD and TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD an EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1). ..... 117

Table 36: Performance statistics between CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1). ..... 117

Table 37: Performance statistics between CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1). ..... 118

Table 38: Performance statistics between CLMS OLCI LSP versus PEP725 database: SOSD versus PEP725 Phases 10, 11 and 223; and EOSD versus PEP725 Phases 95 and 205. Analysis for 2024 (season 1). ..... 145

Table 39: Performance statistics between CLMS OLCI LSP versus TEMPO database: SOSD versus TEMPO Phases 0 and 1; and EOSD versus TEMPO Phase 9. Analysis for 2024 (season 1).. 147

Table 40: Performance statistics between CLMS OLCI LSP SOSD and EOSD with NEON (2024) breaking&emerging, colored and falling leaves/buds. ..... 149

Table 41: Performance statistics between CLMS OLCI LSP SOSD and EOSD with PhenoCam V3 equivalent phases defined at 10%, 25% and 50% for 2024 (season 1). ..... 151

## List of Acronyms

| | |
|---|---|
| AD | Applicable Document |
| AG | Agriculture |
| AMPL | Seasonal amplitude |
| AOI | Area Of Interest |
| APAR | Absorbed Photosynthetically Active Radiation |
| APU | Accuracy, Precision and Uncertainty |
| ATBD | Algorithm Theoretical Basis Document |
| B | Bias |
| BBCH | Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie |
| BPLUT | Biome Properties Look-Up Table |
| BRDF | Bidirectional Reflectance Distribution Function |
| CAL/VAL | CALibration/VALidation |
| CCI | Climate Change Initiative |
| CEOS | Committee on Earth Observation Satellites |
| CGLOPS | Copernicus Global Land OPerationS |
| CLC | Corine Land Cover |
| CLMS | Copernicus Land Monitoring Service |
| COG | Cloud Optimized GeoTiff |
| CUE | Carbon Use Efficiency |
| CUL | CULtivated |
| DB | Deciduous Broadleaf Forest |
| DBF | Deciduous Broadleaf Forest |
| DEOS | Date End Of Season |
| DMP | Dry Matter Productivity |
| DN | Deciduous Needleleaf Forest |
| DSOS | Date Start Of Season |
| EB | Evergreen Broadleaf Forest |
| EBF | Evergreen Broadleaf Forest |
| ECMWF | European Centre for Medium-Range Weather Forecasts |
| EN | Evergreen Needleleaf Forest |
| EOSD | End Of Season Date |
| EOSV | Vegetation index value at EOSD |
| ERA5 | ECMWF Reanalysis 5 |
| ESA | European Space Agency |
| EVAL | Evaluation newtork |
| EVI2 | 2-band Enhanced Vegetation Index |

| | |
|---|---|
| FAPAR | Fraction of Absorbed Photosynthetically Active Radiation |
| FLUXNET | Fluxes Network |
| FPAR | Fraction of Photosynthetically Active Radiation |
| FWHM | Full Width at Half Maximum |
| GBOV | Ground Based Observations for Validation |
| GCC | Green Chromatic Coordinate |
| GCOS | Global Climate Observing System |
| GDMP | Gross Dry Matter Productivity |
| GLC-100m | Global Land Cover at 100m resolution |
| GMAO | Global Modelling and Assimilation Office |
| GPP | Gross Primary Productivity |
| GR | Grasslands |
| HER | HERbaceous |
| HPLM | Hybrid Piecewise Logistic Model |
| HR-VPP | High Resolution Vegetation Phenology and Productivity |
| INSPIRE | INfrastructure for SPatial InfoRmation in Europe |
| IPCC | Intergovernmental Panel on Climate Change |
| IUPAC | International Union of Pure and Applied Chemistry |
| JCGM | Joint Committee for Guides in Metrology |
| JRC | Joint Research Centre |
| LAI | Leaf Area Index |
| LANDVAL | Land Validation |
| LENGTH | Length of season |
| LCD | Land Cover Dynamics |
| LINT | Large Integral |
| LPV | Land Product Validation subgroup of CEOS |
| LSLOPE | Slope of the green-up period |
| LSP | Land Surface Phenology |
| LUE | Light Use Efficiency |
| MAD | Median Absolute Deviation |
| MAR | Major Axis Regression |
| MAXD | Day of the maximum of season |
| MAXV | Maximum value of the season at MAXD |
| MD | Median Deviation |
| MINV | Minimum value of the season |
| MMU | Minimum Mapping Unit |
| MOD | MODIS products based on TERRA data |
| MODIS | Moderate Resolution Imaging Spectroradiometer |

| | |
|---|---|
| MR | Maintenance Respiration |
| MR-VPP | Medium Resolution Vegetation Phenology and Productivity |
| MSI | MultiSpectral Instrument |
| N | Number of samples |
| NASA | National Aeronautics and Space Agency |
| NBAR | Nadir BRDF-corrected TOC Adjusted Reflectance |
| NDVI | Normalized Difference Vegetation Index |
| NEON | National Ecological Observatory Network |
| NetCDF | Network Common Data Form |
| NIR | Near InfraRed |
| NLF | Needle-Leaf Forest |
| NPN | National Phenology Network |
| NPP | Net Primary Productivity |
| NRT | Near Real Time |
| NV | No Vegetated |
| OF | Other Forests |
| OLCI | Ocean and Land Colour Instrument |
| OLS | Ordinary Least Squares |
| PEP725 | Pan-European Phenology Project |
| PGQ | Proportion of Good Quality |
| PhenoCam | Ecosystem Phenology Camera |
| PPI | Plant Phenology Index |
| PROBA-V (or PBV) | Project for On Board Autonomy – Vegetation |
| PQAD | Product Quality Assurance Document |
| PSF | Point Spread Functions |
| PSN | Net Photosynthesis |
| PUM | Product User Manual |
| QAR | Quality Assessment Report |
| QC | Quality Control |
| QF | Quality Flag |
| R | Correlation coefficient |
| ReBeLS | Regularised BRDF inversion for Land Surface |
| RMSD | Root Mean Square Deviation |
| RMSE | Root Mean Square Error |
| ROI | Region of Interest |
| RSLOPE | Slope of the green-down period |
| SBA | Sparse and Bare Areas |
| SH | Shrublands |

| | |
|---|---|
| SHR | SHRublands |
| SNPP | Suomi National Polar-orbiting Partnership |
| SOSD | Start of the season date |
| SOSV | Vegetation index value at SOSD |
| SPROD | Seasonal Productivity |
| STD | Standard deviation |
| SWIR | Short-Wave InfraRed |
| TIMESAT | A program for analysing time-series of satellite sensor data |
| TN | Tundra |
| TOC | Top-of-Canopy |
| TPROD | Total Productivity |
| UM | User Manual |
| UNFCCC | United Nations Framework Convention on Climate Change |
| V1 | Version 1 |
| V3 | Version 3 |
| VIIRS | Visible/Infrared Imager Radiometer Suite |
| VNP | SNPP/VIIRS suite of products |
| VPP | Vegetation Phenology and Productivity |
| VR | Validation Report |
| WGS84 | World Geodetic System 1984 |
| WMO | World Meteorological Organization |
| XX | Landscape Region |

## EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters are to take place in a timely manner and are complemented by the constitution of long-term time series.

This report presents the quality assessment of the Ocean and Land Colour Instrument (OLCI) Land Surface Phenology (LSP) products (300 m, Version 1) which are derived from Bidirectional Reflectance Distribution Function (BRDF)-adjusted top-of-canopy (TOC) reflectance data. LSP products provide a total of 13 vegetation phenology and productivity (VPP) parameters, derived for up to two growing seasons. The parameters are derived from smooth seasonal trajectories of the plant phenology index (PPI). The validation methodology is described in the CLMS Product Quality Assurance Document [CGLOPS1_PQAD_I2.30]. Two main validation approaches are defined: direct validation (i.e., comparison of satellite products with in situ measurements) and indirect validation or product intercomparison. Several criteria of performance are evaluated, including the visualization of the product content, and the evaluation of the product completeness, spatial consistency and error (accuracy, precision and uncertainty).

The direct validation of Start Of the Season Date (SOSD) and the End Of the Season Date (EOSD) is performed with Pan-European Phenology Project (PEP725), TEMPO and National Ecological Observatory Network (NEON) datasets, which provide phenological observations. The Ecosystem Phenology Camera (PhenoCam) V3 dataset, which infers phenometrics from digital cameras, is also used for direct comparison. The product intercomparison with global satellite-based products includes comparison with phenology (SOSD and EOSD vs. MODIS V6.1, VIIRS V2 equivalent phenometrics) and productivity (TPROD/SPROD vs. MOD17A2HGF V6.1 and CLMS OLCI V1.1 Gross Primary Productivity (GPP) products). Finally, the comparison over Europe with Medium Resolution and High Resolution VPP (MR-VPP and HR-VPP) products is also performed for start and end of season dates and for productivity layers.

CLMS OLCI LSP 2023 products are reliable at global scale and consistent with ground references and equivalent estimates of other satellite products, except for the non-expected values over desert areas. The spatial precision over homogeneous areas covering different biomes is good, with Standard Deviation (STD) and Median Absolute Deviation (MAD) typically lower than 4 and 7days for SOSD and EOSD (except for Evergreen Broadleaf Forests (EBF)). The spatial variability found for productivity SPROD and TPROD layers ranges typically (both STD and MAD values) between 10% and 25%, respectively. The variations of SOSD and EOSD per latitude (latitudinal transect from Norway to South Africa) show the expected pattern and consistent trend values with MODIS V6.1 and VIIRS V2 products. The exception was over tropical forest, which exhibits the lowest agreement between products as well as lower fraction of good quality retrievals. The comparison with MODIS

V6.1 phenology product shows the best agreement with MidGreenup (MD = 1 day, RMSD = 25 days) for SOSD and with MidGreendown (MD = 5 days, RMSD = 32 days) for EOSD with the expected delay due to the definition (both are defined at 50% amplitude threshold level). The comparison of OLCI LSP SOSD and EOSD with VIIRS V2 also shows the best agreement with Date_Mid_Greenup_Phase (MD = -2 days, RMSD = 19 days) and Date_Mid_Senescence_Phase (MD = 11 days, RMSD = 36 days) phenological metrics, with expected anticipated start and delayed end of season due to the difference in phenometric definitions.

CLMS OLCI LSP provides bias close to zero or slight delay (up to 5 days) in the comparison with PEP725, TEMPO and NEON ground phenophases related to the first stages of the growing season and anticipated SOSD (B between -7 and -12 days) with phenophases related to more developed leaf growth. In case of EOSD, bias typically between -33 (i.e., anticipated EOSD) to 4 days is found compared with PEP725, TEMPO and NEON datasets. SOSD cases within goal requirements typically ranges between 10% and 20% (25%-40% for threshold) and EOSD between 15% and 20% (30%-45% for threshold). CLMS OLCI LSP typically provides shorter seasons as compared with near-ground-based Phenocam V3 (B from 3 to 12 days for SOSD, B=from -11 to 0 days for EOSD) and satellite datasets MR-VPP (B=18 and -2 days for SOSD and EOSD) and HR-VPP (B=14 and 6 days for SOSD and EOSD).

The comparisons of SPROD and TPROD show high correlations with MR-VPP and HR-VPP, but higher productivity (positive bias around 25%-30%) despite of the fact the CLMS seasons are typically shorter. Comparisons of CLMS TPROD and SPROD with other global productivity GPP products show consistent values: $R \geq 0.76$ with CLMS OLCI GPP and $R \geq 0.67$ with MOD17A2HGF V6.1.

The validation of SOSD and EOSD for 2024 compared with PEP725, TEMPO and NEON ground references and near-ground-based Phenocam data showed similar results than in 2023, with some exceptions mainly explained with the different samplings of different years.

# 1 BACKGROUND OF THE DOCUMENT

## 1.1 SCOPE AND OBJECTIVES

The document presents the results of the quality assessment of Land Surface Phenology 300m Version 1.0 products based on Sentinel-3/OLCI observations (OLCI LSP hereafter). The quality assessment is focused on the year 2023. An annual update, including direct validation results, for the year 2024 was also performed (Annex IV).

The OLCI product for 2023 was produced using, as input data, the OLCI BRDF-corrected TOC Adjusted Reflectance (NBAR) from 2023.01.01 to 2024.02.29. The OLCI product for 2024 was produced using, as input data, the OLCI NBAR from 2023.01.01 to 2025.02.28.

The objective is to evaluate the scientific quality of OLCI LSP products at global scale.

## 1.2 CONTENT OF THE DOCUMENT

This document is structured as follows. Chapter 2 recalls the requirements. Chapter 3 describes the methodology for quality assessment, the metrics and the criteria of evaluation. Chapter 4 presents the results of the analysis and Chapter 5 summarizes the main conclusions of the study.

## 1.3 RELATED DOCUMENTS

### 1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

### 1.3.2 Input documents

| Document ID | Descriptor |
|---|---|
| CGLOPS1_PQAD_I2.30 | Product Quality Assurance Document "Vegetation & Energy". |
| CGLOPS1_ATBD_LSP300m-V1 | Algorithm Theoretical Basis Document of Land Surface Phenology 300m Version 1 |

| | |
|---|---|
| CGLOPS1_ATBD_BRDFCorrection300m-V1 | Algorithm Theoretical Basis Document of the Version 1 of the correction of the Bidirectional Reflectance Distribution Function |
| CGLOPS1_ATBD_DMP-NPP300m-V1.1 | Algorithm Theoretical Basis Document of 300m Dry Matter Productivity and Net Primary Production Version 1.1 product |
| CGLOPS1_ATBD_FAPAR300m-V1.1 | Algorithm Theoretical Basis Document of 300m LAI, FAPAR, FCOVER Version 1.1 product, derived from Sentinel-3/OLCI data |
| CGLOPS1_QAR_DMP-NPP300m-V1.1 | Quality Assessment Report of DMP, GDMP, NPP and GPP 300m Version 1.1 |

Available at https://land.copernicus.eu/en/technical-library

### 1.3.3 Output documents

| Document ID | Descriptor |
|---|---|
| CGLOPS1_PUM_LSP300m-V1 | Product User Manual of Land Surface Phenology 300m Version 1. |

Available at https://land.copernicus.eu/en/technical-library

### 1.3.4 External documents

| Document ID | Descriptor |
|---|---|
| GCOS#245 | The 2022 GCOS ECVs Requirements |

https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

| | |
|---|---|
| GCOS#200 | The Global Observing System for Climate: Implementation needs. World Meteorological Organization, Geneva, Switzerland (2016) |

https://library.wmo.int/opac/doc_num.php?explnum_id=3417

| | |
|---|---|
| JCGM, 2014 | JCGM, 2014. International Vocabulary of Metrology–Basic and General Concepts and Associated Terms, Chemistry International -- Newsmagazine for IUPAC. Walter de Gruyter GmbH. |

https://doi.org/10.1515/ci.2008.30.6.21

| | |
|---|---|
| HR-VPP_ST&VPP_ATBD | Copernicus Land Monitoring Service. High-Resolution Vegetation Phenology and Productivity (HR-VPP), Seasonal Trajectories and VPP parameters. Algorithm Theoretical Basis Document |
| HR-VPP_ST&VPP_UM | Copernicus Land Monitoring Service. High-Resolution Vegetation Phenology and Productivity (HR-VPP), Seasonal Trajectories and VPP parameters. User Manual |
| HR-VPP_ST&VPP_VR | Copernicus Land Monitoring Service. High-Resolution Vegetation Phenology and Productivity (HR-VPP), Seasonal Trajectories and VPP parameters. Validation Report 2021-2022 |
| MR-VPP_PPI&VPP_UM | Copernicus Land Monitoring Service. Medium-Resolution Vegetation Phenology and Productivity (MR-VPP), Plant Phenology Index and VPP parameters. User Manual |

Available at https://land.copernicus.eu/en/technical-library

| | |
|---|---|
| MOD12Q2_User_Guide_V61 | User Guide to Collection 6.1 MODIS Land Cover Dynamics (MCD12Q2) Product. |

https://lpdaac.usgs.gov/documents/1417/MCD12Q2_User_Guide_V61.pdf

| | |
|---|---|
| VNP22_User_Guide_V1 | VIIRS Global Land Surface Phenology Product User Guide.Version 1.1 |

https://lpdaac.usgs.gov/documents/637/VNP22_User_Guide_V1.pdf

| | |
|---|---|
| MOD17_User_Guide_V61 | User's Guide Daily GPP and Annual NPP (MOD17A2H/A3H) and Year-end GapFilled (MOD17A2HGF/A3HGF) Products. NASA Earth Observing System MODIS Land Algorithm (For Collection 6.1) |

https://lpdaac.usgs.gov/documents/972/MOD17_User_Guide_V61.pdf

| | |
|---|---|
| IPCC 2003 | Intergovernmental Panel on Climate Change - Good Practice Guidance for Land Use, Land-Use Change and Forestry. |

https://www.ipcc.ch/site/assets/uploads/2018/03/GPG_LULUCF_FULLEN.pdf

# 2 REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for the Land Surface Phenology (LSP) are described below.

## 2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

| PRODUCT SPECIFICATION | |
|---|---|
| Geometric Properties: | |
| Spatial resolution | 300 m |
| Geolocation precision | Better than 0.5 pixels |
| Coordinate position | Centre of the pixel |
| Geodetic datum | WGS84 |
| Geographic projection | Regular latitude/longitude grid |
| Geographic coverage | Global |
| **LSP – LAND SURFACE PHENOLOGY** | |
| LSP indicates the timing, date of occurrence, and corresponding value of the vegetation parameter, of key phases of the vegetation development cycle;<br>Phenological metrics related to the start, end of growing season, length of the season and maximum development, seasonal total and partly integrals (as productivity parameters), as well as growth and senescence rates, intra-annual coefficient of variation, etc. | |
| Time span | Yearly |
| Timeliness | after one month of last year observation |
| Uncertainty (2-sigma) | Dates related metrics:<br>Threshold: 10 days,<br>Goal: 5 days |
| Stability (per decade, i.e. 10 years) | Threshold: 10%<br>Goal: 5% |

## 2.2 FURTHER REQUIREMENTS

### 2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

### 2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

### 2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the CA during the FWC.

### 2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

### 2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

### 2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.
The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

# 3 QUALITY ASSESSMENT METHOD

## 3.1 OVERVALL PROCEDURE

The Quality Assessment follows the procedures described in the PQAD [CGLOPS1_PQAD_I2.30]. The procedures are mainly based on the CEOS LPV validation protocol for biophysical variables (Fernandes et al., 2014) and on the state-of-the-art for validation of satellite-based phenology products (Bolton et al., 2020; Melaas et al., 2016; Moon et al., 2021; Peng et al., 2017; Rodriguez-Galiano et al., 2015; Zhang et al., 2018b). The proposed methodology relies on direct validation and product intercomparison approaches.

- The **direct validation,** comparing satellite products to ground measurements of the corresponding biophysical variables. The direct validation is usually limited by a small number of reference sites. PEP725 (see section 3.3.1), TEMPO (section 3.3.2) and NEON (section 3.3.3) datasets provide phenological observations and are used to validate the start and end of the growing season. In addition, PhenoCam V3 (section 3.3.4) infers phenometrics from digital cameras, which are also used for direct comparison with near-ground-based estimates.

- The **indirect validation** includes intercomparison with similar satellite-based products. The indirect validation allows extending the evaluation over large areas or time periods but only relative discrepancies between existing datasets can be assessed. Reference global products providing phenology metrics (MODIS V6.1 and VIIRS V2, see section 3.2.2) and productivity parameters (CLMS OLCI GPP, MOD17A2HGF V6.1, see section 3.2.3) are used in this study, as well as the CLMS Medium and High-Resolution Vegetation Phenology and Productivity (MR-VPP, HR-VPP) products (sections 3.2.5 and 3.2.4) covering Europe. Two different samplings are used depending on the reference product:
    - For global products, the LAND VALidation version 2 (LANDVAL V2) network of sites (Figure 1, left side) is used (Martínez-Sánchez et al., 2024), which provides a global sampling representative of global conditions in term of land cover, continental area and ecoregion. LANDVAL V2 counts with a total of 2000 sites and is based on Copernicus Global Land Cover based on PROBA-V (Buchhorn et al., 2020) and World Wild LiFe Terrestrial Ecoregion of the World classification (Olson et al., 2001). The following quality criteria to select the coordinates along the globe are: i) same land cover and ecoregion (>90% homogeneity) over an area of 3 km x 3 km; ii) distance between samples at least of 2 degrees on the hypotenuse in Plate Carrée projection; and iii) slope within the 3 km x 3 km area lower than 100 m. LANDVAL V2 provides almost identical distribution than the global case, improving the global

representative of LANDVAL V1 used in several validation studies for vegetation LAI/FAPAR or surface albedo products (Fuster et al., 2020; Sánchez-Zapero et al., 2023, 2020).

- For the comparison over Europe with HR-VPP and MR-VPP, the EVALuation network (EVAL) composed by 3800 evaluation sites is used (Camacho et al., 2024). This sampling, representative of the European region, was defined for the validation of HR-VPP products [HR-VPP_ST&VPP_VR]. Sites were defined to be homogeneous over areas of, at least, 700 $m^2$. The sites are well distributed across the European region, as shown in Figure 1 (right side).

These analyses are achieved per aggregated land cover class based on the 8 generic classes derived from the Global Land Cover 100m (GLC-100m) classification (Buchhorn et al., 2020): Evergreen Broadleaf Forest (EBF), Deciduous Broadleaf Forest (DBF), Needle-Leaf Forest (NLF), Other Forests (OF), Cultivated (CUL), Herbaceous (HER), Shrublands (SHR) and Sparse and Bare areas (SBA).

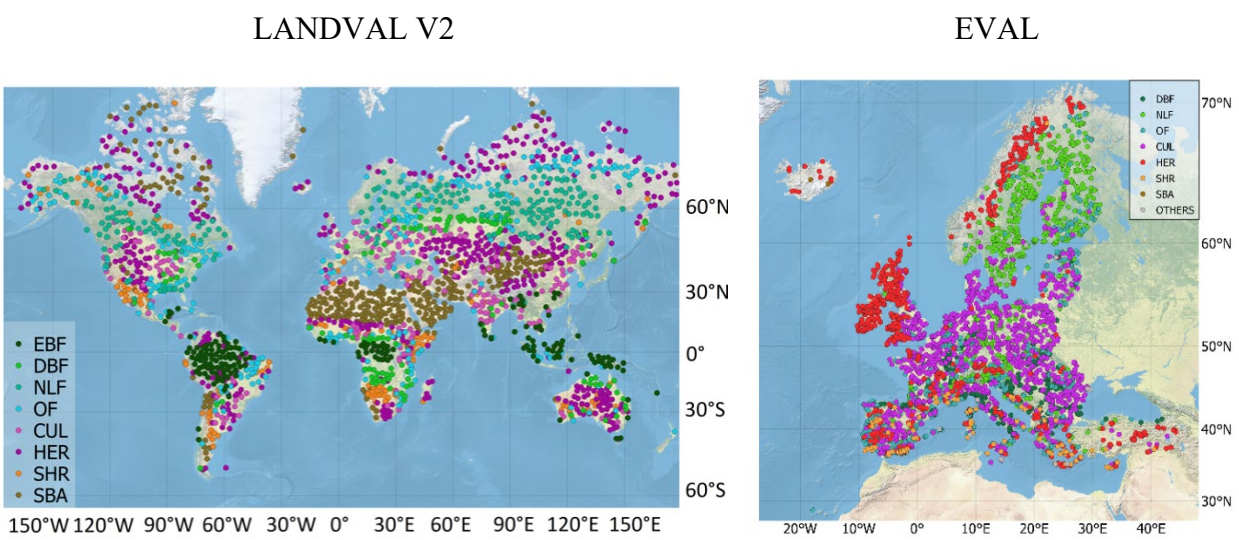



**Figure 1: Sampling strategy for product intercomparison with reference global (LANDVAL V2, left side) and European (EVAL right side) products.**

The following criteria are analysed: product completeness, spatial consistency and error evaluation, which involves the quantification of Accuracy, Precision and Uncertainty (APU). In addition, the conformity test on uncertainty (Chapter 2) is also performed.

### 3.1.1 Product completeness

Completeness corresponds to the absence of spatial and temporal gaps in the data. Missing data are mainly due to cloud or snow contamination, poor atmospheric conditions or technical problems during the acquisition of the images and is generally considered by users as a severe limitation of a given product. It is therefore mandatory to document the completeness of the product. Yearly global maps and percentages of valid and missing data are provided.

### 3.1.2 Spatial consistency

Spatial consistency refers to the realism and repeatability of the spatial distribution of retrievals over the globe. A first qualitative check of the realism and repeatability of spatial distribution of retrievals and the absence of strange patterns or artefacts (e.g., missing values, stripes, unrealistic values, etc.) can be achieved through systematic visual analysis of all global maps based on the expert knowledge of the scientist. The spatial consistency can be quantitatively assessed by comparing the spatial distribution of a reference validated product with the product under study.

- The methodology for visual analysis includes the visualization of the global maps of the product content at reduced resolution (i.e., 1/48), and the zooms over 6 subcontinental areas of 10º latitude x 15º longitude and 6 Areas of Interest (AOIs, see Table 1) of 20 km x 20 km over Europe at full resolution (see Figure 2).
- The spatial consistency with reference products includes:
    - Global maps and histograms of differences with global reference satellite products. Furthermore, global distribution of pixels within the predefined uncertainty levels is estimated.
    - Comparison of maps and histograms with HR-VPP over the 6 AOIs over Europe.
- Additionally, the phenological trend as a function of the latitude is analysed. A latitudinal transect is defined as the geodesic line from Norway (70.315ºN, 20.119ºE) to South Africa (34.872ºS, 20.119ºE) (see Figure 2) with a step of 2 km between samples.
- The spatial variability is computed using the STD and MAD metrics (i.e., indicative of precision) over 6 selected homogeneous Regions of Interest (ROIs) of 5 km x 5 km (Table 2). One ROI for each main biome type (EBF, DBF, NLF, crop, herbaceous, shrublands) is selected for the analysis.

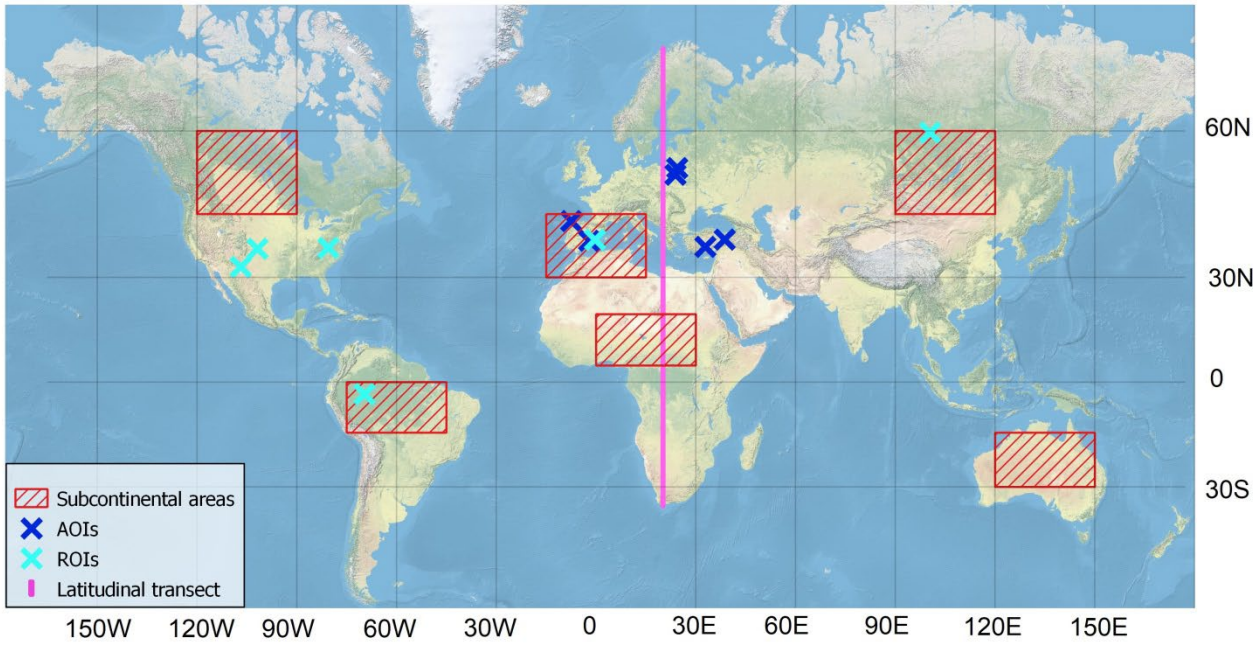
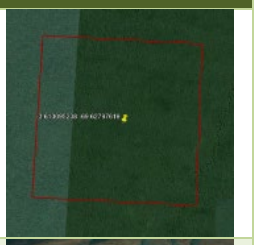
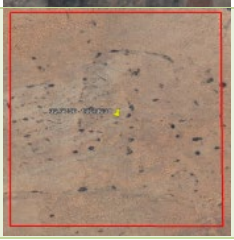



**Figure 2: Location of the subcontinental areas (red boxes), AOIs (dark blue crosses), ROIs (light blue crosses) and latitudinal transect (pink vertical line) used for spatial consistency.**

**Table 1: Summary of central coordinates of selected AOIs (20km x 20km) for spatial consistency.**

| No. | Name | Biome | Central coordinates | |
|---|---|---|---|---|
| | | | Latitude | Longitude |
| **1** | Barrax | CUL | 39.05437° N | -2.10068° E |
| **2** | LUCAS 2777 | DBF | 43.497616° N | -7.482012° E |
| **3** | Bialowieza National Park | DBF | 52.749876° N | 23.867391° E |
| **4** | LUCAS 1208 | NLF | 53.942444° N | 24.406548° E |
| **5** | EOLAB 53 | CUL | 37.619363° N | 32.86563° E |
| **6** | Spar 113 | HER | 39.326087° N | 38.681288° E |

**Table 2: Relevant information and central coordinates of selected ROIs (5 km x 5 km).**

| ROI # | Name | Country | Biome | Central coordinates | | Google Earth View |
|---|---|---|---|---|---|---|
| | | | | Latitude | Longitude | |
| **1** | Evare | Brazil | EBF | -3.6131° N | -69.6280° E |  |
| **2** | Mountain Lake | USA (Virginia) | DBF | 37.3783° N | -80.5248° E | |
| **3** | Oskoba | Russia | NLF | 59.7619° N | 100.4345° E | |
| **4** | Albufera | Spain | Crop (rice) | 39.2734° N | -0.3188° E | |
| **5** | Cimarron National Grassland | USA (Kansas) | Herbaceous | 37.1369° N | -101.8065° E | |
| **6** | Jornada | USA (New Mexico) | Shrublands | 32.6756 | -106.8631° E |  |


© C-GLOPS Lot1 consortium

### 3.1.3 Error evaluation

Accuracy is the degree of the "*closeness of the agreement between the result of a measurement and a true value of the measurand*" [JCGM, 2014]. Commonly, accuracy represents systematic errors and often is computed as the statistical mean bias (B), i. e., the difference between the short-term average measured value of a variable and the true value. The short-term average is the average of a sufficient number of successive measurements of the variable under identical conditions, such that the random error is negligible relative to the systematic error. The latter can be introduced by instrument biases or through the choice of remote sensing retrieval schemes [GCOS#200].

Precision or repeatability is the "*closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement*" [JCGM, 2014]. Commonly, precision represents the dispersion of product retrievals around their expected value and can be estimated by the standard deviation of the difference (STD) between retrieved satellite product and the corresponding reference estimates.

Uncertainty is a "*parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand*" [JCGM, 2014]. Uncertainty includes systematic and random errors and can be estimated by the Root Mean Square Deviation (RMSD).

In addition to these metrics, other statistics are useful to evaluate the goodness of fit between two datasets including linear model fits. For this purpose, Major Axis Regression (MAR) is computed instead Ordinary Least Squares (OLS) because is specifically formulated to handle error in both of the x and y variables (Harper, 2014). Other metrics are used, such as number of samples (N), indicative of the power of the validation, or the correlation coefficient (R, estimated as Pearson coefficient), which indicates descriptive power of the linear accuracy test.

It should be noted that strong and/or multiple outliers affect the classical metrics described above (i.e., B and STD): in such cases using the median deviation (MD) instead of the mean bias to estimate systematic error and the median absolute deviation (MAD) as a measure of precision is preferable.

APU quantities are evaluated by several metrics (Table 3) reporting the goodness of fit between the products and the corresponding reference dataset. Scatter-plots and validation metrics (Table 3) versus references are produced. This analysis is complemented per main biome type.

**Table 3: Validation metrics. 'x' and 'y' refer to reference and evaluated products.**

| Statistics | Comment |
|---|---|
| **N** | Number of samples. Indicative of the power of the validation |
| **B** | Mean Bias. Difference between average values of y and x (y-x). Indicative of accuracy and offset.<br>Bias (%) is the relative mean bias between the average of x and y. |
| **MD** | Median deviation between y and x. Best practice reporting the accuracy.<br>MD (%) is the relative MD between the average of x and y. |
| **STD** | Standard deviation of the pair differences. Indicates precision.<br>STD (%) is the relative STD between the average of x and y. |
| **MAD** | Median absolute deviation between x and y. Best practice reporting the precision.<br>MAD (%) is the relative MAD between the average of x and y. |
| **RMSD** | Root Mean Square Deviation. RMSD is the square root of the average of squared errors between x and y. Best practice reporting uncertainty.<br>RMSD (%) is the relative RMSD between the average of x and y. |
| **MAR** | Slope and offset of the Major Axis Regression linear fit. Indicates some possible bias |
| **R** | Correlation coefficient. Indicates descriptive power of the linear accuracy test.<br>Pearson coefficient is used. |

### 3.1.4 Summary of the Quality Assessment Procedure

The quality assessment is mainly focused on the two key phenological metrics (SOSD and EOSD) for which ground observations are available and user requirements defined. Moreover, the two productivity (SPROD and TPROD) layers are evaluated against similar satellite-based references. For the other CLMS OLCI LSP layers, the completeness and the product content are also analyzed. The summary of the quality assessment procedures for each criterion is provided in Table 4.

**Table 4: Summary of the QA procedure for CLMS OLCI LSP validation.**

| Quality Criteria | Evaluated CLMS OLCI LSP layer | Reference Product | Coverage |
|---|---|---|---|
| **Completeness** | All | MODIS LCD V6.1<br>VIIRS LCD V2 | Global |
| | Global maps of valid and missing data. Percentage of valid and missing data. | | |
| **Spatial Consistency** | All | - | Global and subcontinental regions |
| | Maps of product content layers. | | |
| | SOSD, EOSD<br>SPROD, TPROD | MR-VPP | Europe |
| | Global maps and histograms of differences. Pixels within user requirements. | | |
| | SOSD, EOSD,<br>SPROD, TPROD | HR-VPP | AOIs |
| | Visual inspection of maps and histograms of values. | | |
| | SOSD, EOSD | MODIS LCD V6.1<br>VIIRS LCD V2 | Global<br>Latitudinal transect |
| | Global maps of differences and phenological trend as a function of the latitude. | | |
| | SOSD, EOSD,<br>SPROD, TPROD | - | ROIs |
| | Visual inspection of maps and spatial variability (precision) | | |
| **Error evaluation (direct validation)** | SOSD, EOSD | PEP725, TEMPO,<br>NEON, Phenocam | PEP725, TEMPO,<br>NEON, Phenocam |
| | Scatter-plots and validation metrics. Conformity test. | | |
| **Error evaluation (product intercomparison)** | SOSD, EOSD | MR-VPP, HR-VPP<br>MODIS LCD V6.1, VIIRS LCD V2 | EVAL<br>LANDVAL V2 |
| | Scatter-plots and validation metrics. Conformity test. Analysis per biome type. | | |
| | SPROD, TPROD | CLMS OLCI GPP<br>MODIS GPP<br>MR-VPP, HR-VPP | LANDVAL V2<br>LANDVAL V2<br>EVAL |
| | Scatter-plots and validation metrics. Conformity test. Analysis per biome type. | | |

## 3.2 SATELLITE PRODUCTS

This section provides an overview of the retrieval algorithms of the satellite-based products used in this exercise. The main characteristics of the satellite products intercomparison are summarized per two families: phenology (Table 5) and productivity (Table 6). Summary of retrieval algorithms are provided hereafter.

**Table 5: Characteristics of the remote sensing phenology products used in this exercise.**

| Product | Satellite / Sensor | Resolution / Period | Coverage | Input: | Fitting / Smoothing methods | Phenology Transition | Reference |
|---|---|---|---|---|---|---|---|
| CLMS OLCI LSP* | Sentinel-3 / OLCI | 300m / 2023 | Global | BRDF-normalized PPI | Cubic spline and double-logistic fitting | 25% and 15% of amplitude for green-up and green-down | [CGLOPS1_ATBD_LSP 300m-V1] |
| MODIS LCD V6.1 | Terra / MODIS | 500m / 2001-current | Global | NBAR EVI2 | Cubic spline fitting | 15%, 50%, and 90% of amplitude for green-up and green-down | (Zhang et al., 2006) [MOD12Q2_User_Guide_V61] |
| VIIRS LCD V2 | -SNPP / VIIRS | 500m / 2013-current | Global | NBAR EVI2 | Stavisky-Golay filter, mean/median filter, logistic fitting | Third derivative (extreme curvature change rate) | (Zhang et al., 2018b) [VNP22_User_Guide_V1] |
| MR-VPP | Terra+Aqua / MODIS | 500 m / 2000-current | Europe | BRDF-normalized PPI | Cubic spline and double-logistic fitting | 25% and 15% of amplitude for green-up and green-down | [MR-VPP_PPI&VPP_UM] |
| HR-VPP | Sentinel-2A & B / MSI | 10m / 2017-current | Europe | PPI | Cubic spline and double-logistic fitting | 25% and 15% of amplitude for green-up and green-down | [HR-VPP_ST&VPP_ATBD] |

*dataset under evaluation

**Table 6: Characteristics of the remote sensing productivity products used in this exercise.**

| Product | Satellite / Sensor | Resolution / Period | Coverage | Input | Method | Reference ATBD |
|---|---|---|---|---|---|---|
| CLMS OLCI LSP* | Sentinel-3 / OLCI | 300m / 2023 | Global | BRDF-normalized PPI | PPI integral between SOSD and EOSD | [CGLOPS1_ATBD_LSP300m-V1] |
| CLMS OLCI GPP | Sentinel-3 / OLCI | 300m / 2020-present | Global | FAPAR | (Monteith, 1972) | [CGLOPS1_ATBD_DMP-NPP300m-V1.1]. |
| MODIS GPP (MOD17A2HGF V6.1) | Terra / MODIS | 500m / 2001-current | Global | FAPAR | (Monteith, 1972) | [MOD17_User_Guide_V61] |
| MR-VPP | Terra+Aqua / MODIS | 500 m / 2000-current | Europe | BRDF-normalized PPI | PPI integral between SOSD and EOSD | [MR-VPP_PPI&VPP_UM] |
| HR-VPP | Sentinel-2A & B / MSI | 10m / 2017-current | Europe | PPI | PPI integral between SOSD and EOSD | [HR-VPP_ST&VPP_ATBD] |

*dataset under evaluation

The following quality flag information of the different products was used to select the good quality pixels (Table 7).

**Table 7: Quality Flag information used to select good quality pixels.**

| Product | Quality Flag |
|---|---|
| **CLMS OLCI LSP** | QA <= 127 (overall good quality) |
| **MODS LCD V6.1** | QA_ Overall <= 1 (best/good) |
| **VIIRS LCD V2** | GLSP_QC bits 0-2='000' (processed, good quality) |
| **MODIS GPP** | Not used, as recommended in the user guide [MOD17_User_Guide_V61]. |
| **CLMS OLCI GPP** | Not used, as heritage from CLMS FAPAR 300m products [CGLOPS1_ATBD_FAPAR300m-V1.1]. |
| **MR-VPP** | Not provided, as quality filter is used for MODIS input data. [MR-VPP_PPI&VPP_UM] |
| **HR-VPP** | QF >= 7 (high and medium quality data) |

Satellite products must be compared over a similar spatial support. To allow comparison between the products, the same spatial support of 1 $km^2$ is used (i. e., 3x3 pixels in case of CLMS products, 2x2 pixels in case of MODIS and VIIRS based products and 100x100 pixels in case of HR-VPP). Reference products (e.g., MR-VPP/VIIRS) are re-sampled on the *Plate carrée* projection over 1/112º which is the CLMS grid.

For the temporal match-up between products, different approaches are performed:

- In the case of SOSD and EOSD error evaluation (quantitative evaluation) with equivalent products, only retrievals provided for year 2023 are considered, excluding SOSD/EOSD dates that correspond with 2022 or 2024 adjacent years. For the spatial consistency, dates of different years (i.e., 2022, 2024) are used with the aim of covering the spatial gaps of the maps.
- In the case of TPROD and SPROD productivity layers, the productivity of reference CLMS OLCI and MODIS GPP products is computed between the SOSD and EOSD OLCI LSP product through a discrete integral for the season period.

### 3.2.1 Evaluated dataset: CLMS OLCI LSP Collection 300m V1 (OLCI LSP)

The calculation of annual LSP products [CGLOPS1_ATBD_LSP300m-V1] is based on the following four steps:

(1) Data preparation: generating nadir BRDF-corrected TOC Adjusted Reflectance (NBAR) using Regularised BRDF Inversion for Land Surface (ReBeLS) retrieval algorithm [CGLOPS1_ATBD_BRDFCorrection300m-V1] and collecting European Centre for Medium-Range Weather Forecasts Reanalysis v5 (ECMWF ERA5) diffuse fraction of solar radiation.

(2) Generation of the PPI, which is a physically based vegetation index developed from a simplified solution to the radiative transfer equation (Jin and Eklundh, 2014), from TOC BRDF-corrected measurements at 300 m spatial resolution and five-day time step.

(3) To smooth and gap-fill data, PPI seasonal trajectories are generated, which are based on cubic spline functions implemented in a program for analyzing time-series of satellite sensor data (TIMESAT - Jönsson and Eklundh, 2004), version 4.2. These functions adapt smoothly to seasonal and inter-annual vegetation index variations. Cubic spline functions have high fidelity and flexibility to represent satellite measurements of plant phenology (Bolton et al., 2020; Cai et al., 2017).

(4) From the seasonal trajectories (data not stored as output in the CGLOPS product), 13 phenology and productivity parameters for up to two growing seasons per year are extracted (Figure 3). The computation of start and end of seasons is based on defined threshold points on the smoothed seasonal trajectory. TIMESAT determines the timing of start of season (SOSD) and end of season (EOSD) by applying a threshold value in relation to the seasonal amplitude (25% for SOSD and 15% for EOSD). Once start and end dates have been defined, the other parameters are extracted.

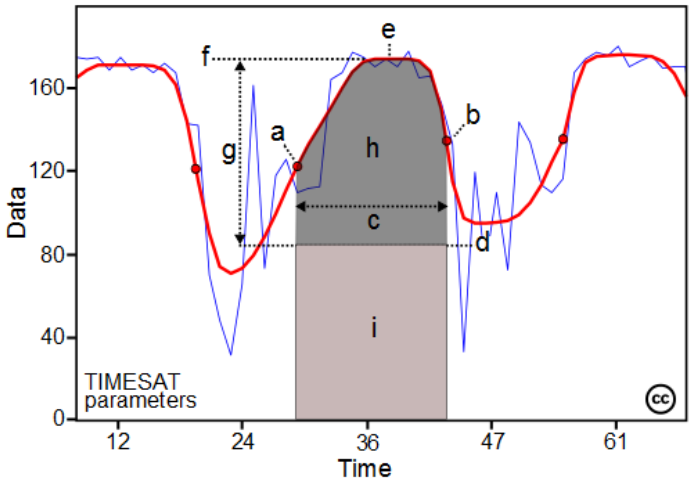



**Figure 3. Examples of TIMESAT-derived phenology and productivity parameters: (a) start of season, (b) end of season, (c) length of season, (d) minimum value, (e) time of maximum, (f) maximum value, (g) amplitude, (h) seasonal productivity, (h+i) total productivity. Source: https://web.nateko.lu.se/timesat/timesat.asp.**

It should be noted that, to generate the LSP products for the year 2023, Sentinel-3/OLCI NBAR (red and near infrared channels) data from January 2023 to February 2024 were used and PPI products were internally produced for this period.

### 3.2.2 Reference phenology global products

#### *3.2.2.1 NASA Land Cover Dynamics Version 6.1 (MODIS LCD V6.1)*

The Terra and Aqua combined Moderate Resolution Imaging Spectroradiometer (MODIS) Land Cover Dynamics (MCD12Q2) Version 6.1 data product[3] provides global maps of phenometrics at 500 m with an annual time step since 2001 (Zhang et al., 2006) [MOD12Q2_User_Guide_V61]. The MCD12Q2 Version 6.1 data product is derived from time series of the 2-band Enhanced Vegetation Index (EVI2) calculated from MODIS Nadir BRDF-NBAR. Vegetation phenology metrics at 500-meter spatial resolution are identified for up to two detected growing cycles per year. For pixels with more than two valid vegetation cycles, the data represent the two cycles with the largest NBAR-EVI2 amplitudes. For areas where the NBAR-EVI2 values are missing due to cloud cover or other reasons, the data gaps are filled with good quality NBAR-EVI2 values from the year directly preceding or following the product year.

The layers provided by the product correspond to the total number of vegetation cycles detected for the product year (up to two cycles/year), the onset of greenness, green-up midpoint, maturity, peak greenness, senescence, green-down midpoint, dormancy, EVI2 minimum, EVI2 amplitude, integrated EVI2 over a vegetation cycle, as well as overall and phenology metric-specific quality information. MCD22Q2 V6.1 provides different phenological dates using 15% (Greenup and Dormancy for start and end of season), 50% (MidGreenup and MidGreendown), and 90% (Maturity and Senescence) EVI2 amplitude.

For the product intercomparison with CLMS LSP, Greenup, MidGreenup and Maturity are compared with SOSD. Senescence, MidGreendown and Dormancy are compared with CLMS LSP EOSD. Additionally, the quality layer (‘QA_detailed’) to identify pixels with ‘poor’ and ‘fair’ quality, and only ‘Best’ and ‘Good’ retrievals are used in the comparison.

Comparison of vegetation phenology retrieved from MODIS with in situ measurements revealed realistic estimates over Harvard Forest (Zhang et al., 2006). More generally, the spatial distribution of phenological metrics estimated from MODIS data were qualitatively realistic, and exhibits strong correspondence with temperature patterns in mid- and high-latitude climates, with rainfall seasonality in seasonally dry climates, and with cropping patterns in agricultural areas (Zhang et al., 2006).

[3] https://lpdaac.usgs.gov/products/mcd12q2v061/ (accessed on 2025/05/30)

#### *3.2.2.2 NASA VIIRS Land Cover Dynamics Version 2 (VIIRS LCD V2)*

The Suomi National Polar-Orbiting Partnership (SNPP) National Aeronautics and Space Agency (NASA) VIIRS Land Cover Dynamics data product[4] generates phenological metrics of individual vegetation growing cycles on the vegetated land surface at a spatial resolution of 500m across the globe (Zhang et al., 2018b) [VNP22_User_Guide_V1]. The VNP22Q2 V2 dataset is derived from time series of 2-band Enhanced Vegetation Index (EVI2) calculated from VIIRS Nadir BRDF-NBAR. Vegetation phenology metrics at 500-meter spatial resolution are identified for up to two detected growing cycles per year. VIIRS V2 products are generated with same algorithm of previous V1 but introduces some improvements or changes:

- Improved calibration algorithm and better coefficients for entire SNPP mission.
- Improved geolocation accuracy and updates to fix outliers around maneuver periods and other events.
- Corrections to the aerosol quantity flag (low, average, high) mainly over brighter surfaces in the mid to high latitudes such as desert and tropical vegetation areas. This has an impact on the retrieval of other downstream data products such as VIIRS Vegetation Indices and BRDF/Albedo.
- Improved cloud mask input product for corrections along coastlines and artifacts from use of coarse resolution climatology data.
- Replaced the land/water mask input product with MODIS heritage seven class land/water mask.

The main changes respect to the predecessor based on MODIS (MCD12Q2 V6.1) are: first, VNP22Q2 reconstructs the EVI2 time series using a physically-based hybrid piecewise logistic model (HPLM), while MCD12Q2 V6.1 fits the EVI2 time series using a penalized cubic smoothing spline. Second, VNP22Q2 identifies phenological transition dates using the curvature change rate derived from the HPLM without any predefined thresholds involved, whereas MCD12Q2 V6.1 detects phenological dates using thresholds of EVI2 amplitude. Third, VNP22Q2 records the phenological metrics occurring within a given calendar year or product year. However, MCD12Q2 V6.1 records the metrics for the vegetation cycles for which the peak date falls within the calendar year of the product year.

The product contains six phenological transition dates: onset of greenness increase, onset of greenness maximum, onset of greenness decrease, onset of greenness minimum, dates of mid-green-up, and

[4] https://lpdaac.usgs.gov/products/vnp22q2v002/ (accessed on 2024/03/06)

mid-senescence phases. The product also includes the growing season length. The greenness related metrics consist of EVI2 onset of greenness increase, EVI2 onset of greenness maximum, EVI2 growing season, rate of greenness increase and rate of greenness decrease. The confidence of phenology detection is provided as greenness agreement growing season, proportion of good quality (PGQ) growing season, PGQ onset greenness increase, PGQ onset greenness maximum, PGQ onset greenness decrease, and PGQ onset greenness minimum. The final layer (GLSP_QC) specifies the overall quality of the product. A low-resolution browse image showing green-up is also available when viewing each VNP22Q2 granule.

For comparison with CLMS LSP, CLMS SOSD is compared with VIIRS onset of greenness increase, mid-greenup and maximum dates whilst CLMS EOSD is compared with VIIRS onset of greenness decrease, mid-senescence and minimum dates. VIIRS V2 pixels with overall quality 'processed, good quality' according to the GLSP_QC layer are used (except for the spatial consistency analysis for sake of completeness).

Comparison of VNP22Q2 resulting phenological metrics over United States with species-specific field phenological observations, Landsat phenology retrievals and the MODIS phenology detections showed that the VIIRS phenometrics are of high quality and in a good agreement with the other independent satellite and field observations (Zhang et al., 2018b). Globally, the phenological dates derived from VIIRS EVI2 were compared with MODIS detections (Zhang et al., 2018a), which showed a good agreement with an average absolute difference less than a week except for the senescence onset. Phenological dates derived from VIIRS EVI2 and PhenoCam were generally comparable for the vegetation green-up phase, but differed considerably for the senescence phase (Zhang et al., 2018a). The continuity between VIIRS V2 and MODIS V6.1 phenometrics was assessed and they agreed well and had little systematic bias (Román et al., 2024).

### 3.2.3 Reference global productivity products

#### *3.2.3.1 CLMS OLCI GPP Collection 300m V1.1 (OLCI GPP)*

The CLMS OLCI GPP is provided in the CLMS OLCI DMP/GDMP & NPP/GPP V1.1 product suite [CGLOPS1_ATBD_DMP-NPP300m-V1.1]. Dry Matter Productivity (DMP) is one of the bio-geophysical products delivered by the CLMS, representing the daily growth of standing biomass. Next to DMP, the Gross Dry Matter Productivity (GDMP) is also provided. GDMP equals DMP without autotrophic respiration. Net Primary Production (NPP) expresses the rate at which plants produce net useful chemical energy expressed in mass of carbon per unit area per day ($gC/m^2/day$). NPP is a rescaling of the DMP product to fulfill the needs of the climate community. Similar to GDMP, also Gross Primary Production (GPP) is provided, which is the NPP without considering the

autotrophic respiration. The product is globally delivered every 10 days with a maximum of 3 days in Near Real Time (NRT) followed by consolidation in the course after 6 dekads.

The algorithm is built from the simplified Monteith (Monteith, 1972) approach but uses as input improved meteorological data, the fraction of photosynthetically active radiation absorbed by the green elements of the canopy (fAPAR) and land cover information. The FAPAR is obtained from CLMS OLCI 300m FAPAR V1.1 10 daily products [CGLOPS1_ATBD_FAPAR300m-V1.1]. The algorithm uses biome specific Light Use Efficiency (LUE) and the information of land cover types is based on European Space Agency (ESA) Climate Change Initiative (CCI) Land Cover Map epoch 2010.

The DMP is an indicator of the dry matter biomass increase or growth rate of the vegetation. The DMP is directly related to Net Primary Productivity (NPP), but customized for agro-statistics and expressed in kilograms of dry matter (kgDM) per hectare per day. GDMP is the total amount of dry matter "fixed" by land plants per unit time through photosynthesis and is directly related to GPP. Both, DMP and GDMP represent the above and below ground part of plants. The main difference between DMP/GDMP and GPP/NPP is that the first ones are expressed in kgDM/ha/day and the second ones in gram carbon per square meter per day ($gC/m^2/day$). For the conversion of Dry Matter (DM) into Carbon (C) the conversion factor defined in the Intergovernmental Panel on Climate Change (IPCC) Good Practice Guide [IPCC 2003] can be used which states that vegetation dry matter generally consists of 50% of carbon. So gC/gDM = 0.5. So, to summarize, NPP and GPP are calculated from DMP and GDMP as follows:

$$NPP\ [gC/m^2/day] = DMP\ [kgDM/ha/day] \times 0.5 \times 0.1$$

$$GPP\ [gC/m^2/day] = GDMP\ [kgDM/ha/day] \times 0.5 \times 0.1$$

For comparison with CLMS LSP SPROD and TPROD, the sum of CLMS GPP values between CLMS LSP SOSD and EOSD is computed, as CLMS LSP TPROD are defined as the sum of all PPI daily values between SOSD and EOSD (SPROD is the same minus the base level value).

QAR [CGLOPS1_QAR_DMP-NPP300m-V1.1] for July 2018 – April 2019 period revealed that, in the overall global comparisons with MODIS, CLMS GDMP OLCI product exceeds MODIS for most biomes. A validation with Fluxes Network (FLUXNET) data spread over 66 sites shows that there is a fairly good overall relation between satellite-based products and the FLUXNET in situ based GDMP, but the uncertainty is high and precision of the relation is low. The results obtained in this report are also valid for NPP and GPP, as these are a simple rescaling from kgDM/ha/day to $gC/m^2/day$, taking into account the Carbon Use Efficiency (CUE), which is taken as a constant term (0.5 according to [IPCC 2003]).

#### *3.2.3.2 NASA MODIS Gross Primary Productivity (MODIS GPP) Version 6.1*

The NASA MODIS GPP (provided in MOD17A2H V6.1 dataset) product[5] consists in a cumulative 8-days composite of values based on the radiation use efficiency concept at 500 m spatial resolution [MOD17_User_Guide_V61]. The dataset includes GPP and Net Photosynthesis (PSN), which is the GPP less the Maintenance Respiration (MR). The dataset also contains PSN Quality control (QC) layer, which is the quality information for GPP and PSN.

The MOD17A2H V6.1 uses as input the satellite-derived Fraction of Photosynthetically Active Radiation (FPAR) (from MOD15), independent estimates of PAR, other surface meteorological fields from NASA Global Modelling and Assimilation Office (GMAO) and estimation of growth respiration and maintenance that are subtracted from GPP to arrive NPP. The 8-day estimate of FPAR from MOD15 and the daily estimated PAR from GMAO are multiplied to produce daily Absorbed Photosynthetically Active Radiation (APAR) per pixel. Based on the land cover product MOD12, a set of biome-specific radiation use efficiency parameters are extracted from the Biome Properties Look-Up Table (BPLUT).The Version 6.1 fixes some errors, contaminations or data gaps included in Version 6 (Zhao et al., 2005).

In this analysis, the gap-filled method (i.e., MOD17A2HGF) is used, which increases the quality of MOD17A2H with the unique limitation that only can be applied at the end of the year, as the gap-filled introduces the climatology of FPAR (i.e., MOD15A2HCL product).

Wang et al., (2017) tested MOD17A2H, version 6 across 18 tower sites in 6 biomes and found that MOD17A2H GPP underestimated flux-derived GPP at most sites. Its performance in estimating annual GPP was poor ($R^2 = 0.62$). Another study of grassland biomes comparing MOD17A2H against flux tower data for grasslands in tropical, temperate and alpine environments showed high correlations and Root Mean Square Error (RMSE) of around 2.5 $gCm^{-2}\,day^{-1}$ (Zhu et al., 2018).

For comparison with CLMS LSP SPROD and TPROD, the sum of MODIS GPP cumulative values between CLMS LSP SOSD and EOSD is computed, as CLMS LSP TPROD are defined as the sum of all PPI daily values between SOSD and EOSD (SPROD is the same minus their base level value).

### 3.2.4 Pan-European High-Resolution Vegetation Phenology and Productivity (HR-VPP)

The High-Resolution Vegetation Phenology and Productivity product suite (HR-VPP), produced within the Pan-European component of the Copernicus Land Monitoring Service (CLMS), are derived from the Sentinel-2 constellation (Sentinel-2A and Sentinel-2B) at high spatial resolution (10m x 10m) with a revisit time 5 days [HR-VPP_ST&VPP_UM]. They are generated over the entire

[5] https://lpdaac.usgs.gov/products/mod17a2hv061/ (accessed on 2024/04/02)

EEA39 (32-member countries, the United Kingdom (UK) and 6 cooperating countries in the Western Balkans) from January 1, 2017 onwards.

The HR-VPP product suite contains 3 product groups:

- The **raw Vegetation Indices** (VIs). This group consists of products generated in near real-time (NRT) providing, for every pixel, the status of the vegetation vigor. The status is provided in four ways: LAI, FAPAR, NDVI and PPI;
- The **Seasonal Trajectories** (STs) products are provided yearly after the end of the vegetation growing season. These are derived as a regular time-series of every 10 days by fitting a smoothing and gap filling function to the raw Plant Phenology Index, generated in the product group VIs. TIMESAT version 4 software package (Jönsson and Eklundh, 2004) is used to produce STs from PPI retrievals;
- The **Vegetation Phenology and Productivity** parameters (VPPs) products bundle are derived from the STs of the PPI index, on a yearly basis, after the end of the growing season. VPP metrics are provided for up to two growing seasons, being e.g. start of the season, end of season, seasonal productivity, etc. These metrics are defined in concordance with CLMS LSP products, following similar algorithm [HR-VPP_ST&VPP_ATBD] (Tian et al., 2021).

CLMS OLCI LSP SOSD/EOSD and TPRO/SPROD are compared with equivalent HR-VPP layers, which have the same definition.

Validation of HR-VPP phenology metrics for 2021-2022 period [HR-VPP_ST&VPP_VR] showed good agreement with existing medium-resolution products (MR-VPP, MOD12Q2 V6.1 and VNP22Q2 V1) over Europe. The accuracy of phenology metrics (SOSD, EOSD) as compared with ground PEP725 observations showed a slight positive bias of 4 days for SOSD (i.e., late start of season) and negative bias of –6 days for EOSD (i.e., early end of season).

### 3.2.5 Pan-European Medium-Resolution Vegetation Phenology and Productivity (MR-VPP) Version 4

The MR-VPP Version4 (V4) dataset is providing the same phenology information using a similar algorithm than for HR-VPP [MR-VPP_PPI&VPP_UM], but at a lower spatial resolution (500 m) and derived from MODIS data. The input products are MCD43A2 (BRDF Albedo LocalSolarNoon) and MCD43A4 (NBAR bands 1 (RED), 2 (NIR) and 6 (SWIR)) with a revisit time of 5 days. They are generated over the entire EEA39 (32 member countries, the UK and 6 cooperating countries in the Western Balkans)

MR-VPP date of start of season (DSOS, 25% of the season peak) and date of end of season (DEOS, 15% of the season peak) are compared with CLMS SOSD and EOSD products. The LINT (Large

INTegral) and SINT (Small INTegral) are similar to the CLMS OLCI TPROD and SPROD respectively.

MR-VPP phenology metrics were validated against PEP725 dataset for 2021-2022 period [HR-VPP_ST&VPP_VR] with anticipated start (B=-5days) and end (B=-16 days) of season than ground references. The comparison of MR-VPP versus HR-VPP over EVAL [HR-VPP_ST&VPP_VR] showed earlier start of season of 3 days and delayed end of season of 8 days in case of MR-VPP (i.e., MR-VPP provided longer season of around 11 days than HR-VPP).

### 3.3 IN-SITU REFERENCE PRODUCTS

This section provides the description of ground references available for 2023 used in direct validation. Figure 4 shows the distribution of 511 PEP725, 383 TEMPO, 37 NEON and 200 Phenocam V3 validation points used for 2023.

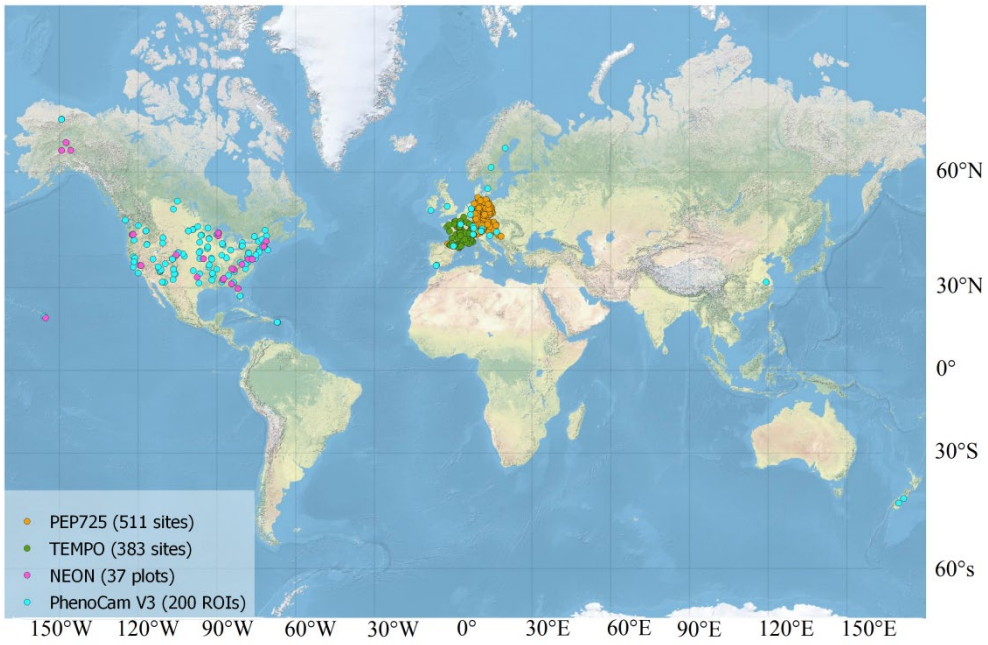



**Figure 4: Distribution of PEP725, TEMPO, NEON and PhenoCam locations used in direct validation for 2023.**

#### 3.3.1 Pan-European Phenology Project (PEP725)

The Pan-European Phenology Project (PEP725, http://www.pep725.eu/) has the objective to establish an open access database of plant phenology for science, education and research. The database presently holds almost 12 million records, about 46 growing stages and 265 plant species (including

cultivars). However, this dataset has some limitations for remote sensing validation purposes as is not evenly distributed around the world and the geolocation accuracy of the observations is often poor (2 – 4 km). For that reason, a strategy using CORINE Land Cover (CLC) was performed similar to that used in Tian et al., 2021. Corine Land Cover 2018 (v2020_20u1) provides inventory of land cover in 44 different classes, with a Minimum Mapping Unit (MMU) of 35 ha for areal phenomena and a minimum width of 100 m for linear phenomena in Europe. Several species were grouped into two vegetation groups (Table 8).

**Table 8: Grouping strategy for linking OLCI LSP with PEP725 phenology observations.**

| Vegetation group | PEP725 genus | CLC classes |
|---|---|---|
| Deciduous broad-leaved trees | Acer, Aesculus, Alnus, Betula, Fagus, Fraxinus, Quercus, Robinia, Salix, Sambucus, Sorbus, Syringa and Tilia. | Broad-leaved forest, Transitional woodland-shrub, Mixed forest, Green urban areas. |
| Evergreen coniferous trees | Picea, Larix. | Coniferous forest. |

For each site, a 4 km x 4 km window around the station is selected, then the CLC is resampled to the OLCI LSP 300m spatial resolution and the percentage of each CLC class is computed. Then, the pixels corresponding to the main vegetation group (i.e., deciduous broad-leaved trees or evergreen coniferous trees) are identified. The median OLCI LSP SOSD and EOSD values for each vegetation group is obtained and only used in the comparison if the vegetation group represents more than 10% of the area. Figure 5 shows an example of aggregated vegetation group for PEP725 site ID=110. In the example, only DBF pixels are used in the comparison, as represent 35.6% in the 4 km x 4 km area. The phases considered in the comparison are the same to that used in Tian et al., 2021 and are displayed in Table 9.

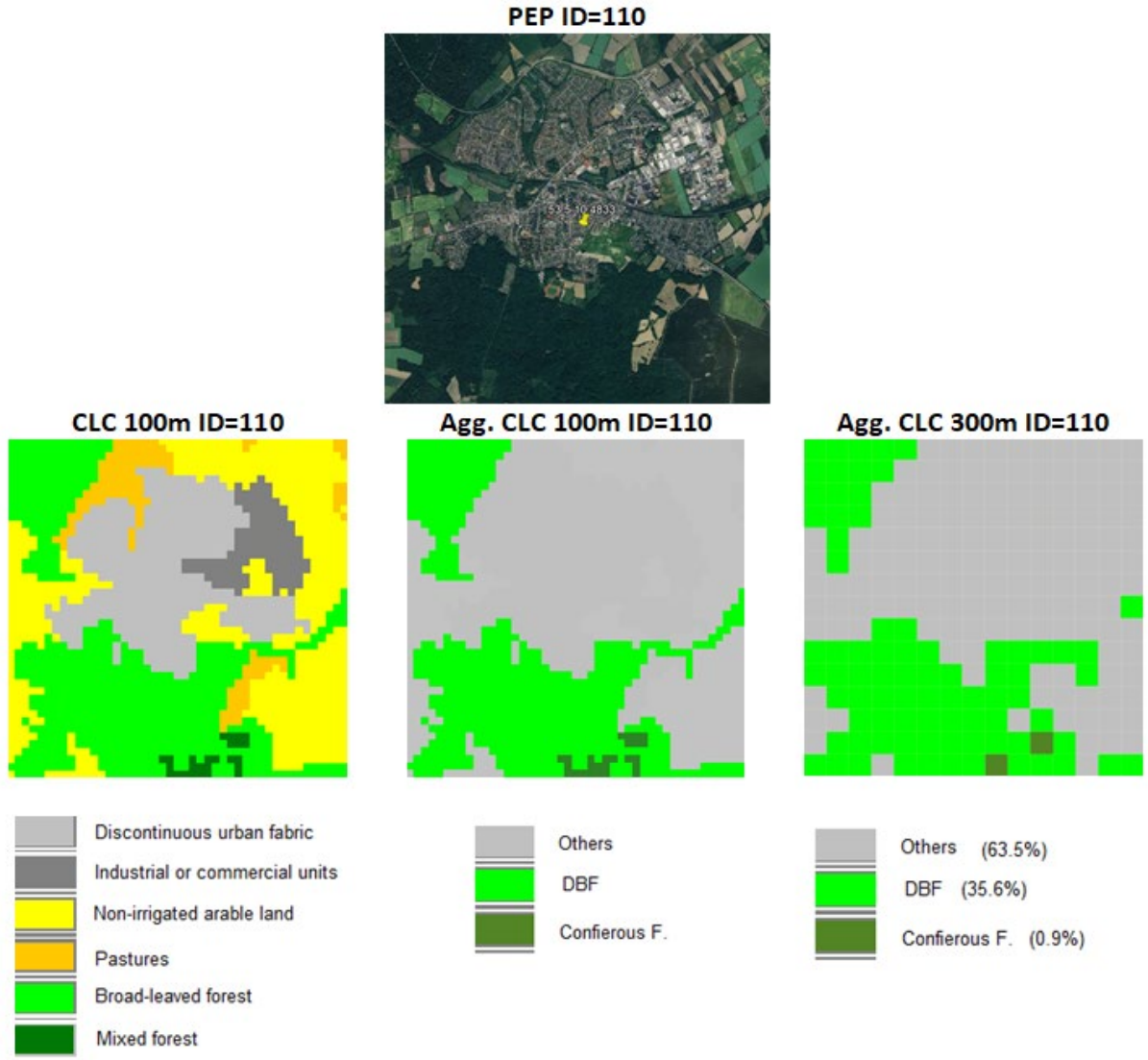



**Figure 5: Vegetation group strategy. Top: Google Earth view of 4 km x 4 km around PEP site ID=110. Bottom: CLC around the site at 100m of spatial resolution (left), aggregated CLC around the site at 100 m resolution (center) and aggregated CLC around the site resample at 300 m resolution (right).**

**Table 9: Phases used for linking CLMS OLCI LSP SOSD and EOSD with PEP725 phenology observations.**

| PEP725 Phase_id | Description | Short name | Metrics |
|---|---|---|---|
| **10** | Gramineae: Fist true leaf emerged from coleoptile, Dicotyledons and Monocotyledons: Cotyledons completely unfolded, Perennial plants: First leaves separated | First leaves separated | SOSD |
| **11** | First true leaf, leaf pair or whorl unfolded, Perennial plants: First leaved unfolded | First leaves unfolded | SOSD |
| **223** | Leaf unfolding (≥50%) | Leaf unfolding >=50% | SOSD |
| **95** | 50% of leaves fallen | 50% of leaves fallen | EOSD |
| **205** | Autumnal coloration ≥50% | Autumnal colouration >=50% | EOSD |

A total of 511 out of 1361 available sites providing data in 2023 are used in the validation after performing the vegetation group selection with availability of selected phenophases.

### 3.3.2 TEMPO

TEMPO (https://tempo.pheno.fr/) (Maury et al., 2023) is the French national network of observatories dedicated to the phenology of the entire living kingdom, whose objective is to answer how climate change influences in the seasonal rhythms of living organism and productivity. TEMPO counts with 10 observatories and more than 95 partners (research units, technical and public institutions, etc.).

As in the comparison with PEP725, several species were grouped into two vegetation groups (Table 10). Phenological stages are recorded following the Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie (BBCH) reference system, which describes annual crops and fruit trees phenological stages. Similar growth stages of each plant species are given the same BBCH code. The phenological stages used as potential start of season are “Germination / sprouting / bud development” and “Leaf development (main shoot)” and the stage used as potential end of season is “Beginning of dormancy, die off of leaves, stem” (see Table 11).

**Table 10: Grouping strategy for linking OLCI LSP SOSD and EOSD with TEMPO phenology observations.**

| Vegetation group | TEMPO genus | CLC classes |
|---|---|---|
| Deciduous broad-leaved trees | Aesculus, Betula, Fagus, Fraxinus, Quercus, Robinia, Sambucus, Sorbus, Syringa, Populus | Broad-leaved forest, Transitional woodland-shrub, Mixed forest, Green urban areas. |
| Evergreen coniferous trees | Larix, Picea, Pinus | Coniferous forest. |

**Table 11: Phases used for linking CLMS OLCI LSP SOSD and EOSD with TEMPO phenology observations.**

| TEMPO phenological event code | Description | Metrics |
|---|---|---|
| **0** | Germination / sprouting / bud development | SOSD |
| **1** | Leaf development (main shoot) | SOSD |
| **9** | Beginning of dormancy, die off of leaves, stem, … | EOSD |

For each site, genus and phenophase, the median values of the TEMPO samples are compared to the satellite products at native resolution. The CLMS LC300 V3 has been used to discard sites classified as urban or water bodies. The current dataset counts with more than 70 species and 607 sites in 2023. A total of 383 sites are available after performing the vegetation group selection with availability of selected phenophases.

### 3.3.3 National Science Foundation's National Ecological Observatory Network (NEON)

The National Science Foundation's National Ecological Observatory Network (NEON, https://www.neonscience.org/) is a continental-scale observation facility designed to collect long-term open access ecological data to better understand ecosystem changes. Neon provides free and open data that characterize plants, animals, soil, nutrients, freshwater and the atmosphere.

The NEON Plant Phenology observations dataset (DP1.10055.001) contains quality-controlled native sampling resolution data from in-situ observations of plant leaf development and reproductive phenophases. Phenophase status and intensities follow those of the USA National Phenology Network (USA-NPN). The status and intensity data are reported per phenophase, individual or patch for every sampled day.

The phenophases used as potential start of season are "breaking leaf/needle buds" and "emerging leaves/needles," and the phenophases used as end of season are "colored leaves/needles" and "falling leaves/needles". Each species of NEON is classified according to their growth form. The available growth forms were aggregated according to Table 12. For each plot, a recount of how many individual species belong to an aggregated growth form has been made. The final phenophase used to compare with satellite products is the median phenophase of the individuals weighted by the percentage of aggregated growth form which they represent.

**Table 12: Growth Forms aggregation for NEON data.**

| Aggregated growth forms | Growth forms |
|---|---|
| DBF (Deciduous Broadleaf Forests) | Deciduous broadleaf<br>Drought deciduous broadleaf |
| DCF (Deciduous Coniferous Forests) | Deciduous conifer |
| EBF (Evergreen Broadleaf Forests) | Evergreen broadleaf<br>Evergreen broadleaf –no leaf buds<br>Evergreen broadleaf – no leaves<br>Evergreen broadleaf – no needles<br>Semi-evergreen broadleaf |
| ECF (Evergreen Coniferous Forests) | Evergreen conifer<br>Pine |
| HER (Herbaceous) | Cactus<br>Graminoid<br>Evergreen forb<br>Forb<br>Semi-evergreen forb |

The whole dataset counts with more than 500 species and has 46 sites and 2 different plots per site. A total of 37 plots, available in 2023, provide data for the growth forms and phenophases selected for validation.

### 3.3.4 Ecosystem Phenology Camera (PhenoCam V3)

The Ecosystem Phenology Camera (PhenoCam, https://phenocam.nau.edu/webcam/) is a network created to provide automated near-surface remote sensing of canopy phenology. For that, high-resolution digital cameras were distributed throughout this region. Images of the cameras are uploaded to PhenoCam server every half hour, extracting quantitative colour information from each picture. Canopy greenness indices provide information about the amount of foliage present and its colour. The image analysis of these cameras provides an objective means by which canopy phenology can be monitored and quantified.

The PhenoCam standard index is the green chromatic coordinate (GCC), which is the relative brightness of the green channel of cameras normalized against the overall brightness of the red, green and blue channels. From GCC time series, PhenoCam also provides an estimation of the green-up and green-down transition dates, which correspond with the start and end of the growing season. To estimate the transition dates, first the maximum and minimum values of greenness are estimated. Then, the 10%, 25% and 50% of the greenness curve range is computed as transition dates.

PhenoCam V3 (pre-release[6]) counts with a total of 1189 ROI's masks and data from 2000 to 2024. Each PhenoCam site can have different ROI's with different vegetation types in the same coordinate. In order to choose the vegetation type ROI more similar to the satellite retrievals, a strategy of selecting ROIs was developed, choosing the ROIs with vegetation similar to the CLMS LC300 V3 aggregated classes (PhenoCam forest vegetation types if LC300 indicates forest and PhenoCam non-forest vegetation types if LC300 indicates non forest, as shown in Table 13). Additionally, a visual filter was applied using Google Earth, filtering out ROI's masks which no correspond to the main dominated biome or specie around an area of 300 m x 300 m. Finally, a total of 200 ROIs are used. Figure 6 shows one example of one selected (top side) and discarded (bottom) ROI based on this visual filter.

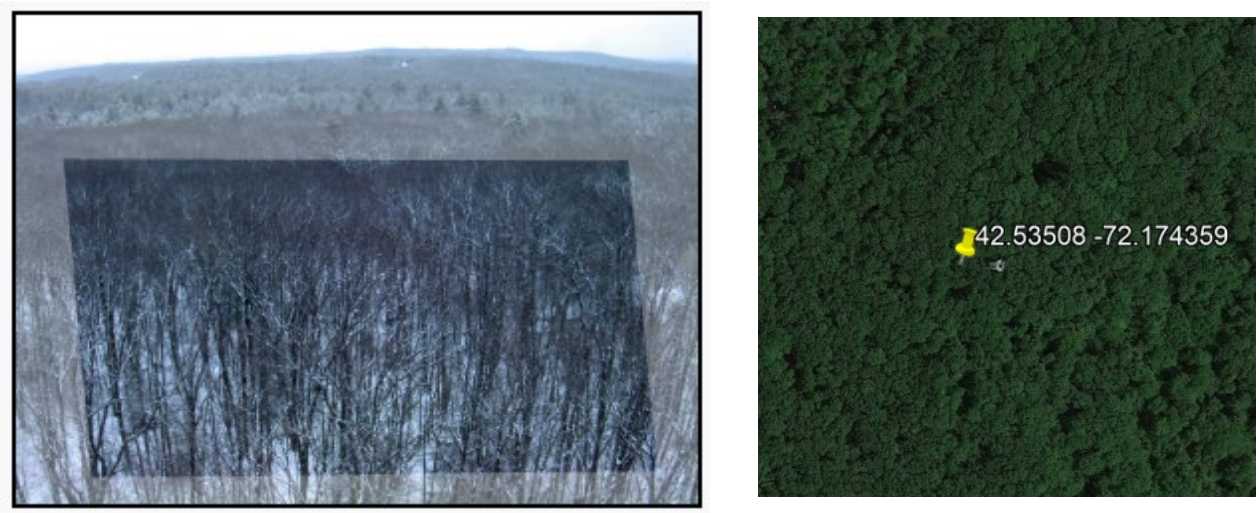
**Table 13: Strategy for choosing PhenoCam ROIs vegetation types according to CLMS LC300 V3.**

| | CLMS LC300 V3 Aggregated classes | PhenoCam Vegegation Types |
|---|---|---|
| **Forests** | EBF (Evergreen Broadleaf Forests)<br>DBF (Deciduous Broadleaf Forests)<br>NLF (Needleleaf Forests)<br>OF (Other Forests) | EB (Evergreen Broadleaf Forests)<br>DB (Deciduous Broadleaf Forests)<br>EN (Evergreen Needleleaf Forests)<br>DN (Deciduous Needleleaf Forests) |
| **Non-forests** | CUL (Cultivated)<br>HER (Herbaceous)<br>SHR (Shrublands)<br>SBA (Sparse and Bare Areas) | AG (Agriculture)<br>GR (Grasslands)<br>TN (Tundra)<br>SH (Shrublands)<br>XX (Landscape Region)<br>NV (No vegetated) |

**bbc1 - ROI 1000 [DB] (42.53508°N, 72.174359°W)**

[6] https://phenocam.nau.edu/phenocam_explorer_prerelease/ (data access 19th April, 2024)

PhenoCam V3 | Google Earth




**arsgacp3 - ROI 1000 [AG] (31.7563°N, 83.7516°W)**

PhenoCam V3 | Google Earth

**Figure 6: Example of two PhenoCam V3 ROIs (left side) and Google Earth views at 300 m x 300 m spatial extension.**

# 4 RESULTS

## 4.1 PRODUCT CONTENT & VISUAL INSPECTION OF GLOBAL MAPS

CLMS LSP product provides information on vegetation phenology and productivity (VPP) up to two growing seasons. Each season includes 13 parameters and is accompanied by a quality flag layer to ensure data reliability (Table 14) [CGLOPS1_PUM_LSP300m-V1].

**Table 14: Vegetation Phenology and Productivity parameters generated in CLMS LSP products.**

| No. | Name | Data type | Scale, offset | Data range | Fill value | Description | Unit |
|---|---|---|---|---|---|---|---|
| **1** | SOSD | INT16 | 1, 0 | -365~730 | -9999 | Day of start-of-season | Day-Of-Year |
| **2** | EOSD | INT16 | 1, 0 | -365~730 | -9999 | Day of end-of-season | |
| **3** | MAXD | INT16 | 1, 0 | 0~366 | -9999 | Day of maximum-of-season | |
| **4** | SOSV | INT16 | 0.001, 0 | 0~5000 | -9999 | Vegetation index value at SOSD | PPI unit $m^2 \cdot m^{-2}$ |
| **5** | EOSV | INT16 | 0.001, 0 | 0~5000 | -9999 | Vegetation index value at EOSD | |
| **6** | MINV | INT16 | 0.001, 0 | 0~5000 | -9999 | Average vegetation index value of minima on left and right sides of each season | |
| **7** | MAXV | INT16 | 0.001, 0 | 0~5000 | -9999 | Vegetation index value at MAXD | |
| **8** | AMPL | INT16 | 0.001, 0 | 0~6000 | -9999 | Season amplitude (MAXV – MINV) | |
| **9** | LENGTH | INT16 | 1, 0 | 0~730 | -9999 | Length of Season (number of days between start and end) | day |
| **10** | LSLOPE | INT16 | 0.001, 0 | -1000~1000 | -9999 | Slope of the green-up period | $m^2 \cdot m^{-2} \cdot day^{-1}$ |
| **11** | RSLOPE | INT16 | 0.001, 0 | -1000~1000 | -9999 | Slope of the green-down period | |
| **12** | TPROD | INT16 | 1, 0 | 0~2000 | -9999 | Total productivity. Growing season integral computed as the sum of all daily values between SOSD and EOSD. | $m^2 \cdot m^{-2} \cdot day$ |
| **13** | SPROD | INT16 | 1, 0 | 0~2000 | -9999 | Seasonal productivity. Growing season integral computed as sum of all daily values minus their base level value. | |
| **Aux** | QA | UINT8 | - | - | 255 | Quality flag | - |

SOSD, EOSD and MAXD layers correspond to the day of year corresponding to start, end and maximum of season, respectively. LENGTH corresponds to the length of the season, defined as the number of days between start and end. These layers are displayed in Figure 7 for season 1 (left side) and 2 (right).

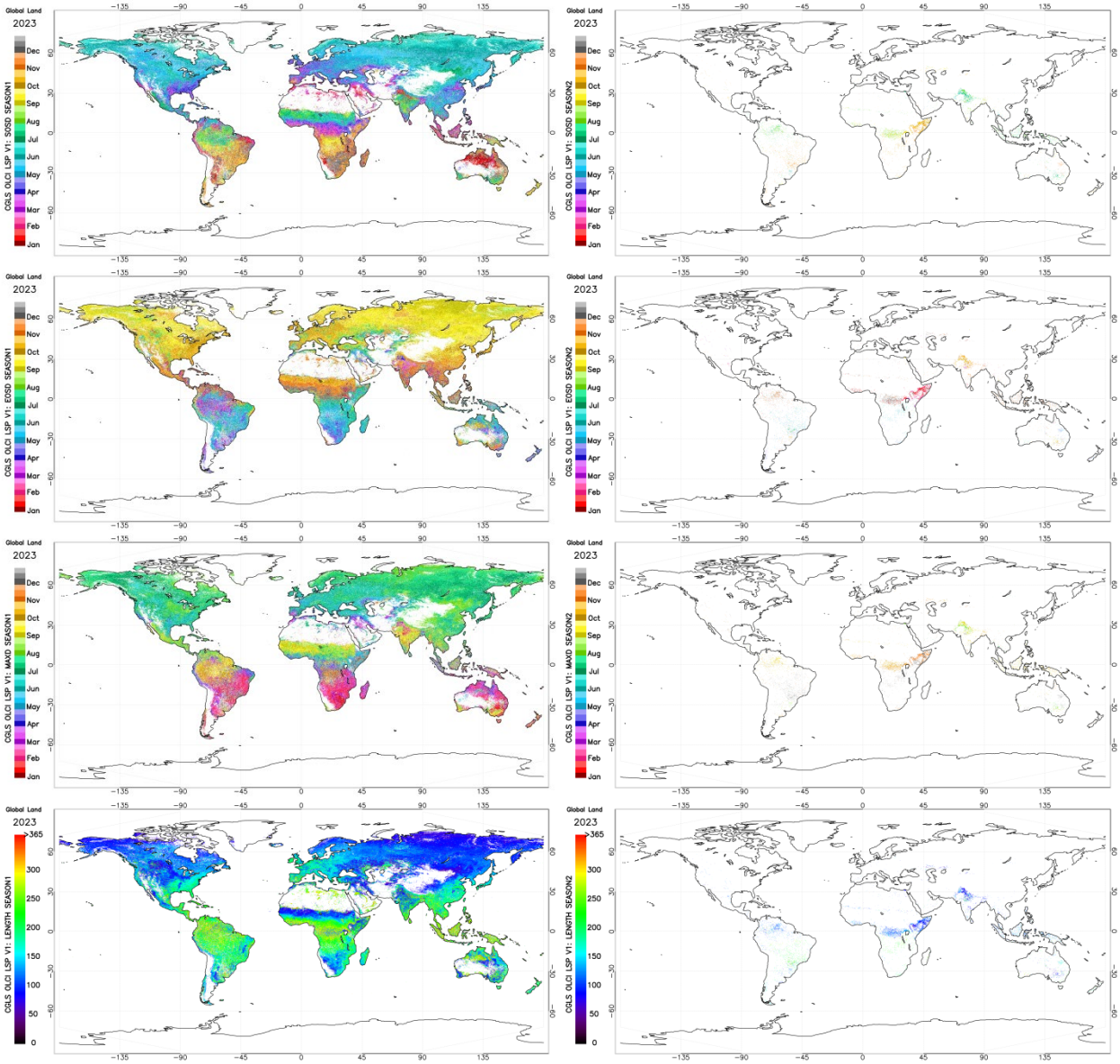



**Figure 7: Global maps of CLMS OLCI LSP following layers for 2023: SOSD, EOSD, MAXD and LENGTH (from top to bottom). Season 1 and season 2 are displayed at left and right sides, respectively.**

The main results of the visual inspection of SOSD, EOSD, MAXD and LENGTH (Figure 7) global maps are:

- Consistent values are generally found over the whole globe. It should be noted that deserts (e.g., Arabia, Sahara or Australian deserts) display seasons over few areas despite of the fact that desert areas should not show vegetation activity.

- Season 2 is provided over few cases over the globe, mainly located over the tropics.

There are several layers which are defined at the level of the vegetation index (i.e., PPI units [$m^2m^{-2}$]) for different phases of the phenological cycle. SOSV and EOSV correspond to the PPI value at SOSD and EOSD whilst MINV and MAXV correspond to average PPI of minima on left and right sides of each season and PPI at MAXD. The season amplitude (i.e., MAXV – MINV) is provided in the AMPL layer. Global maps of these layers are displayed in Figure 8 for 2023 (only season 1 is presented, for sake of brevity).

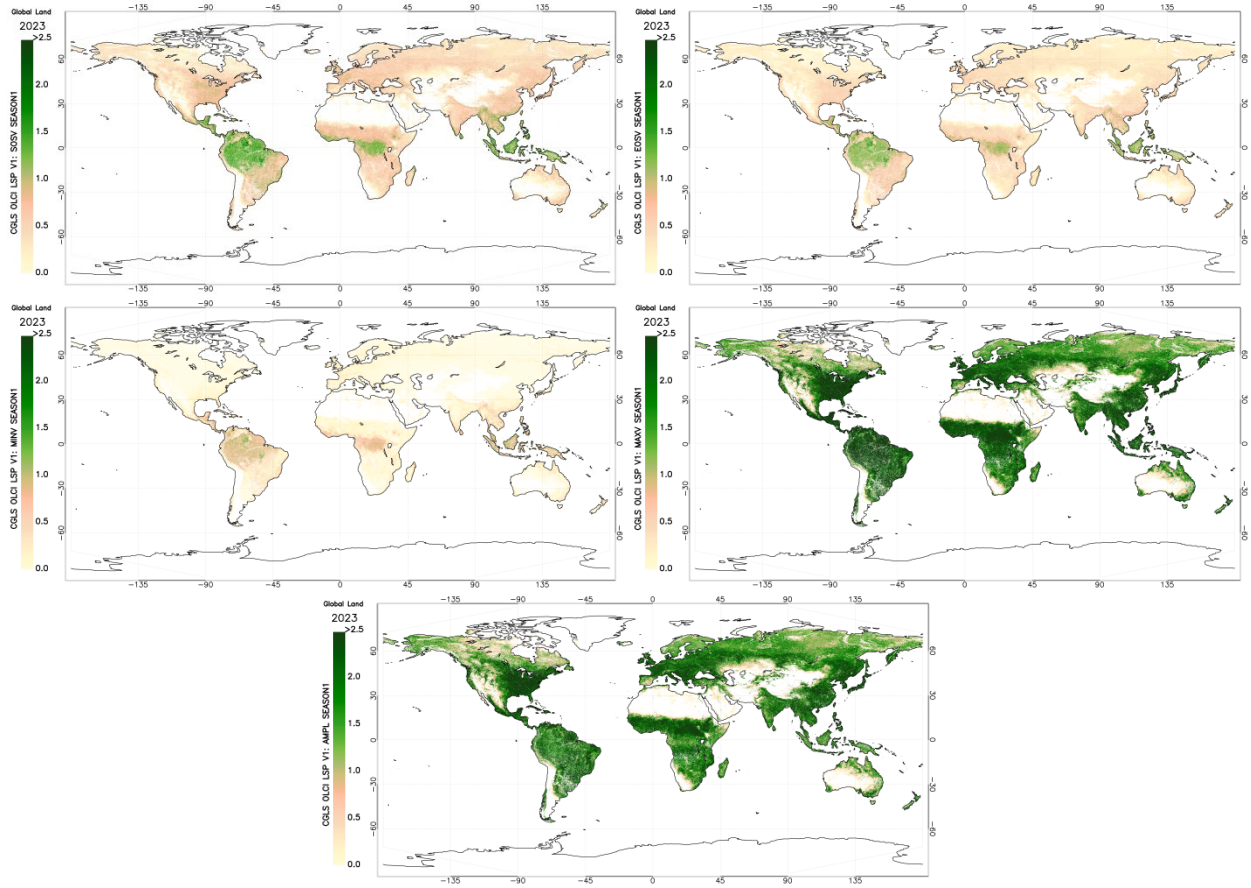



**Figure 8: Global maps of CLMS OLCI LSP following layers for 2023 season 1: SOSV, EOSV, MINV, MAXV and AMPL (from left to right and top to bottom). Unit: $m^2 \cdot m^{-2}$.**

The main findings of Figure 8 are:

- Consistent values are found, with very similar PPI values for SOSV and EOSV and low and high values for MINV and MAXV, as expected. SOSV shows slightly higher values than EOSV which is consistent with the product definition: SOSD and EOSD are defined at 25% and 15% in relation to the seasonal amplitude. Higher values are typically observed over dense vegetated areas such as forest and lower values over sparse areas.
- AMPL, defined as MAXV – MINV, provides similar values than MAXV (slightly lower), as expected.

The slope of the green-up and green-down period [m2·m$^{-2}$·day$^{-1}$] are provided in LSLOPE and RSLOPE layers. Global maps of both layers for 2023 season 1 are presented in Figure 9.

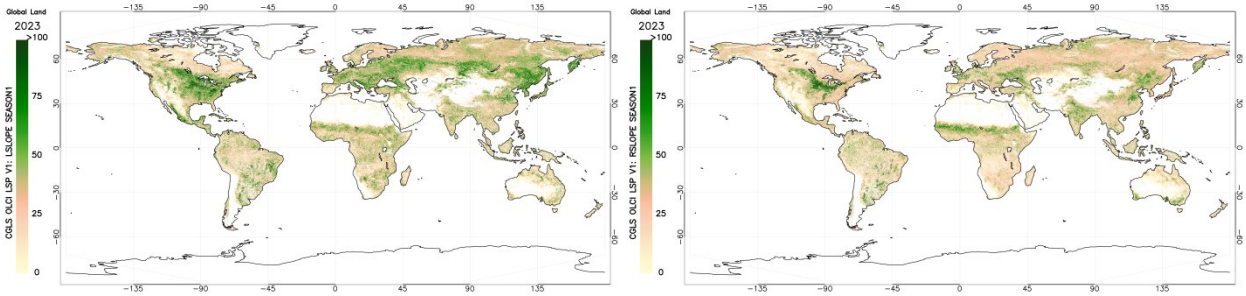



**Figure 9: Global maps of CLMS LSP OLCI LSLOPE (left) and RSLOPE (right) for 2023 season 1. Unit: m²·m$^{-2}$·day$^{-1}$.**

The visual inspection of LSLOPE and RSLOPE layers (Figure 9) shows:

- The slope is typically slightly higher for the green-up (LSLOPE) than for the green-down (RSLOPE) period, as expected.
- RSLOPE is provided as positive slope; therefore, it corresponds to the module of the slope during the green down period (negative slopes).
- Larger slopes are observed over areas dominated by croplands or herbaceous, and lower slopes are found in tropical forest and coniferous forest as expected.

Total and seasonal productivity [m$^2$·m$^{-2}$·day] are provided in TPROD and SPROD layers. TPROD is defined as the growing season integral computed as the sum of all daily values between SOSD and EOSD. SPROD is defined as the TPROD minus their base level value. Global maps of TPROD and SPROD are presented for 2023 season 1 in Figure 10.

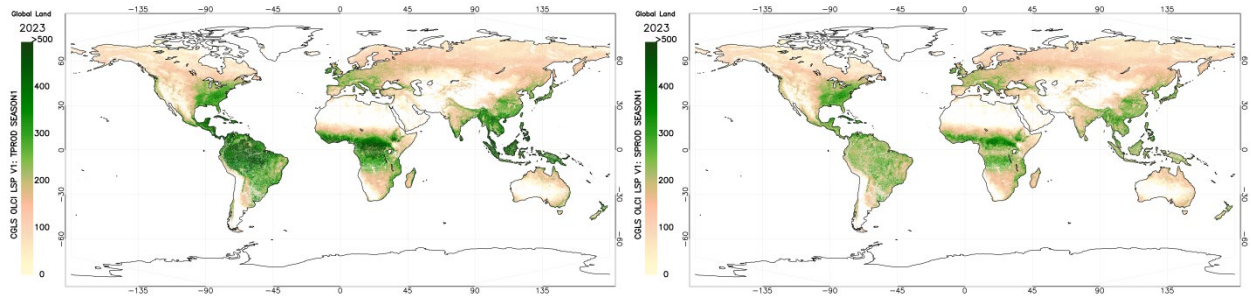



**Figure 10: Global maps of CLMS OLCI LSP TPROD (left) and SPROD (right) for 2023 season 1. Unit: $m^2 \cdot m^{-2} \cdot day$.**

Global maps of TROD and SPROD (Figure 10) show consistent values over the globe with higher productivity over dense vegetated areas (e.g., tropics) and areas with higher amplitude of the season (see the good correspondence of spatial patterns between TPROD/SPROD and AMPL layer in Figure 8).

Finally, the Quality Assurance (QA) layer is assigned based on both the average of PPI data quality and the total number of valid observations in each of phenological phase: green-up (SOS), green peak (peak), and green-down (EOS) of the growing season. Additionally, the overall QA is determined based on the comprehensive evaluation of the quality across all three phases.

QA is 8-bit encoded, where two bits are assigned for each aspect: SOS (bits 0-1), peak (bits 2-3), EOS (bits 4-5) and overall (bits 6-7). The details of bit values and descriptions are listed in Table 15.

**Table 15: CLMS LSP QA flags and descriptions of phenology quality.**

| QA byte | Quality | Description |
|---|---|---|
| **SOS/Peak/EOS quality (two-bit values)** | | |
| **0** | Best | More than 2 valid observations and average best PPI quality |
| **1** | Good | More than 2 valid observations and average good PPI quality |
| **2** | Fair | More than 2 valid observations and average fair PPI quality |
| **3** | Poor | Average poor PPI data quality, or fewer than 3 observations, or more than 24 observations* |
| **Overall quality (two-bit values)** | | |
| **0** | Best | No poor quality in three phenology phases |
| **1** | Good | Maximum one phase with poor quality |
| **2** | Fair | Maximum two phases with poor quality |
| **3** | Poor | All three phases with poor quality** |
| **Summary of QA (byte values)** | | |
| **QA<127** | Good | Overall good quality, no more than one poor results in three phases |
| **QA>127** | Poor | Overall poor quality, more than one poor results in three phases |
| **QA=255** | Failure | No input, or no phenology estimation, or all poor estimates |

* Includes three situations: 1) fewer than 3 valid observations, 2) phenology phase shorter than 10 days, and 3) phenology phase longer than 4 months.
** Overlaps QA=255 for situations of no inputs, or no phenology estimation. In any case, the LSP retrieval is considered a failure.

Figure 11 shows the global maps of SOS, peak, EOS and overall quality provided in QA layer. Main results are:

- The maps of the quality of each phase (SOS/Peak/EOS quality) show that best or good quality pixels typically cover between 90% and 95% of global cases. Poor quality only represents 6.9%, 8.8% and 4.1% of the global cases, observed over different areas among SOS, Peak and EOS.
- In case of the overall quality, best and good quality (i.e., maximum one phase with poor quality) represents around 97% of the cases without finding cases with poor quality (i.e., all three phases with poor quality). This is the condition used to select good quality pixels (see Table 7).

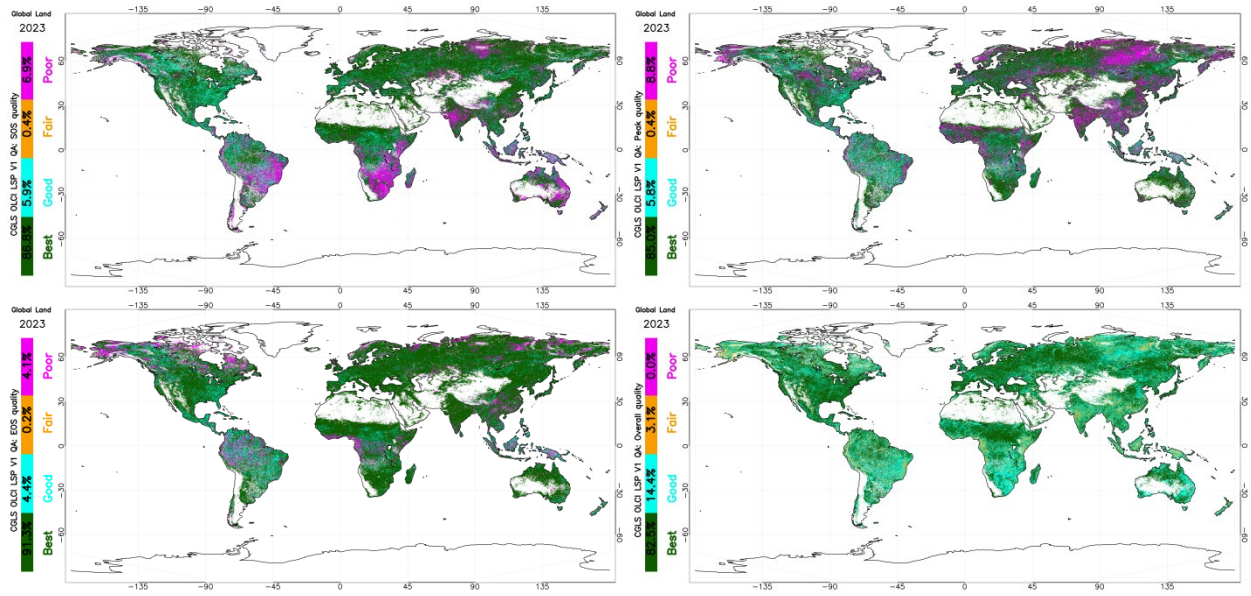



**Figure 11: Global maps of CLMS OLCI LSP QA layers for 2023 season 1: SOS quality, Peak quality, EOSD quality and Overall quality (from left to right and top to bottom).**

## 4.2 PRODUCT COMPLETENESS

This section provides the global maps of valid observations of CLMS OLCI LSP, MODIS V6.1 and VIIRS V2 products for the evaluated year (2023). For sake of brevity, only maps of SOSD are displayed but same results are observed for the rest of layers. Figure 12 shows the global maps of valid (blue) and missing (yellow) data for season 1. Note that, at the right side, the quality flags are used to remove suboptimal quality pixels (see Table 7). Maps for season 2 are displayed in Figure 13.

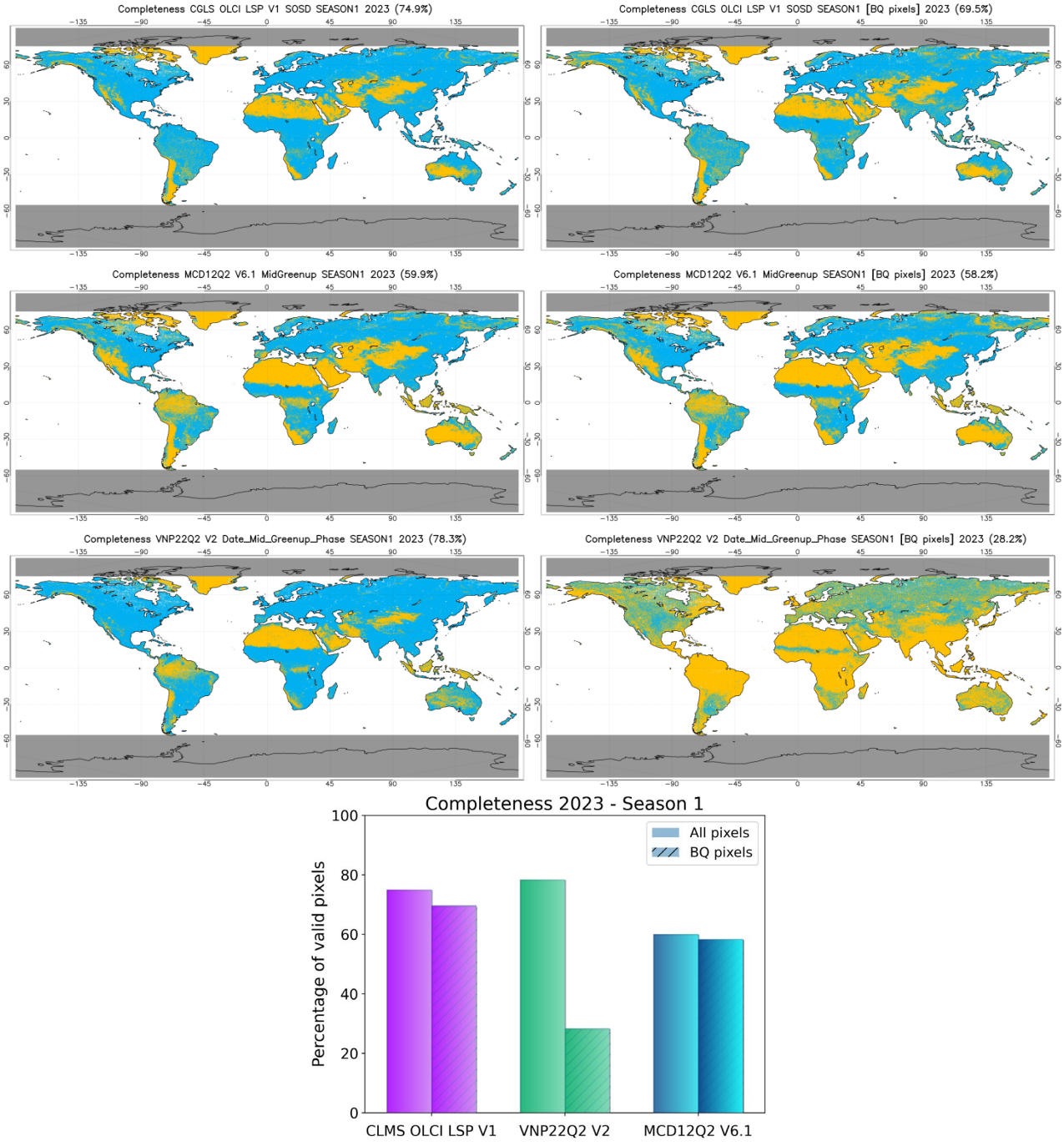
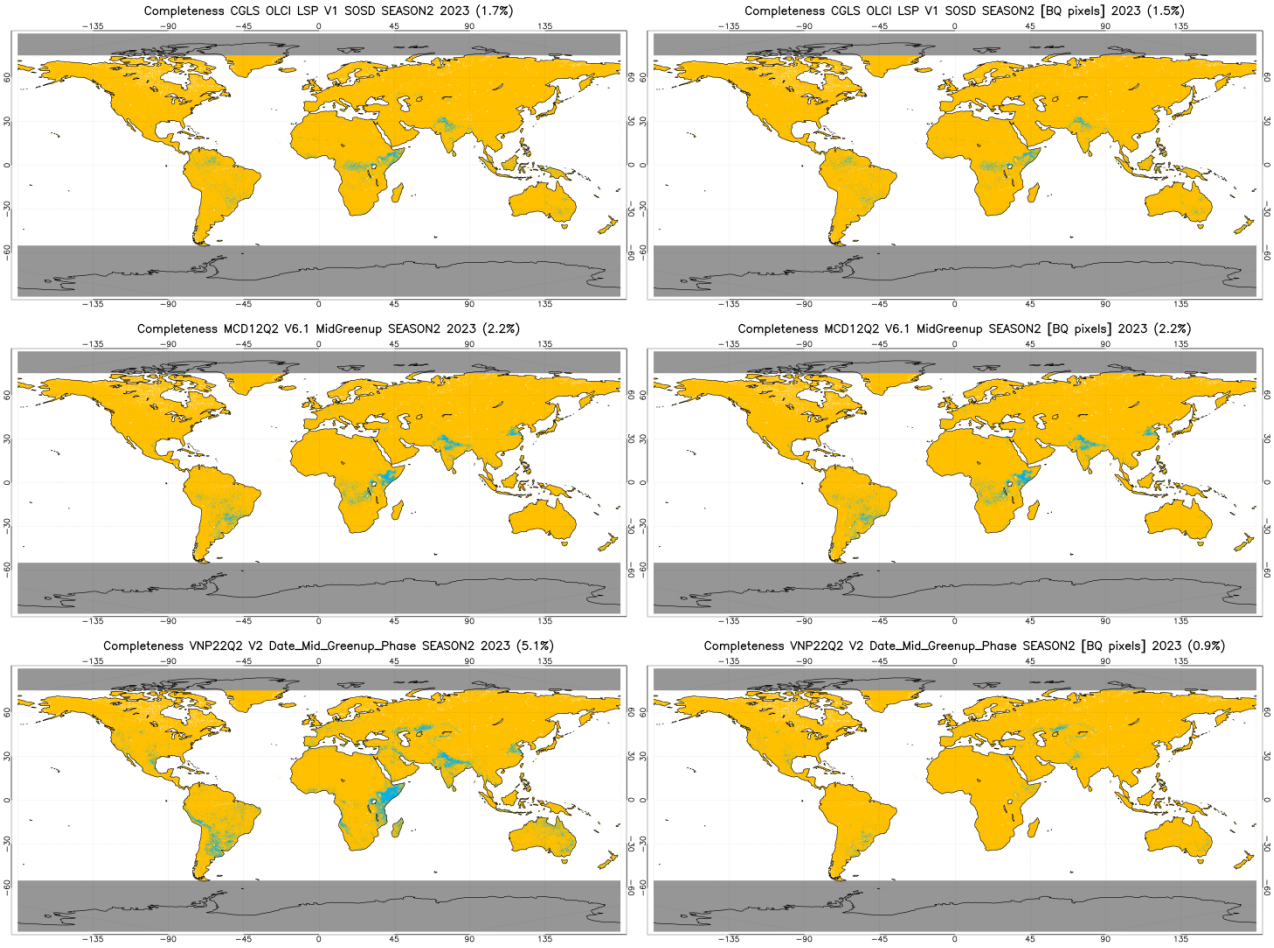



**Figure 12: Global maps of valid (blue) and non-valid (yellow) observations of start of season layer of CLMS OLCI LSP (1st row), MidGreenup layer of MODIS LCD V6.1 (2nd row) and Date_Mid_Greenup_Phase layer of VIIRS LCD V2 (3rd row) products for season 1 (2023). Maps considering all pixels are displayed at left side and considering 'best quality' pixels using quality flags are displayed at right side. Grey stripes at northern and southern latitudes correspond to the areas where CLMS products are not provided. Percentage of valid pixels ('all' and 'best quality (BQ)' cases) is displayed at the bottom side.**



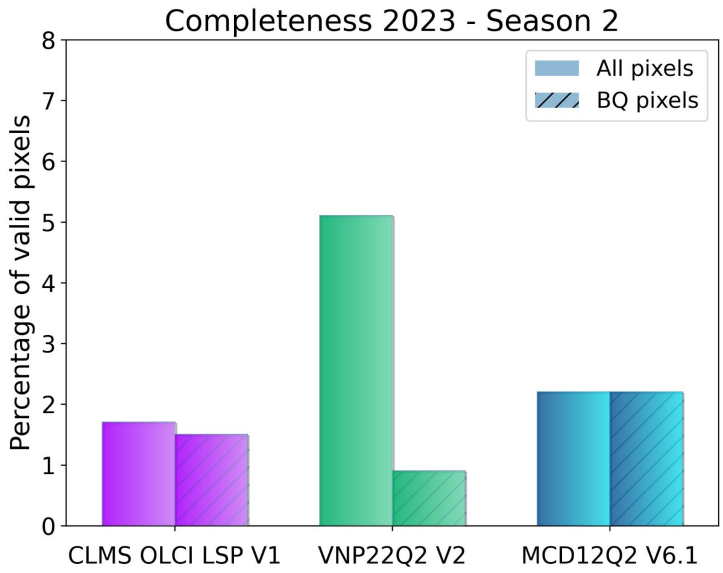




**Figure 13: Same as in Figure 12 for season 2.**

Main results are:

- OLCI LSP shows similar completeness than VIIRS LCD V2 (season 1), with around 75% of valid data. OLCI LSP improves the completeness of VIIRS product over tropical areas. However, VIIRS product provides better completeness over some sparse vegetated areas (e.g., West and South of America and north of Asia). OLCI LSP provides unexpected values over some desert areas (e.g., Sahara or Arabian deserts), as opposite to VIIRS product. MODIS V6.1 shows worse completeness, with around 60% of valid data.
- Considering the quality flag, around 5% of global data is filtered out in the OLCI LSP for season 1. VIIRS V2 has a quality flag very restrictive for best-quality retrievals, going from 78.3% to 28.2% of completeness (valid data) when best-quality is used (i.e., around 50% of data is removed). The quality flag of MODIS V6.1 filters out less than 2% of global data.
- All products provide very few valid cases for season 2, with only 1.8% (OLCI), 2.2% (MODIS) and 5.1% (VIIRS) of global cases.

## 4.3 SPATIAL CONSISTENCY

### 4.3.1 Visual inspection over subcontinental areas

In this section, a visual evaluation of the maps of six subcontinental areas (Figure 2) at the full resolution of CLMS OLCI LSP products is performed. Bellow the maps for SOSD (Figure 14), EOSD (Figure 15), SPROD (Figure 16) and TPROD (Figure 17).

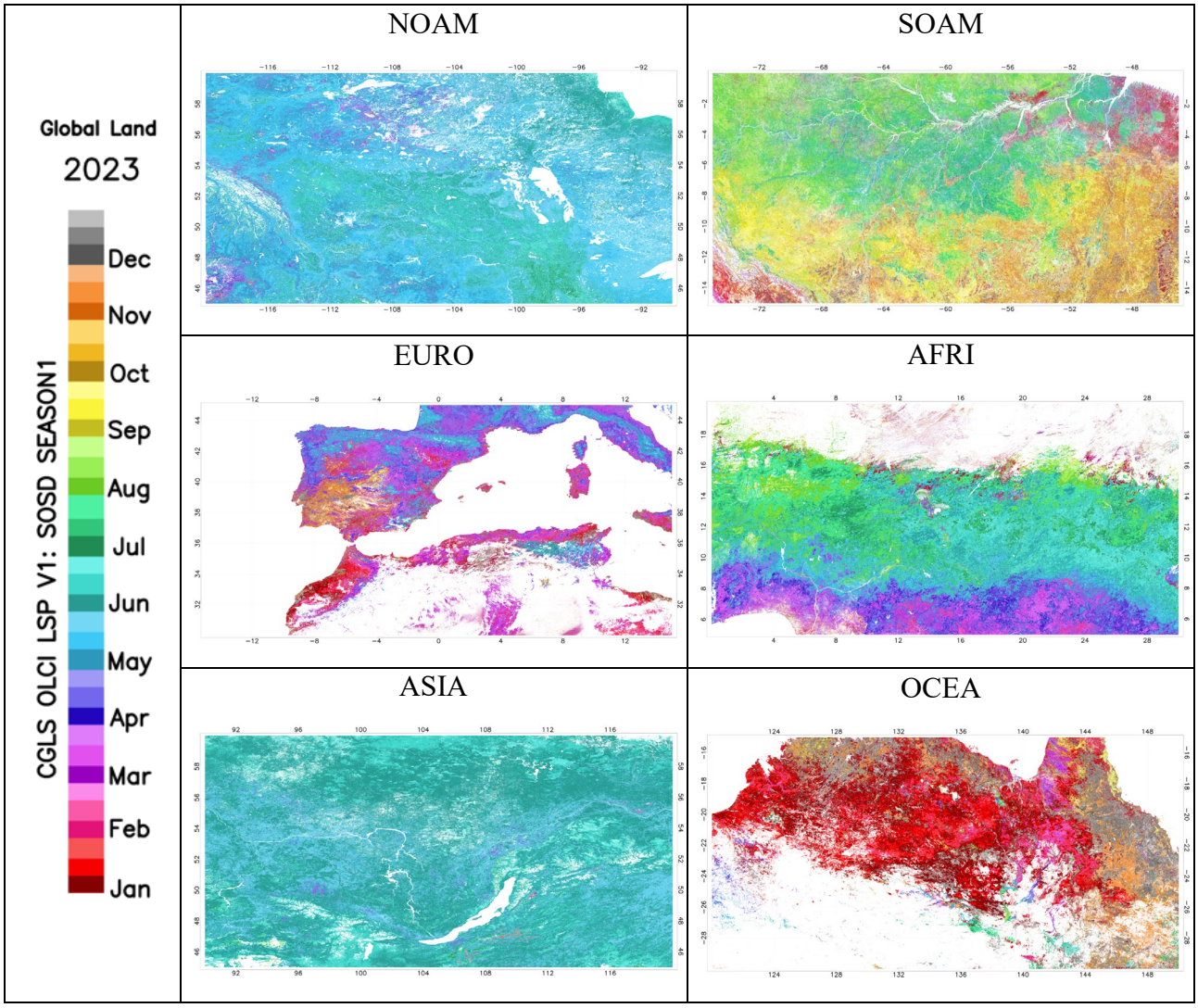



**Figure 14: Maps of CLMS OLCI LSP SOSD for 2023 season 1 over six sub-continental areas (10° latitude x 15° longitude).**


© C-GLOPS Lot1 consortium

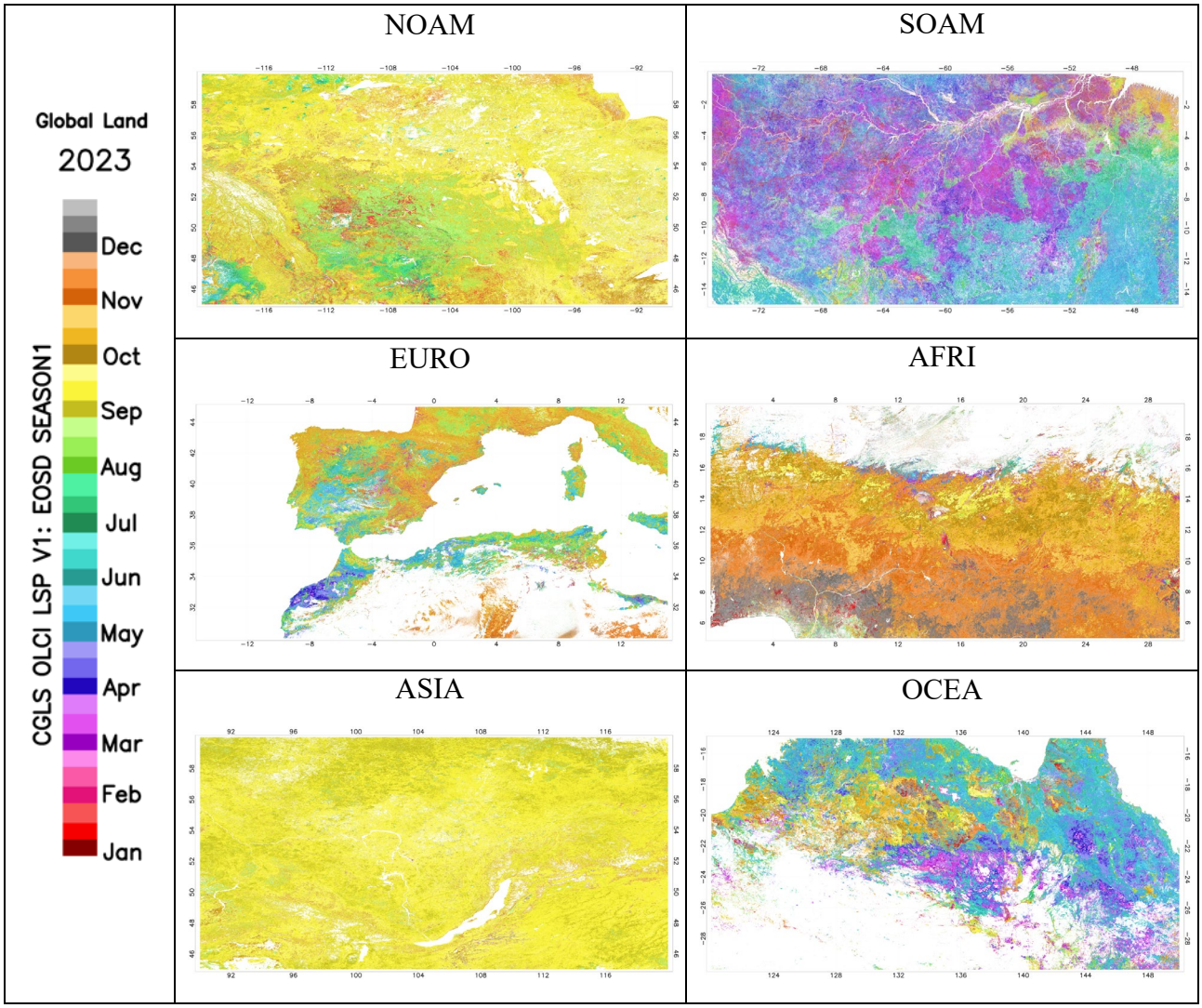



**Figure 15: Same as in Figure 14 but for EOSD.**


© C-GLOPS Lot1 consortium

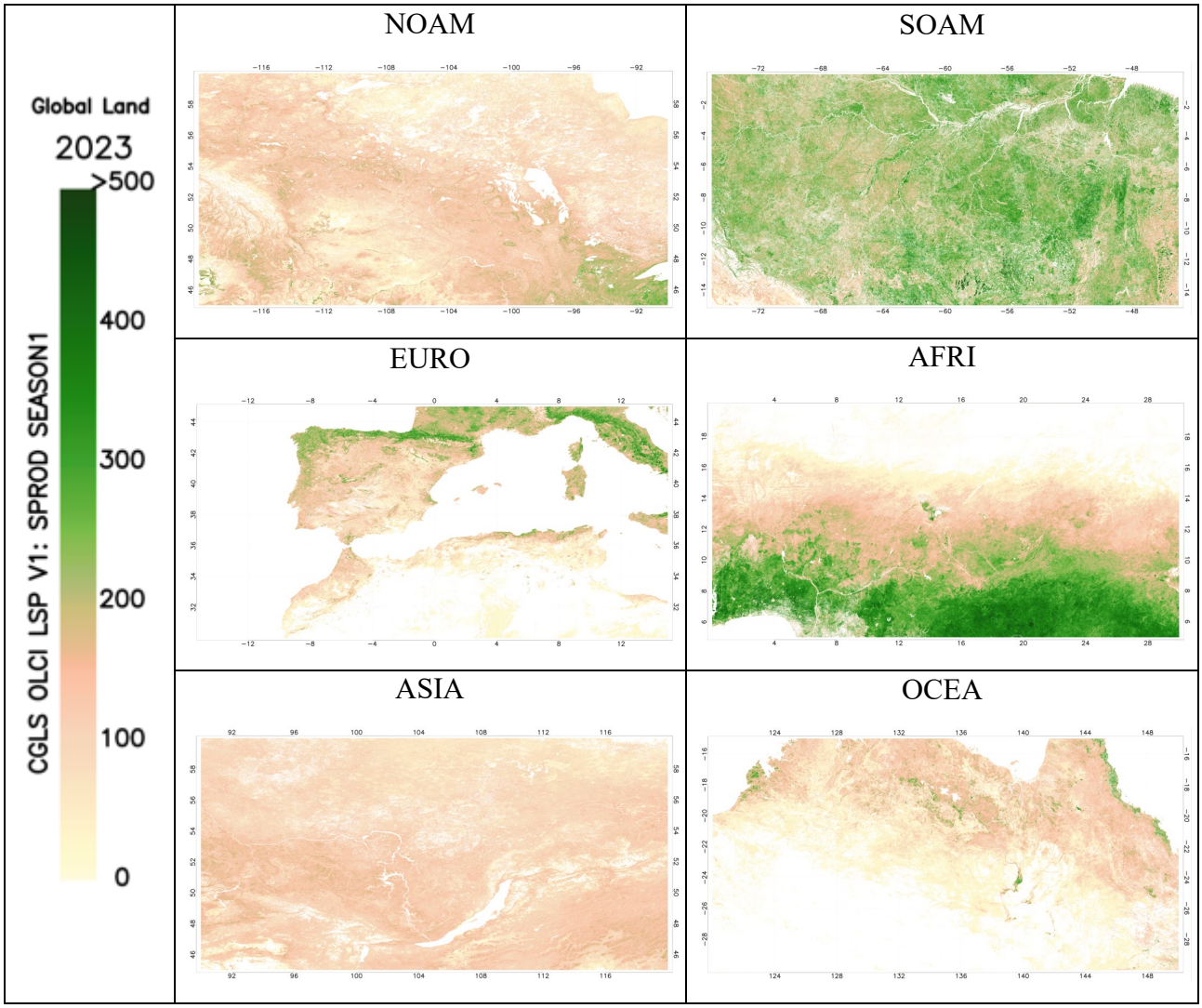



**Figure 16: Same as in Figure 14 but for SPROD. Unit: $m^2 \cdot m^{-2} \cdot day$.**


© C-GLOPS Lot1 consortium

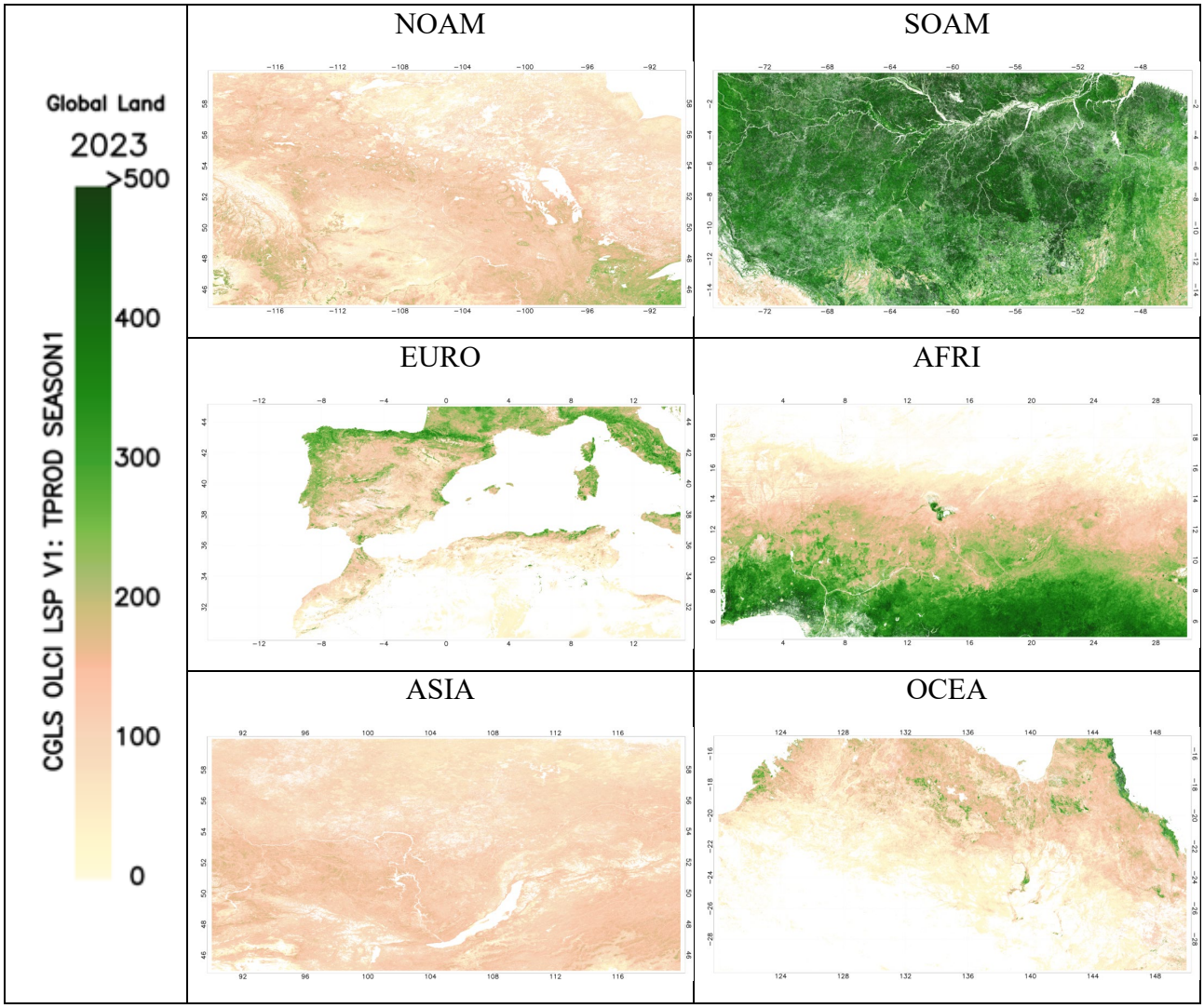



**Figure 17: Same as in Figure 14 but for TPROD. Unit: $m^2 \cdot m^{-2} \cdot day$.**

Main results of the inspection of subcontinental maps at full resolution:

- Good spatial consistency of CLGS OLCI LSP is found, reproducing properly the transitions of vegetation. The exceptions are areas of Sahara Desert present in EURO or AFRI windows where seasons could not be realistic. Similarly, over sparse vegetation areas in OCEA large variability in EOSD is observed.
- No spatial artifacts are found.

### 4.3.2 Maps of residuals with MR-VPP

This section shows the spatial consistency of CLMS OLCI LSP (SOSD, EOSD, SPROD and TPROD layers) versus MR-VPP equivalent layers over Europe for season 1.

Maps and histograms of residuals for SOSD and EOSD are presented along with the maps of pixels within user requirements (i.e., 5 and 10 days corresponding to goal and threshold levels) in Figure 18 (start of season) and Figure 19 (end of season). The spatial consistency is evaluated using the residuals, which remove the existing systematic differences between these two products. Additionally, the difference maps between CLMS OLCI LSP and MR-VPP are also presented in Annex I.

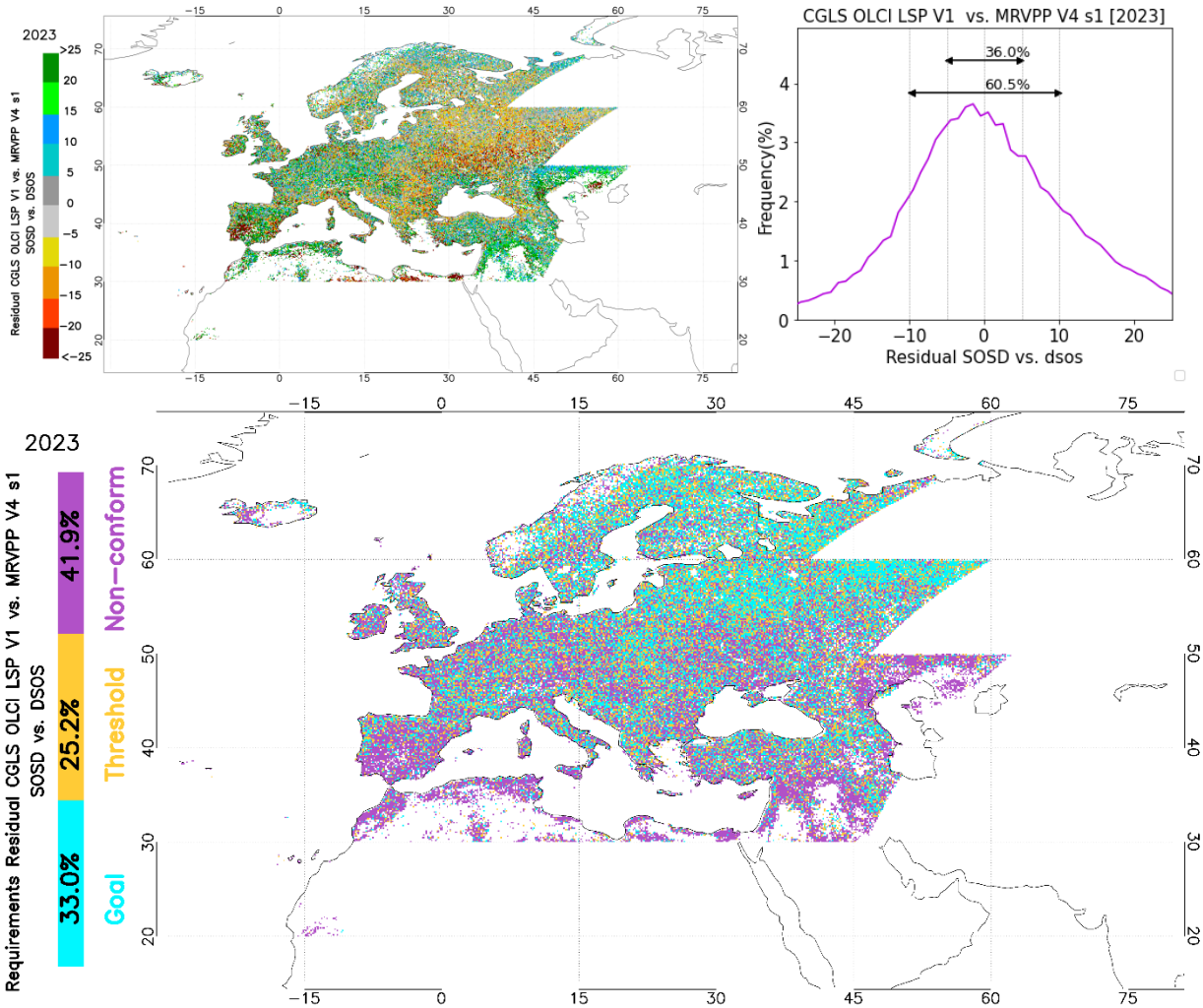



**Figure 18: Top: Maps and histograms of residuals between CLMS OLCI LSP SOSD and MR-VPP DSOS for 2023 (season 1). Bottom: percentage of global cases within goal (blue), threshold (yellow) user requirements and non-conform (purple). Unit: days.**

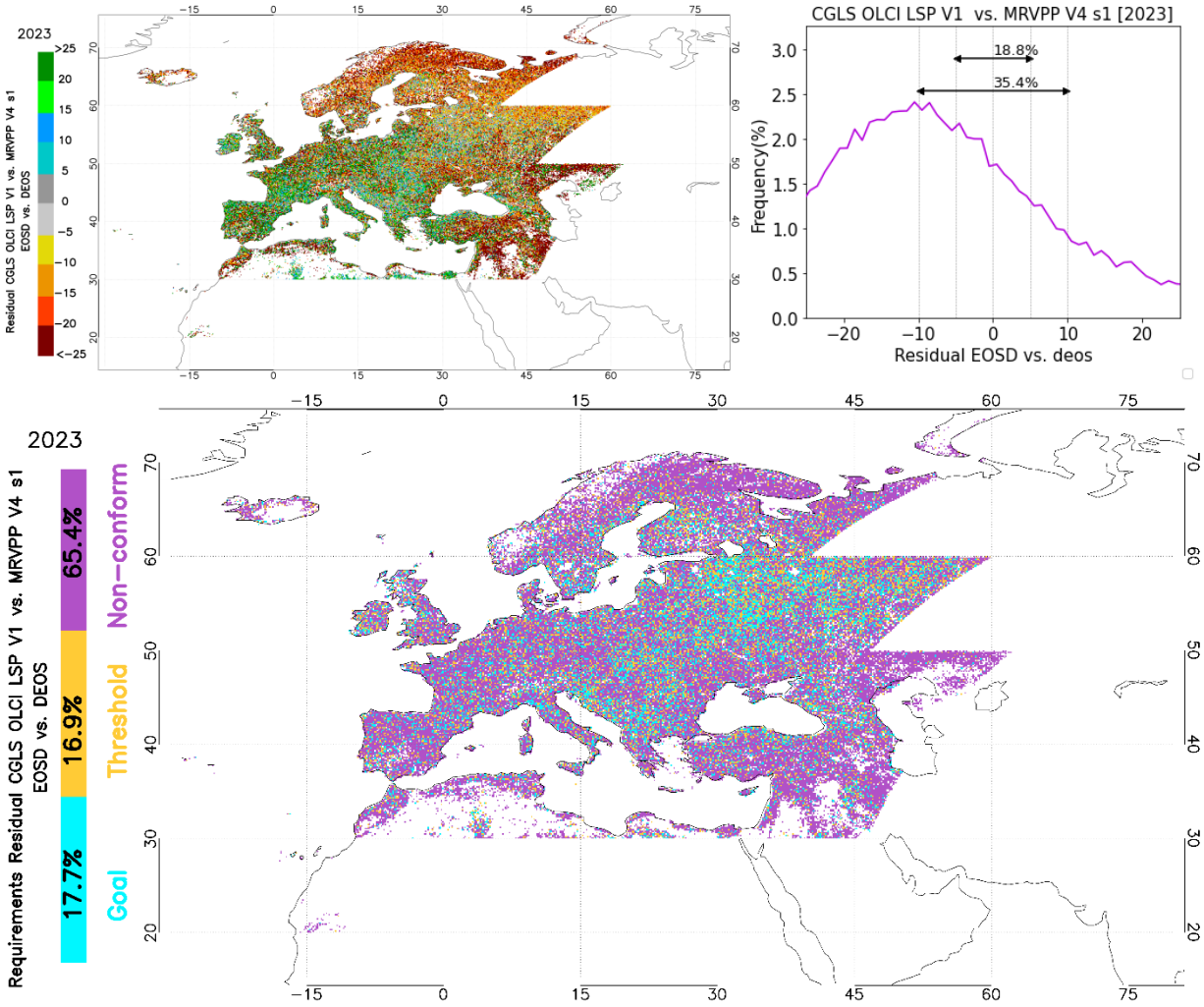



**Figure 19: Top: Maps and histograms of residuals between CLMS OLCI LSP EOSD and MR-VPP DEOS for 2023 (season 1). Bottom: percentage of global cases within goal (blue), threshold (yellow) user requirements and non-conform (purple). Unit: days.**

Main conclusions of the spatial consistency of CLMS OLCI LSP and MR-VPP for SOSD and EOSD layers are:

- For SOSD, both products are spatially consistent over large areas, with residuals typically within ±10 days and around 60% of pixels within the threshold requirement. The main spatial discrepancies are typically found over southern regions mainly dominated by sparse vegetation. However, maps of differences (see Annex I) show a clear tendency of CLMS OLCI LSP to provide late start of season compared with MR-VPP with positive bias (histogram centred at 20 days) over almost the whole area.

- Both products show larger spatial inconsistencies in case of EOSD, with 65% of cases non-conforming uncertainty requirements (i.e., residuals higher than 10 days). The difference maps (see Annex I) show similar results to the residuals indicating no systematic trend between both products for EOSD.

Maps and histograms of residuals between CLMS OLCI LSP and MR-VPP for SPROD and TPROD are presented in Figure 20 and Figure 21. Additionally, the difference maps are also presented in Annex I.

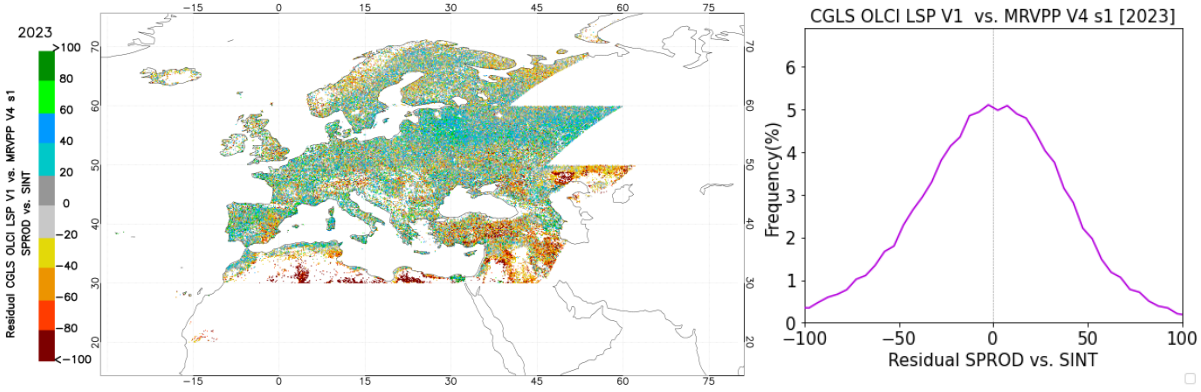



**Figure 20: Maps and histograms of residuals between CLMS OLCI LSP SPROD and MR-VPP SINT for 2023 (season 1). Unit: m².m⁻².day.**

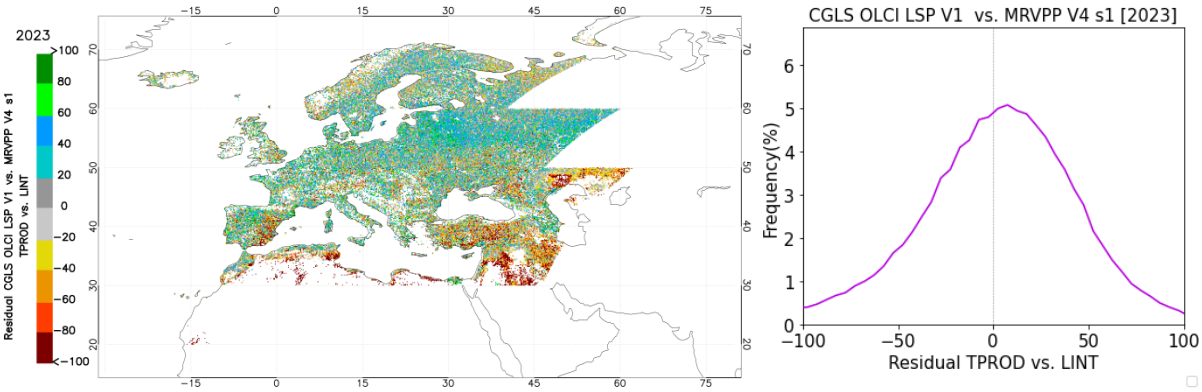



**Figure 21: Maps and histograms of residuals between CLMS OLCI LSP TPROD and MR-VPP LINT for 2023 (season 1). Unit: m².m⁻².day.**

Main results of the spatial consistency of CLMS OLCI LSP and MR-VPP V4 for SPROD and TPROD layers are:

- Both products show residuals typically within ±40 $m^2 \cdot m^{-2} \cdot$day and histograms of residuals centred on around zero. The main spatial discrepancies are typically found over southern regions mainly dominated by sparse vegetation.
- Maps of differences (see Annex I) show a clear tendency of CLMS OLCI LSP to provide higher productivity values than MR-VPP over almost the whole European region, even when OLCI LSP provides shorter seasons than MR-VPP. The exceptions are southern sparse vegetated areas where OLCI LSP provides lower productivity. The CLMS OLCI higher productivity respect to MR-VPP is explained in the maximum PPI values for each product, which are around 5 $m^2 \cdot m^{-2}$ in case of OLCI and 3 $m^2 \cdot m^{-2}$ in case of MR-VPP. Annex I (Figure 63) shows the differences of maximum PPI values (MAXV) for both products, with histograms of differences centred around 1 $m^2 \cdot m^{-2}$.

### 4.3.3 Spatial consistency between CLMS OLCI LSP vs. HR-VPP over AOIs

In this section, the spatial consistency between CLMS OLCI LSP phenological SOSD and EOSD and productivity SPROD and TPROD and HR-VPP equivalent layers is evaluated. The analysis is performed using six AOIs (see location over global map in Figure 2), which are centred at selected EVAL sites (see Table 1) and are defined over an extension of 20 km x 20 km.

To be able to compare medium (CLMS, 300 m) and high-resolution (HR-VPP, 10 m) products, HR-VPP were aggregated to the spatial resolution and projection of CLMS OLCI LSP using the effective Point Spread Functions (PSFs) which improves the performance of the evaluation as compared to ordinary average (Mira et al., 2015). The PSF is approximated by the convolution of a Gaussian function characterizing the PSF from optical sensor (Nestola et al., 2017). For each AOI, an iterative approach was performed to find the extension of the pixel size and the Full Width at Half Maximum (FWHM) in each direction which maximizes the correlation between HR-VPP and CLMS 300m products.

Figure 22 and Figure 23 show, as an example, the results of the spatial consistency between CLMS OLCI LSP and HR-VPP SOSD and EOSD for *Barrax* cultivated and *Bialowieza National Park* DBF sites. Similarly, results for SPROD and TROPD are presented in Figure 24 and Figure 25. The results for the rest of areas are displayed in Annex II.

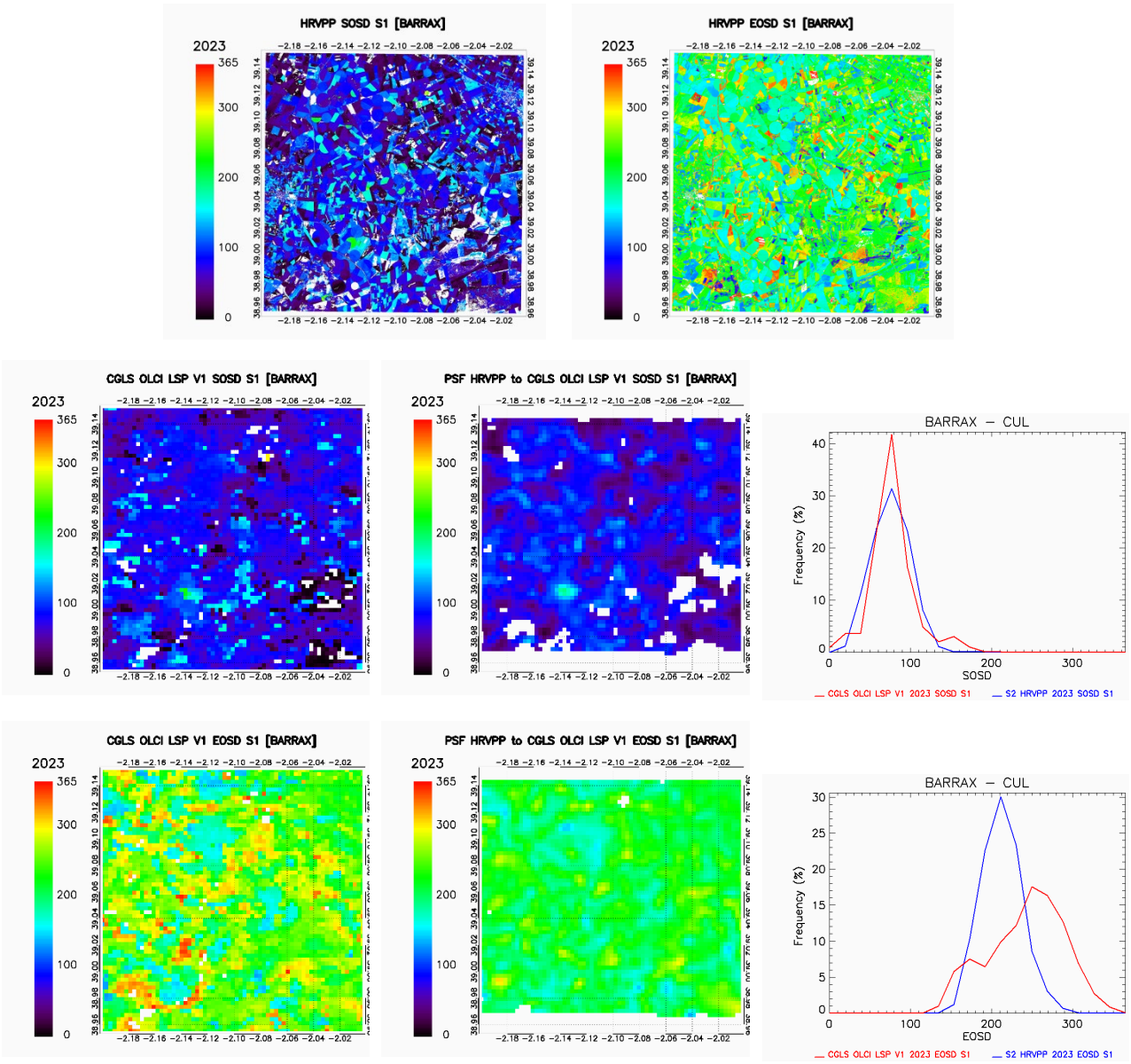



**Figure 22: Spatial consistency over Barrax cultivated site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SOSD (left) and EOSD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SOSD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for EOSD.**

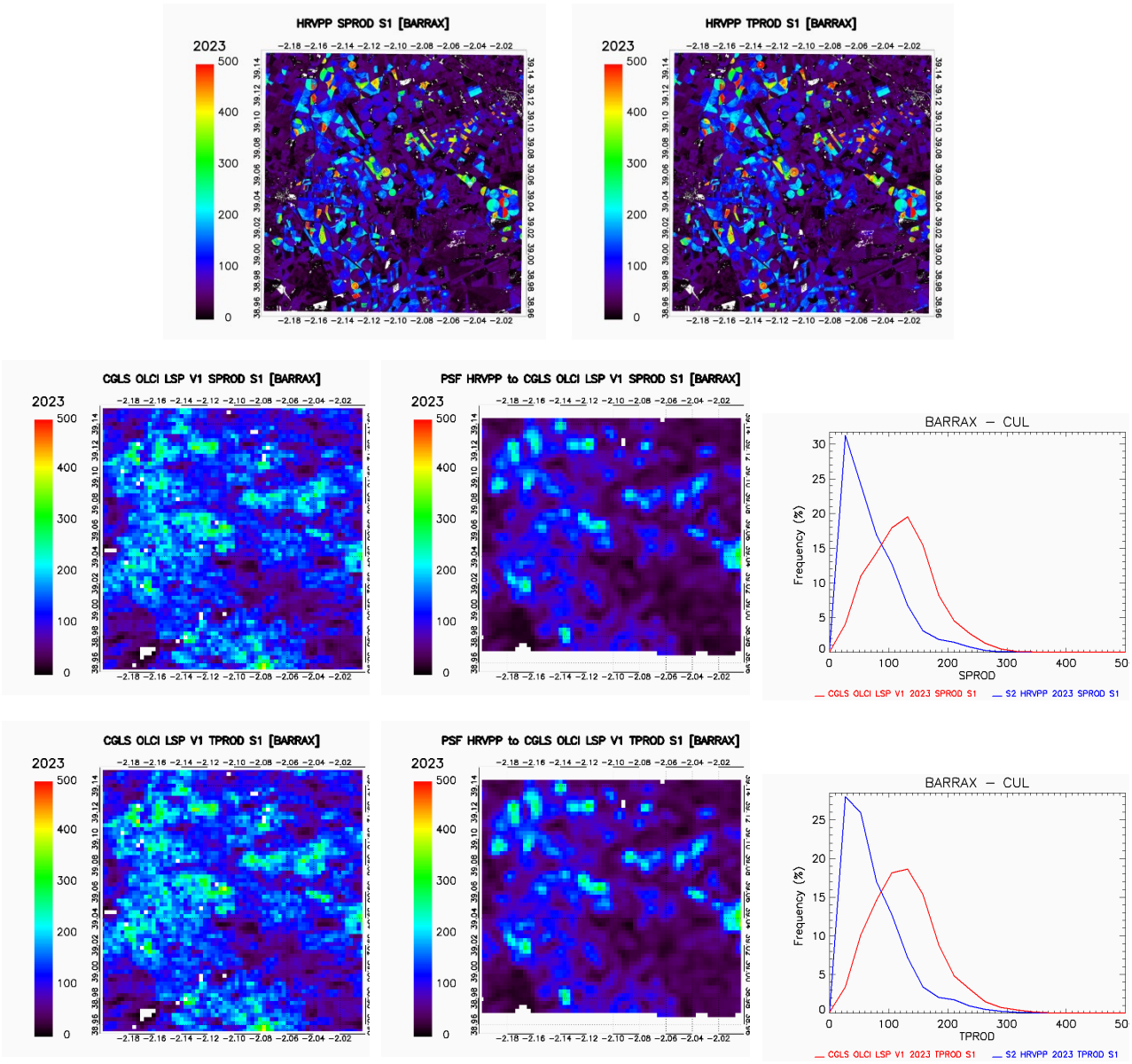



**Figure 23: Spatial consistency over Barrax cultivated site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SPROD (left) and TPROD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SPROD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for TPROD.**

Main findings of the spatial consistency between CLMS OLCI LSP and HR-VPP are:

- The spatial distributions of OLCI LSP SOSD, EOSD, SPROD and TPROD values are consistent with HR-VPP, showing similar spatial patterns for most cases.

- OLCI LSP typically tends to provide similar or slightly higher SOSD values than HR-VPP. In case of EOSD, OLCI LSP shows higher values than HR-VPP over cultivated areas and almost identical distribution of values in other biomes (forests and herbaceous).
- In case of SPROD and TPROD, OLCI LSP provides higher values than HR-VPP in all the investigated AOIs. Again, the cause is the maximum PPI values for each product (5 $m^2 \cdot m^{-2}$ for OLCI and 3 $m^2 \cdot m^{-2}$ for HR-VPP).

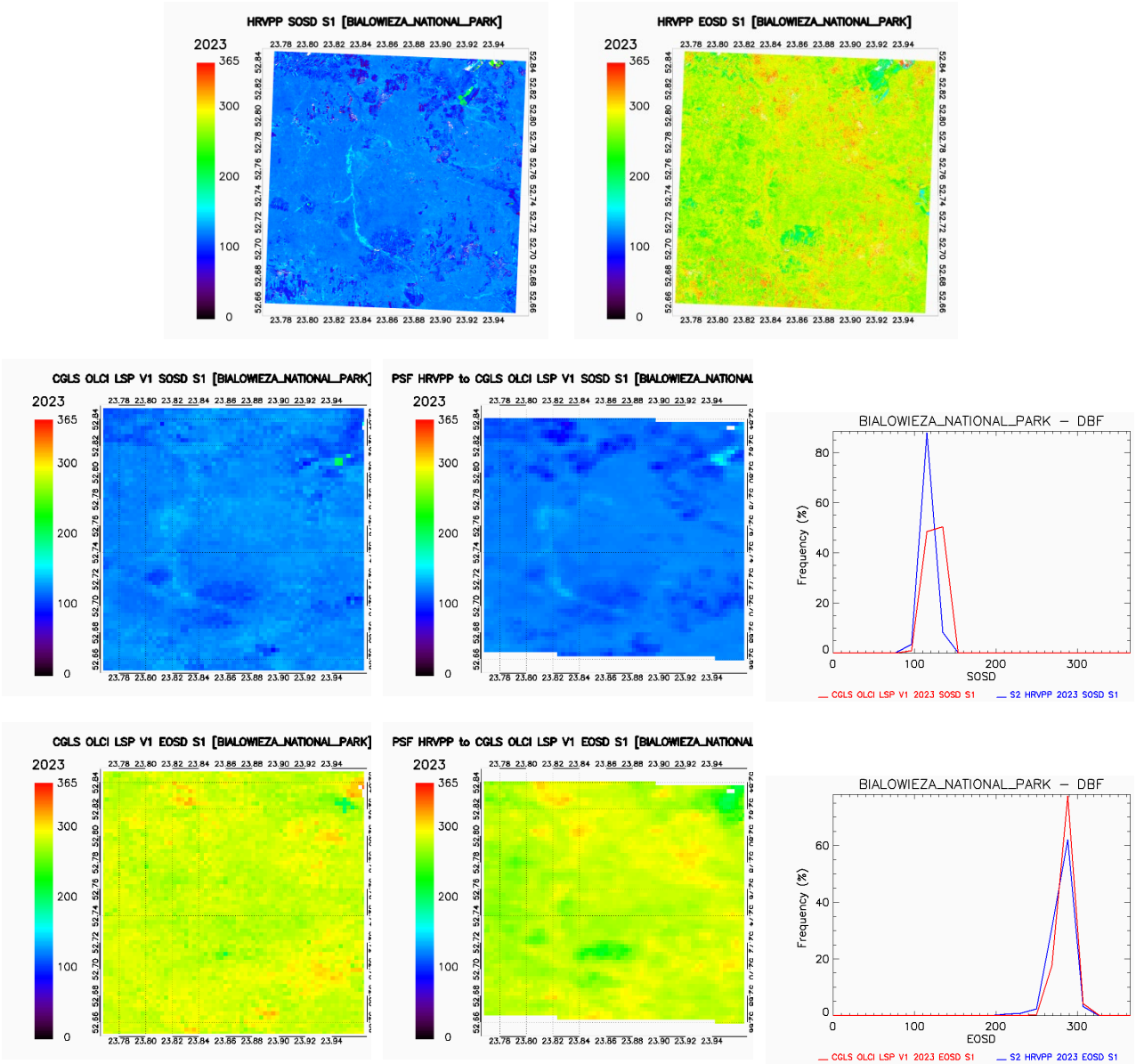



**Figure 24: Same as in Figure 22, but for Bialowieza National Park DBF site.**

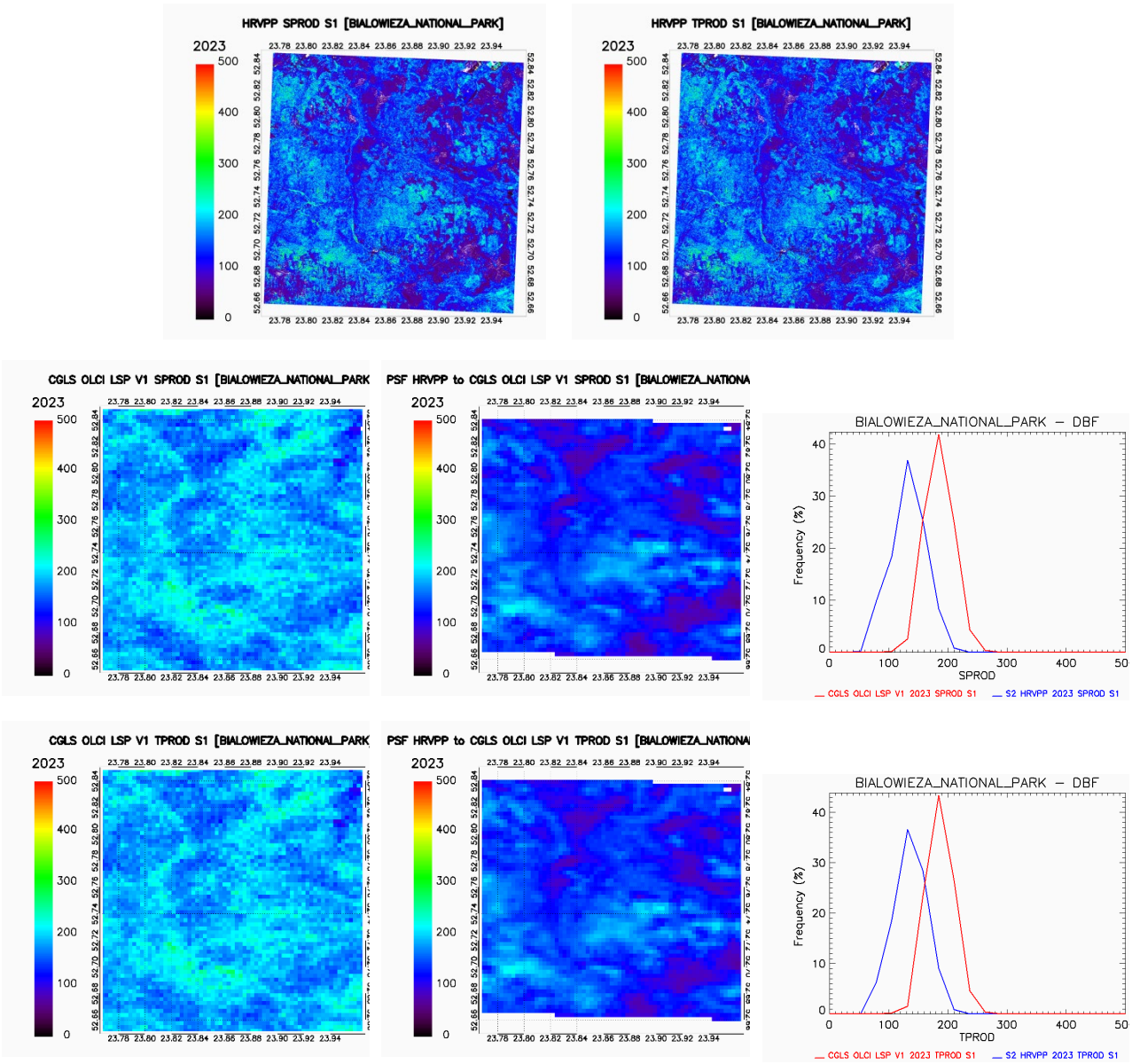



**Figure 25: Same as in Figure 23, but for Bialowieza National Park DBF site.**

### 4.3.4 Maps of differences with global LSP products

This section shows the spatial consistency of CLMS OLCI LSP SOSD and EOSD with MODIS V6.1 and VIIRS V2 LCD products for season 1. Global maps and histograms of differences are presented.

Note that the percentage of pixels within user requirements are not computed here due to different phenometric definitions between CLMS LSP and NASA LCD products.

#### *4.3.4.1 CLMS OLCI LSP vs. MODIS LCD V6.1*

The results of the spatial consistency between CLMS OLCI LSP (SOSD, EOSD) and MODIS LCD V6.1 (MidGreenup, MidGreendown) are presented in Figure 26 and Figure 27. It should be noted that MidGreenup and MidGreendown layers are used in case of MODIS LCD V6.1 as they are those which better agree with OLCI LSP SOSD and EOSD (see section 4.4.5.1).

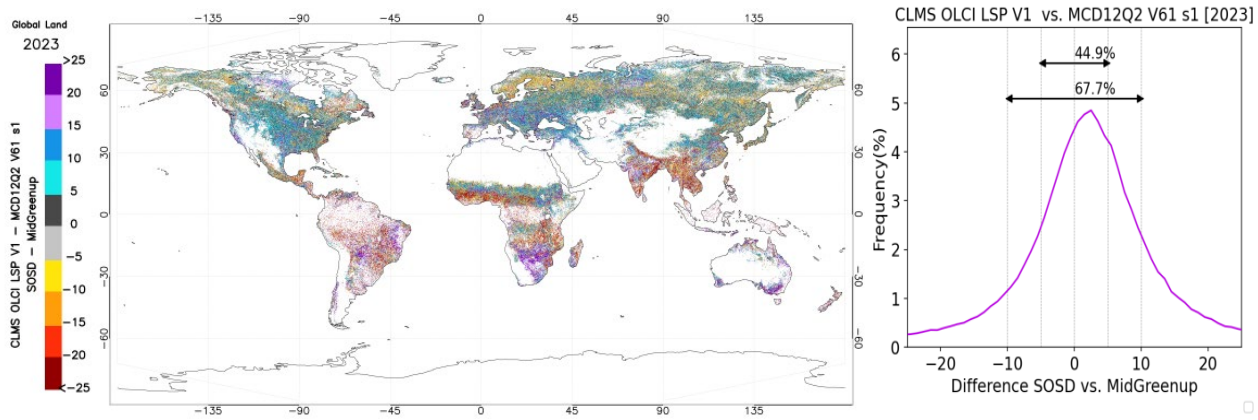



**Figure 26: Global map (left) and histogram (right) of differences between CLMS OLCI LSP SOSD and MODIS V6.1 MidGreenup for 2023 (season 1) considering best-quality pixels.**

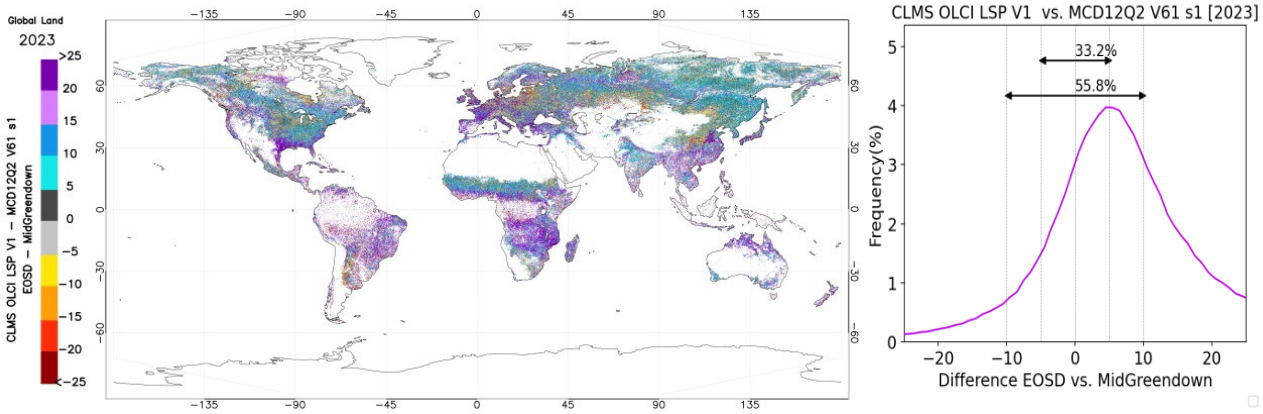



**Figure 27: Global map (left) and histogram (right) of differences between CLMS OLCI LSP EOSD and MODIS V6.1 MidGreendown for 2023 (season 1) considering best-quality pixels.**

Main results of the spatial consistency between OLCI LSP 300m V1 and MODIS LCD V6.1 are:

- OLCI LSP SOSD and MODIS MidGreenup show differences typically within ±15 days over large areas, in particular, for the northern hemisphere. Over tropical areas larger discrepancies can be observed with MODIS showing higher values (>20 days) for the MidGreenup. Large differences (>20 days) are also found over southern hemisphere, but showing OLCI LSP higher values than MODIS. The histogram of difference is centred around 3 days (slightly higher values for OLCI LSP SOSD despite being defined at a lower threshold value than MODIS MidGreenup).
- The global map of differences between OLCI LSP EOSD and MODIS MidGreendown shows, as expected, systematically positive differences (OLCI > MODIS) which reveals that they could be considered spatially consistent over most areas of the globe. The histogram of difference is centred around 5 days, with slightly higher values for OLCI LSP EOSD in agreement with the differences in their definition.

#### *4.3.4.2 CLMS OLCI LSP vs. VIIRS LCD V2*

The results of the spatial consistency between CLMS OLCI LSP (SOSD, EOSD) and VIIRS LCD V2 are presented in Figure 28 and Figure 29. As done for MODIS, Date_Mid_Greenup_Phase, Date_Mid_Senescence_Phase are used since they are the VIIRS V2 phenometrics which better agree with OLCI LSP SOSD and EOSD (see section 4.4.5.2). It should be noted that in case of VIIRS, all pixels are considered since the quality flag removes most of the observations around the whole globe.

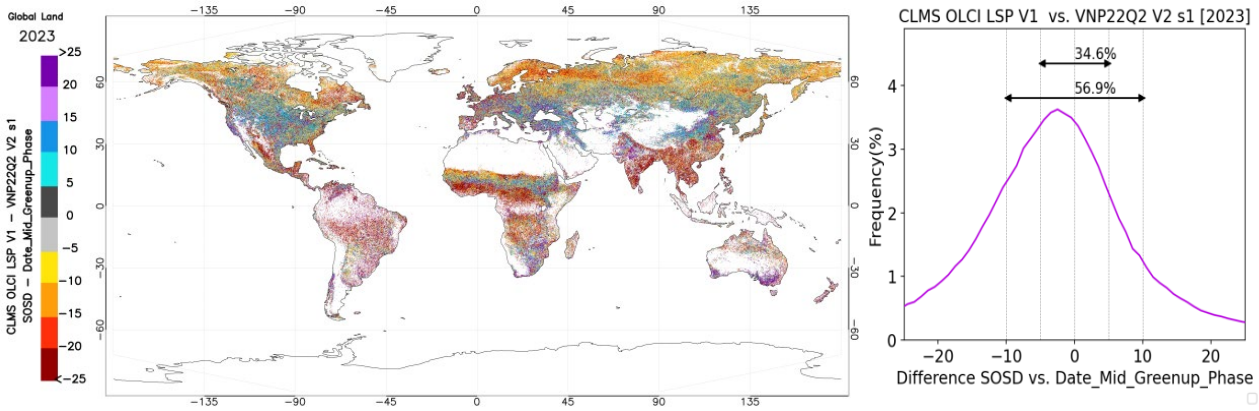



**Figure 28: Global map (left) and histogram (right) of differences between CLMS OLCI LSP SOSD and VIIRS V2 Date_Mid_Greenup_Phase (all pixels) for 2023 (season 1).**

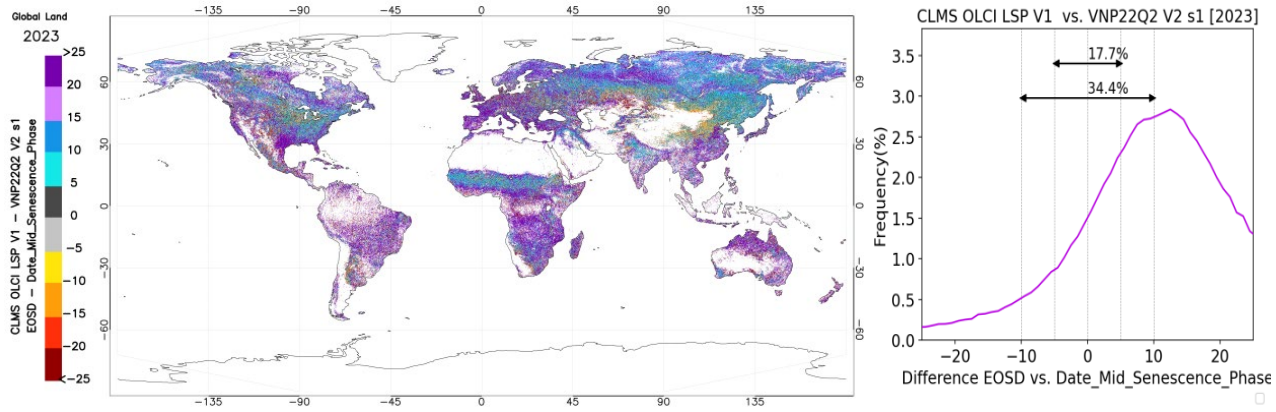



**Figure 29: Global map (left) and histogram (right) of differences between CLMS OLCI LSP EOSD and VIIRS V2 Date_Mid_Senescence_Phase (all pixels) for 2023 (season 1).**

Main results of the spatial consistency between CLMS OLCI LSP and VIIRS V2 are:

- For SOSD, CLMS LSP tends to provide mostly lower values (i.e., anticipated start of the season) than VIIRS V2 Date_Mid_Greenup_Phase as expected due to the differences in product definition. However, the opposite is observed over temperate latitudes.
- For EOSD, CLMS shows as expected higher values (i.e. late EOSD) as compared to VIIRS V2 Date_Mid_Senescence_Phase product over most areas around the globe.
- Histograms of differences are centred of around -3 and 12 days for SOSD and EOSD, respectively.

### 4.3.5 Latitudinal transect

Figure 30 shows the variation of CLMS OLCI LSP SOSD and EOSD as a function of the latitude for the latitudinal transect defined as the geodesic line from Norway (70.315ºN, 20.119ºE) to South Africa (34.872ºS, 20.119ºE) with a step of 2 km. For benchmarking, the latitudinal trends of MODIS LCD V6.1 and VIIRS LCD V2 dates of mid-green-up, and mid-senescence (or mid-green-down) phases are included in the analysis. Only good quality pixels according to quality ancillary layers (Table 7) are considered (except in case of VIIRS V2 where all pixels are considered).

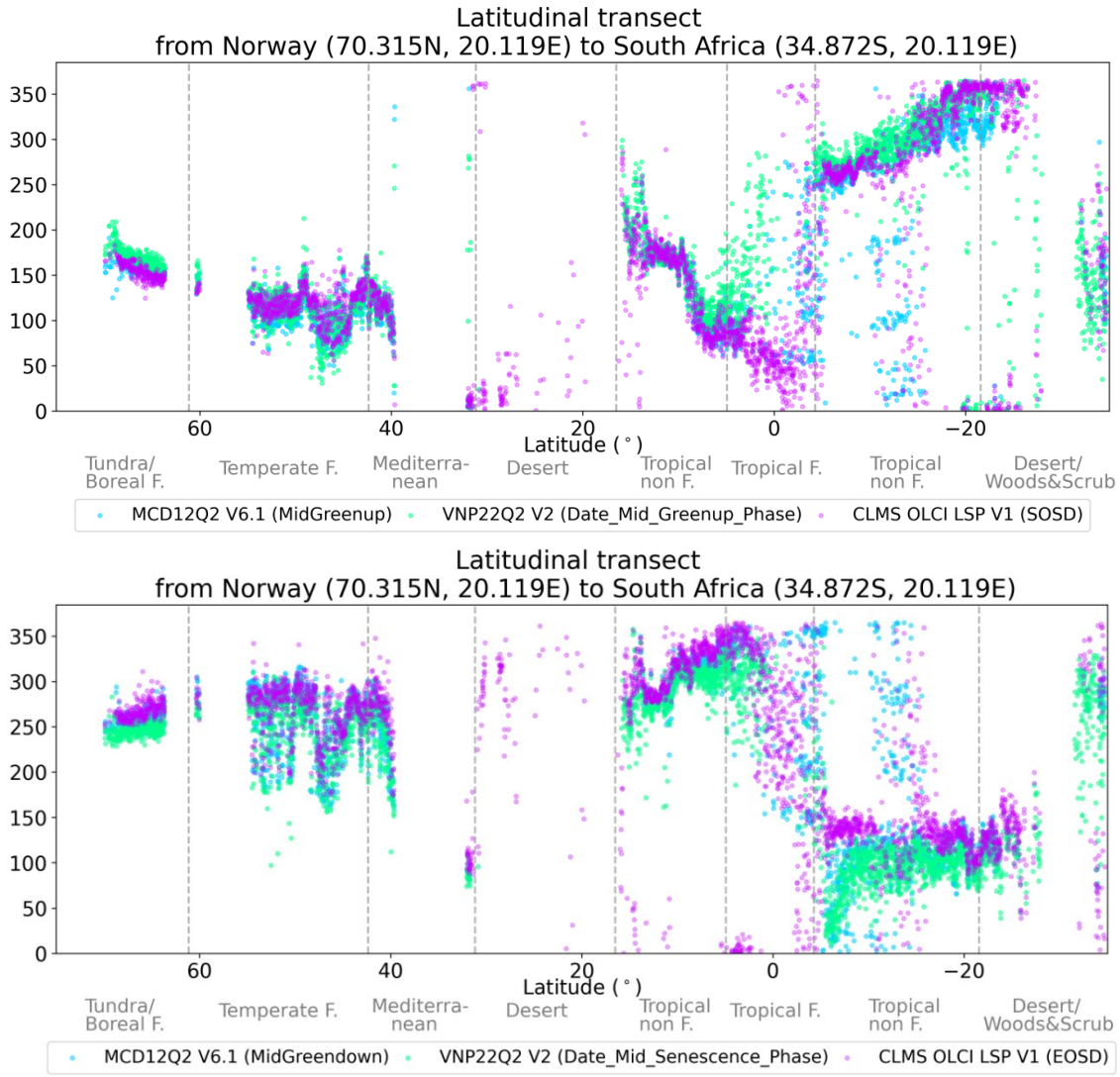



**Figure 30: Latitudinal transect of CLMS OLCI LSP (magenta points) SOSD (top) and EOSD (bottom) during 2023 (season 1) and MODIS LCD V6.1 (blue points) and VIIRS LCD V2 (green points) similar phenometrics from Norway (70.315ºN, 20.119ºE) to South Africa (34.872ºS, 20.119ºE).**

Main results are:

- Over Europe, the three products display the same trends for both SOSD and EOSD. As expected, the SOSD decreases from north to south. Beyond 60°N, VIIRS (green dots) product shows slightly higher SOSD and lower EOSD, i.e. a shorter growing season, than OLCI and MODIS products.
- Over Sahara desert, OLCI displays some unrealistic noisy trends for both SOSD and EOSD while MODIS and VIIRS does not provide estimates on these areas.

- Over tropical non forest region of the north hemisphere, all products agree and display the same trends: decrease of the SOSD (from July to February) and an increase of the EOSD (from September to December) from north to south i.e. about a 2-months season at the desert border to almost 9-months growing season for the equatorial forest.
- Larger discrepancies are observed over tropical forest around the equator for the SOSD for VIIRS products.
- Over the southern part of the equatorial zone, OLCI provides areas with large variability. OLCI LSP SOSD increases from September to December while EOSD is more stable in April/May. Note as well than MODIS (blue dots) display large variability around 12°/15°South, like CLMS, while VIIRS not. At the opposite, VIIRS display similar inconsistency for mid-greenup between 20°-25°South while MODIS and CLMS not.

### 4.3.6 Spatial variability (precision)

The spatial variability of CLMS OLCI LSP SOSD and EOSD phenological metrics and SPROD and TPROD productivity parameters are analysed for 6 ROIs over 5 km x 5 km area at full resolution (Table 2) for 2023. The STD and MAD are used as indicator of precision to evaluate the spatial variability.

Table 16 shows the precision indicators (STD, MAD) for SOSD and EOSD for the 6 ROIs, whilst the indicators for SPROD and TPROD are presented in Table 17. Relative values in percentage respect to the mean value over the area are also presented for SPROD and TPROD.

The maps of CLMS OLCI LSP (2023 season 1, good quality pixels) SOSD, EOSD, SPROD and TPROD layers over the 5 km x 5 km area for the 6 ROIs are presented in Annex III. In order to better understand the results, temporal profiles of OLCI PPI over four selected pixels for each ROI are also presented in Annex III.

**Table 16: Summary of spatial variability STD and MAD (i.e., indicators of precision) of CLMS OLCI LSP SOSD and EOSD (season1) evaluated over 5 km x 5 km ROIs during 2023.**

| | | STD (days) | | MAD (days) | |
|---|---|---|---|---|---|
| ROI | Biome | SOSD | EOSD | SOSD | EOSD |
| Evare | EBF | 55 | 76 | 20 | 28 |
| Mountain Lake | DBF | 4 | 7 | 2 | 4 |
| Oskoba | NLF | 12 | 23 | 3 | 4 |
| Albufera | CUL | 4 | 5 | 2 | 2 |
| Cimarron N.P. | HER | 4 | 6 | 3 | 4 |
| Jornada | SHR | 4 | 23 | 3 | 20 |

**Table 17: Summary of spatial variability STD (precision) and MAD indicators of CLMS OLCI LSP SPROD and TPROD evaluated over 5 km x 5 km ROIs during 2023. Relative values in percentage are computed respect to the mean values of the whole ROI.**

| | | STD ($m^2.m^{-2}.day$) | | MAD ($m^2.m^{-2}.day$) | |
|---|---|---|---|---|---|
| ROI | Biome | SPROD | TPROD | SPROD | TPROD |
| Evare | EBF | 50 (22.1%) | 81 (17.7%) | 27 (12%) | 35 (7.7%) |
| Mountain Lake | DBF | 39 (13.7%) | 41 (14.3%) | 26 (9.3%) | 26 (9%) |
| Oskoba | NLF | 20 (24.8%) | 20 (24.8%) | 12 (14.8%) | 12 (15.4%) |
| Albufera | CUL | 22 (12.9%) | 22 (12.9%) | 16 (9.3%) | 16 (9.3%) |
| Cimarron N.P. | HER | 14 (10.7%) | 14 (10.7%) | 8 (6%) | 8 (6%) |
| Jornada | SHR | 7 (45.8%) | 7 (45.8%) | 5 (36.8%) | 5 (36.8%) |

Main results are:

- For SOSD, remarkably good precision is found for all ROIs except for the EBF case, with STD better than 4 days (except for NLF, with 12 days) and MAD better than 3 days. Similarly, good spatial precision (slightly worse) is also found for EOSD, with STD and MAD lower typically than 7 days except for EBF, NLF and SHR. Additionally:
    - In case of *Evare* EBF area (see Figure 72 in Annex III), the bad spatial precision could be partly explained by the temporal noise of PPI temporal trajectories due to frequent cloud contamination, plus inherent limitations of spectral indices to capture seasonal variations in very dense rainforests, where field measurements reveals vertically and horizontally diverse phenology (Nunes et al., 2022). Season start either in around January-February (see temporal profiles B and D in Figure 72) or, in most cases, during the second half of the year see temporal profiles A and C in Figure 72). For the last cases, the EOSD are in the next year (i.e., ESOD > 365).
    - For *Mountain Lake* DBF (see Figure 73 in Annex III), SOSD and EOSD are spatially consistent around the whole area (STD<3days, MAD<4 days), with values typically provided for May and September/October. Similarly, for *Oskoba* NLF (see Figure 74 in Annex III), consistent SOSD and EOSD around the whole area are found for most cases (MAD of 3-4 days) with some exceptions where EOSD is provided for next year (e.g., temporal profile B in Figure 74).
    - Remarkably good spatial precision (STD=4-5 days, MAD=2 days) is found for SOSD and EOSD in *Albufera* cultivated rice area (see Figure 75 in Annex III) with SOSD and EOSD typically for June and September. Similar good precision is also found for

*Cimarron National Grassland Park* HER (STD and MAD <6 days, Figure 76 in Annex III).

- Finally, for *Jornada* SHR (Figure 77 in Annex III), SOSD and EOSD are typically provided in May and August-September, but there are some areas where the season is not provided (e.g., temporal profiles A and D in Figure 77). This can be explained in the low PPI values in this area and the selected threshold of the algorithm to detect or discard seasons, which are very sensitive for these low PPI values.

- The spatial variability of TPROD/SPROD is higher, with STD typically between 10% and 25% and MAD between 6% and 16%. SHR displays largest variability in relative terms (STD ~45%, MAD ~36%), mainly attributed to the low productivity values over this area.

## 4.4 ERROR EVALUATION

### 4.4.1 CLMS OLCI LSP versus ground references

#### *4.4.1.1 PEP725*

The results of the comparison of CLMS OLCI LSP SOSD and EOSD versus PEP725 dataset for 2023 are presented in Figure 31 and Figure 32. This analysis is performed with different phenophases provided by PEP725 (Table 9). For CLMS SOSD (Figure 31) the comparison is performed with PEP725 phases 10, 11 and 223 corresponding with 'first leaves separated', 'first leaves unfolded' and 'leaf unfolding ≥ 50%'. CLMS EOSD (Figure 32) is compared with PEP725 phases 95 ('50% of leaves fallen') and 205 ('autumnal colouration ≥ 50%'). The main validation metrics are displayed in Table 18. For benchmarking, the comparisons of MR-VPP versus PEP725 are also included.

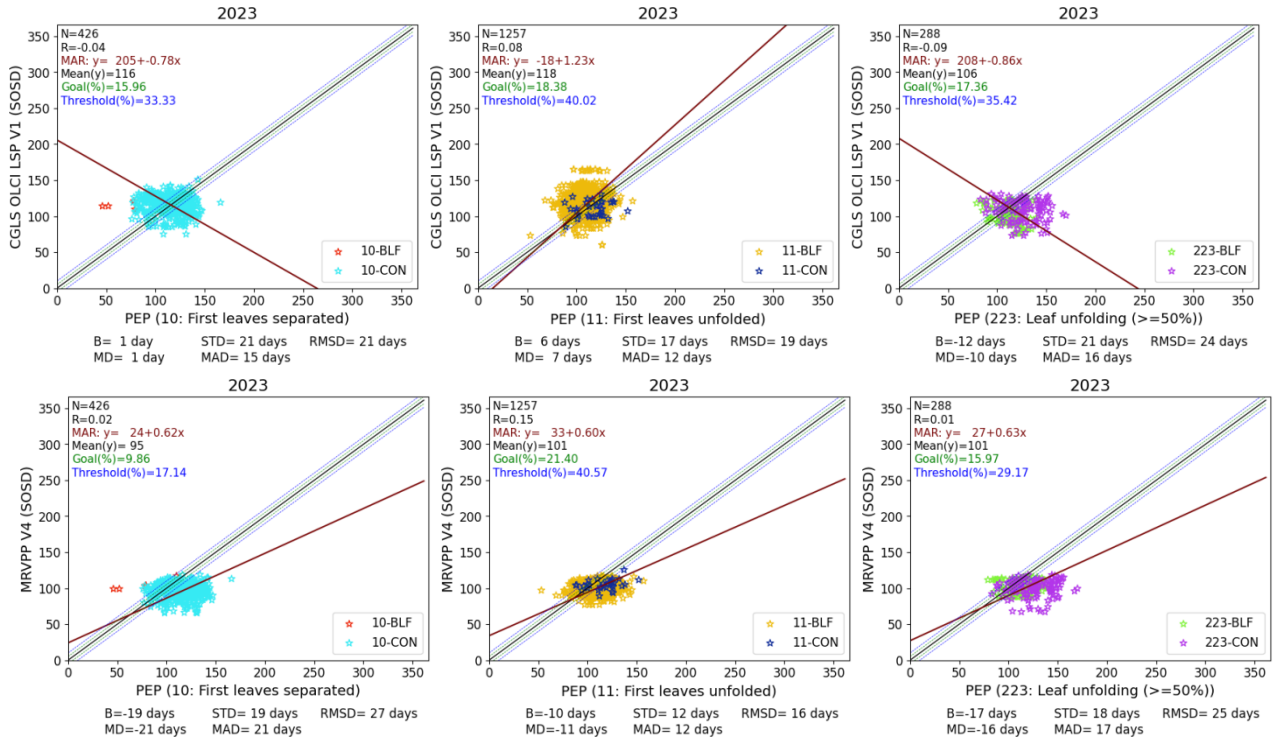



**Figure 31: CLMS OLCI LSP (top) and MR-VPP (bottom) SOSD (good quality pixels) versus PEP725 phases 10 ('first leaves separated', left), 11 ('first leaves unfolded', middle) and 223 ('leaf unfolding ≥ 50%', right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

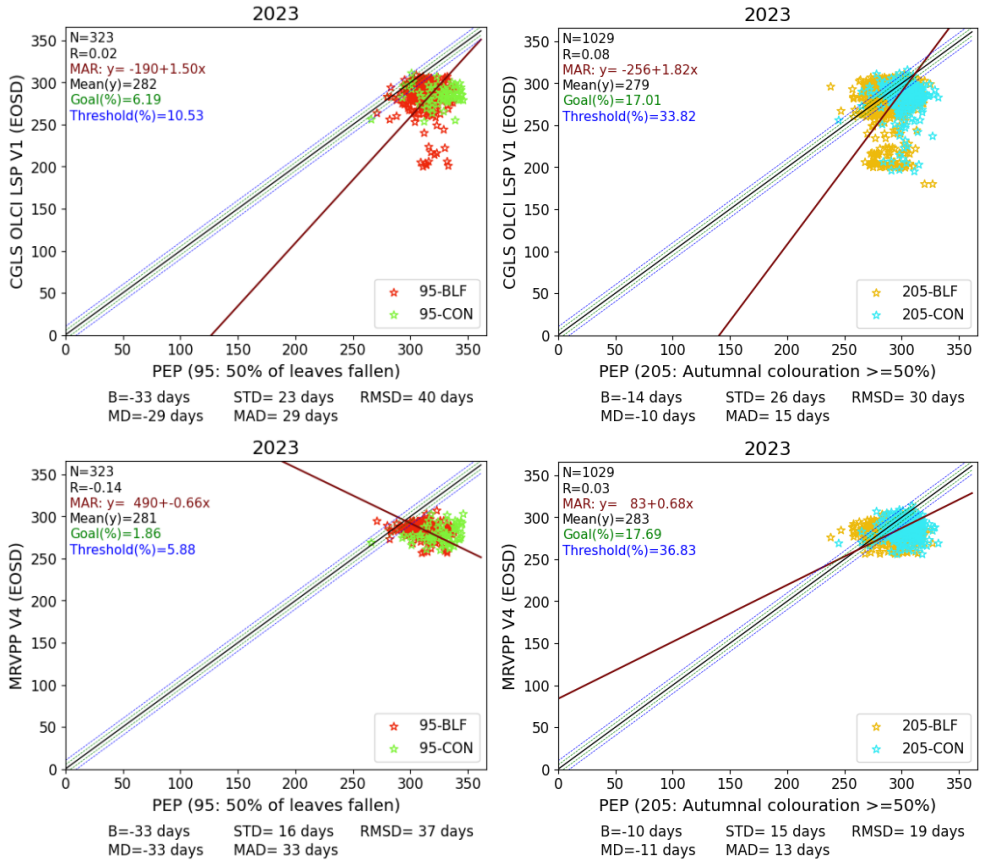



**Figure 32: CLMS OLCI LSP (top) and MR-VPP (bottom) EOSD (good quality pixels) versus PEP725 phases 95 ('50% of leaves fallen', left), and 205 ('autumnal colouration ≥ 50%', right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

**Table 18: Performance statistics between CLMS OLCI LSP and MR-VPP versus PEP725 database: SOSD versus PEP725 Phases 10, 11 and 223; and EOSD versus PEP725 Phases 95 and 205. Analysis for 2023 (season 1).**

| | CLMS OLCI LSP SOSD vs. PEP725 | | | CLMS OLCI LSP EOSD vs. PEP725 | |
|---|---|---|---|---|---|
| | **Phase 10** 'first leaves separated' | **Phase 11** 'first leaves unfolded' | **Phase 223** 'leaf unfolding ≥ 50%' | **Phase 95** '50% of leaves fallen' | **Phase 205** 'autumnal colouration ≥ 50%' |
| N | 426 | 1257 | 288 | 323 | 1029 |
| R | -0.04 | 0.08 | -0.09 | 0.02 | 0.08 |
| Bias (days) | 1 | 6 | -12 | -33 | -14 |
| MD (days) | 1 | 7 | -10 | -29 | -10 |
| STD (days) | 21 | 17 | 21 | 23 | 26 |
| MAD (days) | 15 | 12 | 16 | 29 | 15 |
| RMSD (days) | 21 | 19 | 24 | 40 | 30 |
| MAR | y=205-0.78x | y=-18+1.23x | y=208-0.86x | y=-190-1.5x | y=-256+1.82x |
| % goal | 16.0 | 18.4 | 17.4 | 6.2 | 17.0 |
| % threshold | 33.3 | 40.0 | 35.4 | 10.5 | 33.8 |
| | **MR-VPP SOSD vs. PEP725** | | | **MR-VPP EOSD vs. PEP725** | |
| | **Phase 10** 'first leaves separated' | **Phase 11** 'first leaves unfolded' | **Phase 223** 'leaf unfolding ≥ 50%' | **Phase 95** '50% of leaves fallen' | **Phase 205** 'autumnal colouration ≥ 50%' |
| N | 426 | 1257 | 288 | 323 | 1029 |
| R | 0.02 | 0.15 | 0.01 | -0.14 | 0.03 |
| Bias (days) | -19 | -10 | -17 | -33 | -10 |
| MD (days) | -21 | -11 | -16 | -33 | -11 |
| STD (days) | 19 | 12 | 18 | 16 | 15 |
| MAD (days) | 21 | 12 | 17 | 33 | 13 |
| RMSD (days) | 27 | 16 | 25 | 37 | 19 |
| MAR | y=24+0.62x | y=33+0.60x | y=27+0.63x | y=490-0.66x | y=83+0.68x |
| % goal | 9.9 | 21.4 | 16.0 | 1.9 | 17.7 |
| % threshold | 17.1 | 40.6 | 29.2 | 5.9 | 36.8 |

Main results of the comparison between CLMS OLCI LSP (and MR-VPP) versus PEP725 are:

- CLMS OLCI SOSD shows slight positive bias (i.e., delay of start of season) in the comparison with PEP725 phases 10 and 11, corresponding to first leaves separated (B=1 day) and unfolded (B=6 days). CLMS OLCI SOSD shows anticipated start of season (B = -12 days) with PEP725 223 phase (i.e., leaf unfolding ≥50%), as it can be expected according to the phenophase definition (more than 50% of leaves present). The comparison in terms of uncertainty shows RMSD values of 21, 19 and 24 days for phase 10, 11 and 223, respectively.
- MR-VPP shows anticipated start of season than PEP725 in all cases: B=-19, -10 and -17 days for phase 10, 11 and 223, respectively.
- GCLS OLCI EOSD shows negative bias (i.e., anticipated end of season) of -33 and -14 days in comparison with PEP725 phases 95 (‘50% of leaves fallen’) and 205 (‘autumnal colouration ≥50%’), showing that the OLCI LSP product represents better the ‘colouration’ phenophase than the ‘leaf fallen’. Larger discrepancies are found in terms of uncertainties (RMSD of 40 and 30 days).
- MR-VPP shows similar anticipated end of season and slightly better results in terms of precision (STD, MAD) and uncertainty (RMSD).
- The slightly positive bias for SOSD (compared with ‘first leaves’) and negative for EOSD is translated in shorter season as compared with PEP725 observations. These results are in line to that found for HR-VPP [HR-VPP_ST&VPP_VR] based on Sentinel-2, where positive (4) and negative (-11) biases were found for SOSD and EOSD for the comparison with PEP725 for the 2017-2020 period.
- The compliance of CLMS OLCI LSP SOSD and EOSD layers with user’s requirements ranges between 16-18% of cases for goal and 33-40% for threshold.

#### *4.4.1.2 TEMPO*

The results of the comparison of CLMS OLCI LSP SOSD and EOSD versus TEMPO dataset for 2023 are presented in Figure 33 and Figure 34. CLMS SOSD (Figure 33) is compared with TEMPO phases 0 and 1 corresponding with ‘germination/sprouting/bud development’ and ‘leaf development/main shoot’ (Table 11). In case of CLMS EOSD (Figure 34), the comparison is performed with TEMPO phase 9 ‘beginning of dormancy/die off of leaves’. The main validation metrics are displayed in Table 19. For benchmarking, the comparisons of MR-VPP versus TEMPO are also included.

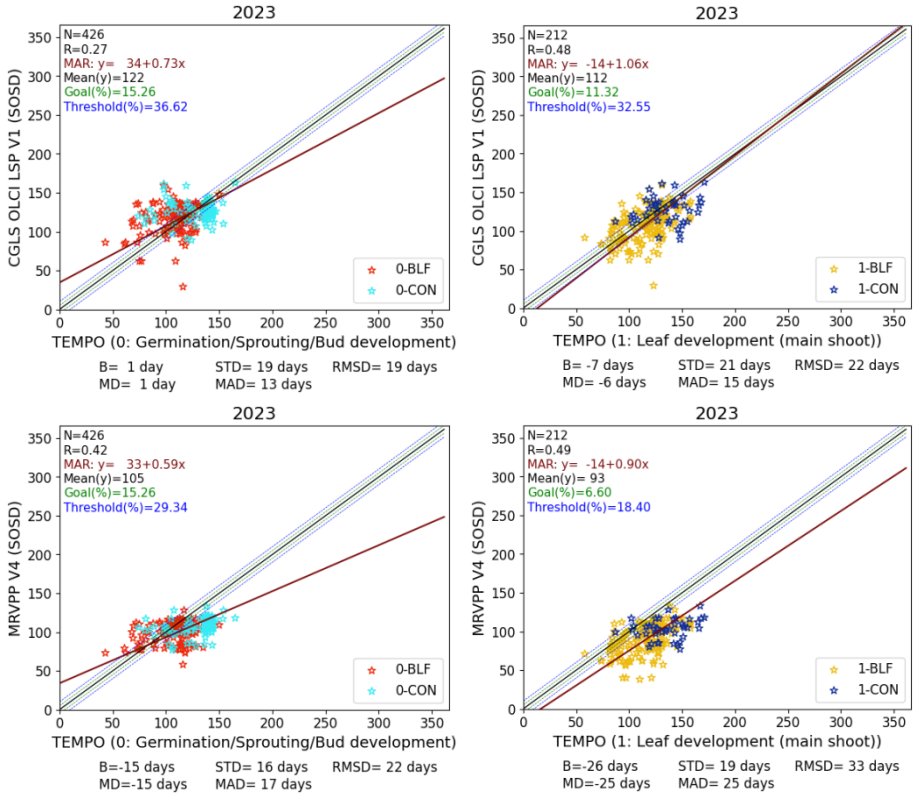



**Figure 33: CLMS OLCI LSP (top) and MR-VPP (bottom) SOSD (good quality pixels) versus TEMPO phases 0 ('germination/sprouting/bud development', left) and 1 ('leaf development/main shoot', right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

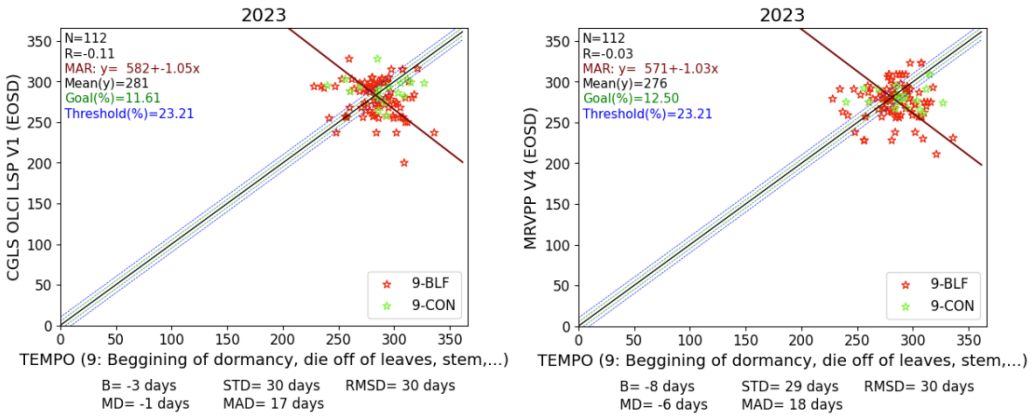



**Figure 34: CLMS OLCI LSP (left) and MR-VPP (right) EOSD (good quality pixels) versus TEMPO phase 9 ('beginning of dormancy/die off of leaves') scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

**Table 19: Performance statistics between CLMS OLCI LSP and MR-VPP versus TEMPO database: SOSD versus TEMPO Phases 0 and 1; and EOSD versus TEMPO Phase 9. Analysis for 2023 (season 1).**

| | CLMS OLCI LSP SOSD vs. TEMPO | | CLMS OLCI LSP EOSD vs. TEMPO |
|---|---|---|---|
| | **Phase 0** 'germination/sprouting/ bud development' | **Phase 1** 'leaf development/ main shoot' | **Phase 9** 'beginning of dormancy/ die off of leaves' |
| N | 426 | 212 | 112 |
| R | 0.27 | 0.48 | -0.11 |
| Bias (days) | 1 | -7 | -3 |
| MD (days) | 1 | -6 | -1 |
| STD (days) | 19 | 21 | 30 |
| MAD (days) | 13 | 15 | 17 |
| RMSD (days) | 19 | 22 | 30 |
| MAR | y=34+0.73x | y=-14+1.06x | y=582-1.05x |
| % goal | 15.2 | 11.3 | 11.6 |
| % threshold | 36.6 | 32.6 | 23.2 |
| | **MR-VPP SOSD vs. TEMPO** | | **MR-VPP EOSD vs. TEMPO** |
| | **Phase 0** 'germination/sprouting/ bud development' | **Phase 1** 'leaf development/ main shoot' | **Phase 9** 'beginning of dormancy/ die off of leaves' |
| N | 426 | 212 | 112 |
| R | 0.42 | 0.49 | -0.03 |
| Bias (days) | -15 | -26 | -8 |
| MD (days) | -15 | -25 | -6 |
| STD (days) | 16 | 19 | 29 |
| MAD (days) | 17 | 25 | 18 |
| RMSD (days) | 22 | 33 | 30 |
| MAR | y=33+0.59x | y=-14+0.90x | y=571-1.03x |
| % goal | 15.3 | 6.6 | 12.5 |
| % threshold | 29.3 | 18.4 | 23.2 |

Main results of the comparison between CLMS OLCI LSP (and MR-VPP) versus TEMPO are:

- CLMS OLCI SOSD shows almost no mean bias (B=1 day) with TEMPO phase 0 (i.e., germination) and anticipated start of season (B=-7 days) with TEMPO phase 1 (leaf development), in line to that found in the comparison with PEP725 with phases related to 'first leaves' and 'leaf unfolding ≥50%'. Correlations are higher (up to 0.48) than with PEP725 data. The comparison in terms of uncertainty shows RMSD values of 19 and 22 days.

- MR-VPP shows worse uncertainty (RMSD of 22 and 33 days) and anticipated start of season than TEMPO: B=-15 and -26 days compared with 'germination' and 'leaf development'.
- GCLS OLCI EOSD shows negative bias (i.e., anticipated end of season) of -3 days in comparison with TEMPO phase 9 ('beginning of dormancy') and overall uncertainty (RMSD) of 30 days.
- MR-VPP shows larger anticipated end of season (B=-8 days) and same results in terms of uncertainty.
- The compliance of CLMS OLCI LSP SOSD and EOSD layers with user's requirements ranges between 11-16% of cases for goal and 23-37% for threshold.

#### *4.4.1.3 NEON*

The results of the comparison on CLMS OLCI LSP SOSD and EOSD versus NEON dataset for 2023 are presented in Figure 35 and Figure 36. CLMS OLCI LSP SOSD is compared with both NEON phases 'breaking' and 'emerging' leaves/buds. In case of CLMS EOSD, the comparison is performed with NEON phases 'colored' and 'falling' leaves/buds. The main validation metrics are summarized in Table 20.

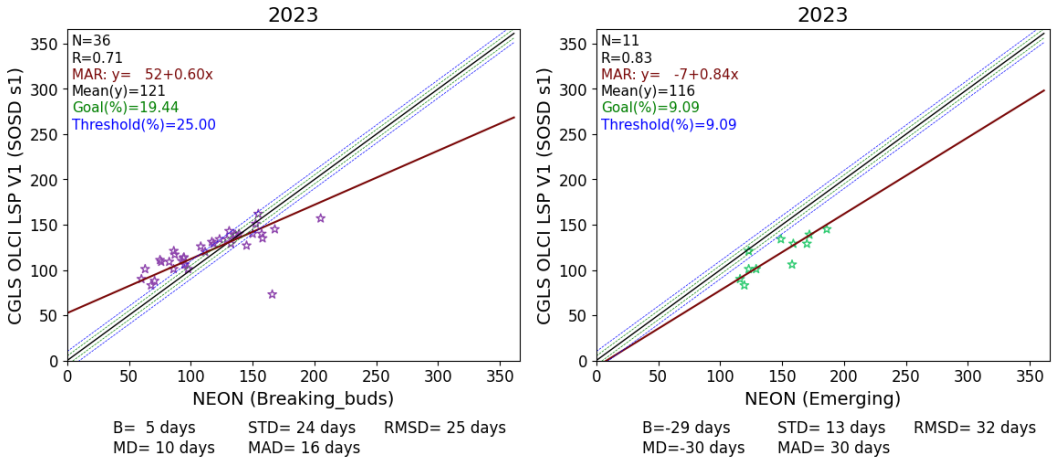



**Figure 35: CLMS OLCI LSP SOSD (good quality pixels) versus NEON phases breaking leaves/buds (left) and emerging leaves/buds (right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

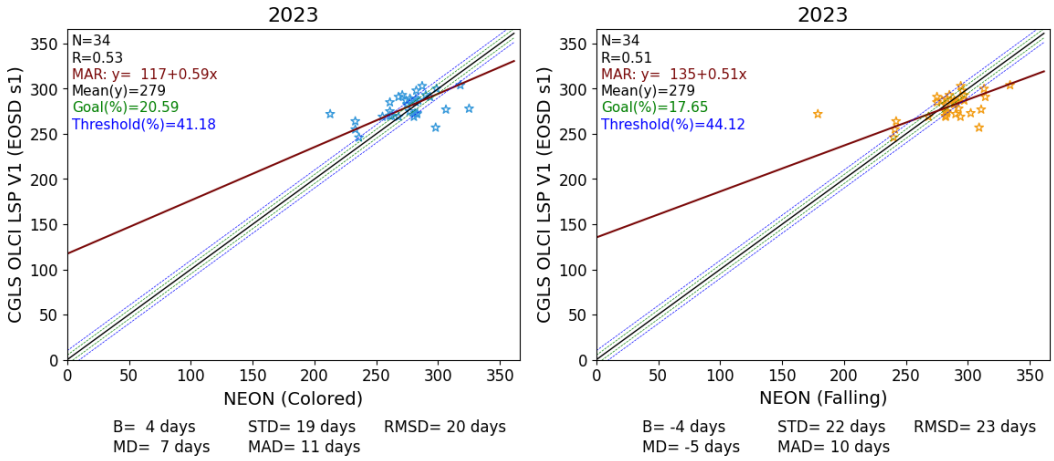



**Figure 36: CLMS OLCI LSP EOSD (good quality pixels) versus NEON phases colored leaves/buds (left), and falling leaves/buds (right) scatter-plots during 2023 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

**Table 20: Performance statistics between CLMS OLCI LSP SOSD and EOSD with NEON breaking&emerging, colored and falling leaves/buds.**

| | CLMS OLCI LSP SOSD vs. NEON | | CLMS OLCI LSP EOSD vs. NEON | |
|---|---|---|---|---|
| | breaking leaves/buds | emerging leaves/buds | colored leaves/buds | falling leaves/buds |
| N | 36 | 11 | 34 | 34 |
| R | 0.71 | 0.83 | 0.53 | 0.51 |
| Bias (days) | 5 | -29 | 4 | -4 |
| MD (days) | 10 | -30 | 7 | -5 |
| STD (days) | 24 | 13 | 19 | 22 |
| MAD (days) | 16 | 30 | 11 | 10 |
| RMSD (days) | 25 | 32 | 20 | 23 |
| MAR | y=52+0.60x | y=-7+0.84x | y=117+0.59x | y=135+0.51x |
| % goal | 19.4 | 9.1 | 20.6 | 17.7 |
| % threshold | 25.0 | 9.1 | 41.2 | 44.1 |

Main results of the comparison between CLMS OLCI LSP and NEON are:

- In terms of accuracy, SOSD shows slight positive bias of 5 days (i.e., delay) compared with NEON ‘breaking leaves/buds’ phase and large negative bias (i.e., anticipated start of season) compared with NEON ‘emerging’ leaves/buds. It should be noted the low number of samples in the case of ‘emerging’ leaves which makes less confident the result. The comparison in

terms of uncertainty shows RMSD of 25 and 32 days, showing remarkably good correlations with Pearson correlations of 0.71 and 0.83.

- EOSD shows positive bias of 4 days compared with 'colored leaves/buds' and negative bias of -4 days in comparison with NEON 'falling leaves/buds'. That means that OLCI LSP EOSD is between the 'colored leaves' and 'falling leaves' which seem to be quite realistic. In terms of uncertainties, RMSD values of 20 and 23 days are found.
- The compliance of CLMS OLCI LSP SOSD and EOSD layers with user's requirements ranges between 9-21% of cases for goal and up to 44% for threshold.

#### *4.4.1.4 Summary of direct validation*

Figure 37 summarizes the results of the accuracy (Bias) and uncertainty (RMSD) of CLMS OLCI LSP SOSD and EOSD versus different phenophases of ground references from PEP725, TEMPO and NEON databases. The summary of the percentages of cases within goal and threshold user requirements is provided in Figure 38.

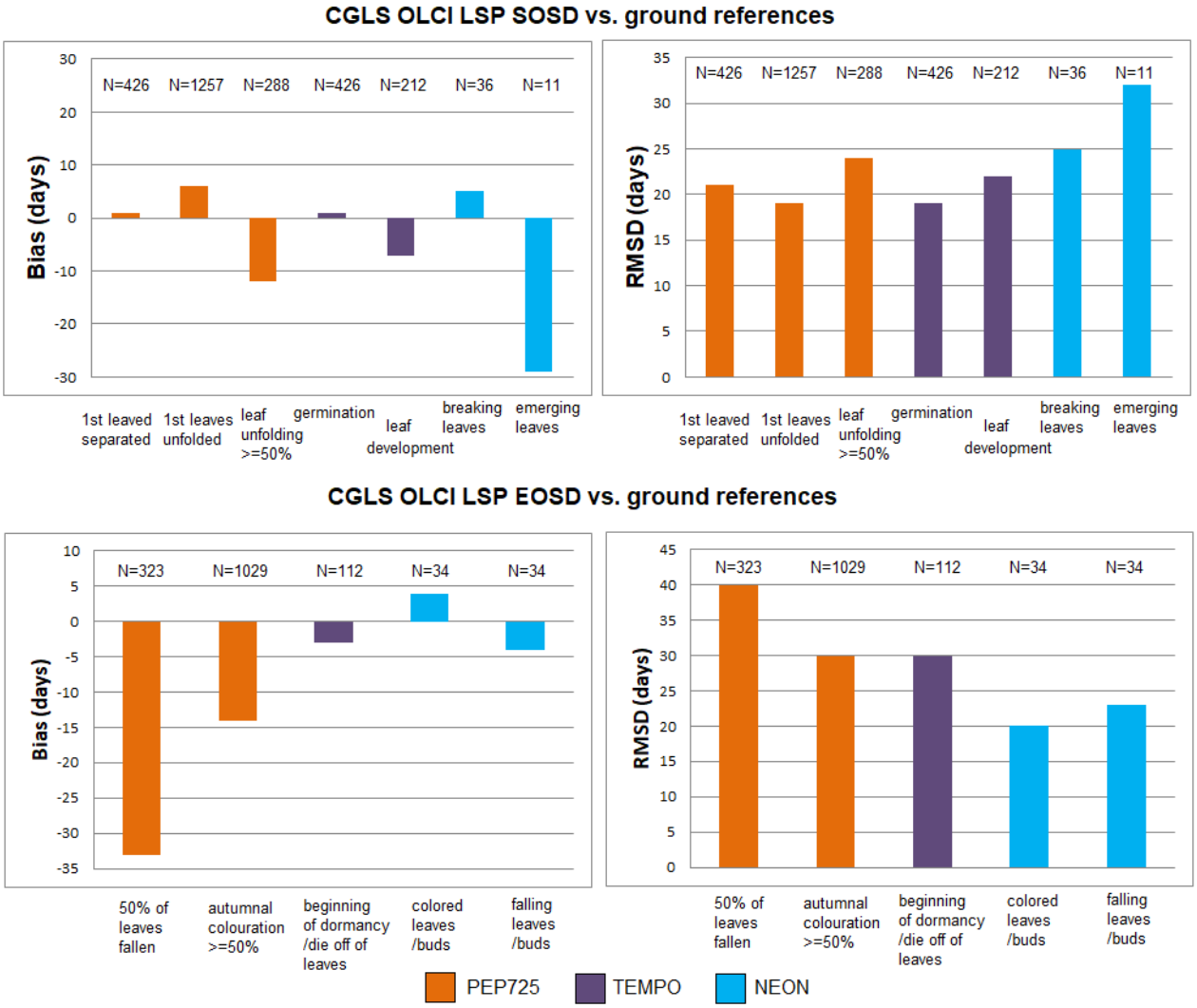



**Figure 37: Diagram bars of Bias (left) and RMSD (right) of the comparison of CLMS OLCI LSP SOSD (top) and EOSD (bottom) versus different phenophases of PEP725, TEMPO and NEON ground databases. Analysis for 2023 (season 1). N indicates the number of samples for each comparison.**

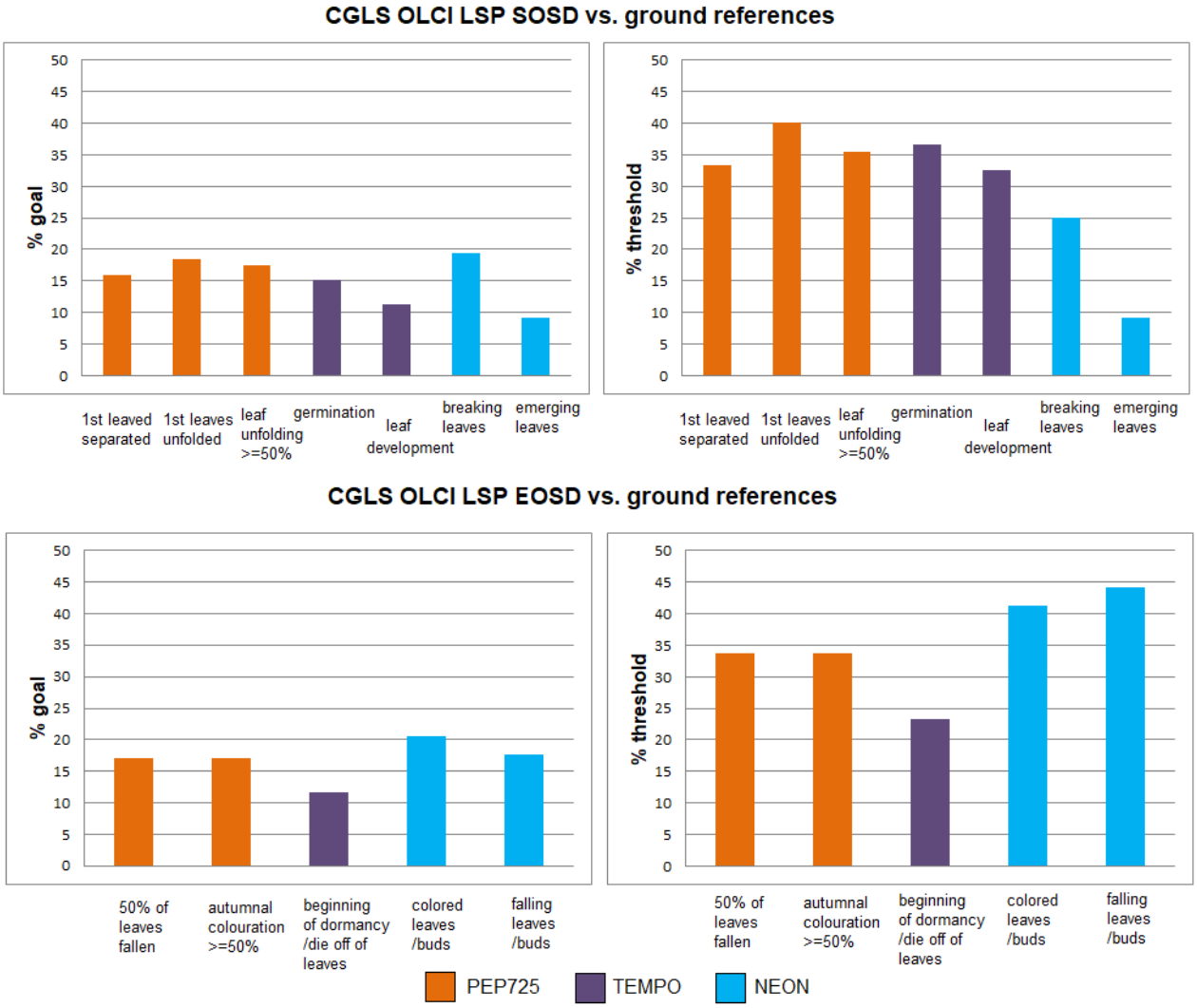



**Figure 38: Diagram bars of percentage of cases within goal (left) and threshold (right) user requirements of the comparison of CLMS OLCI LSP SOSD (top) and EOSD (bottom) versus different phenophases of PEP725, TEMPO and NEON ground databases. Analysis for 2023 (season 1).**

Main conclusions from Figure 37 and Figure 38 are:

- For SOSD, bias close to zero or slight positive (i.e., delay, up to 5 days) is found in the comparison with ground phenophases related to the first stages of the start the growing season (e.g., 1st leaves separated or unfolded, germination or breaking leaves). However, negative bias (typically between -7 and -12 days) is found for the phenophases related to more developed leaf growth (e.g., leaf unfolding ≥50%, leaf development). Larger negative bias (-29 days) is found for NEON emerging leaves but with limited sampling (N=11).

- The bias for EOSD is mostly negative (i.e., anticipated end of season, up to -33 days) and OLCI LSP is typically closer to phenophases related to the coloration phase than the falling of leaves, as expected.
- RMSD is typically ranging between 20 and 30 days for SOSD and from 20 to 40 days for EOSD.
- The percentage of cases CLMS OLCI LSP within goal requirements typically ranges between 10% and 20% in case of SOSD and between 15% and 20% in case of EOSD.
- In case of threshold user requirement, the compliance typically ranges between 25% and 40% of cases for SOSD (except emerging leaves phenophase in NEON with few samples) and between 30% and 45% for EOSD (except beginning of dormancy in TEMPO).

### 4.4.2 CLMS OLCI LSP versus near-ground-based PhenoCam V3 dataset

The results of the comparison on CLMS OLCI LSP SOSD and EOSD versus PhenoCam V3 near-ground-based dataset for 2023 are presented in Figure 39 and Figure 40. This analysis is performed with PhenoCam V3 start and end of seasons defined at 10%, 25% and 50%. The main validation metrics are summarized in Table 21.

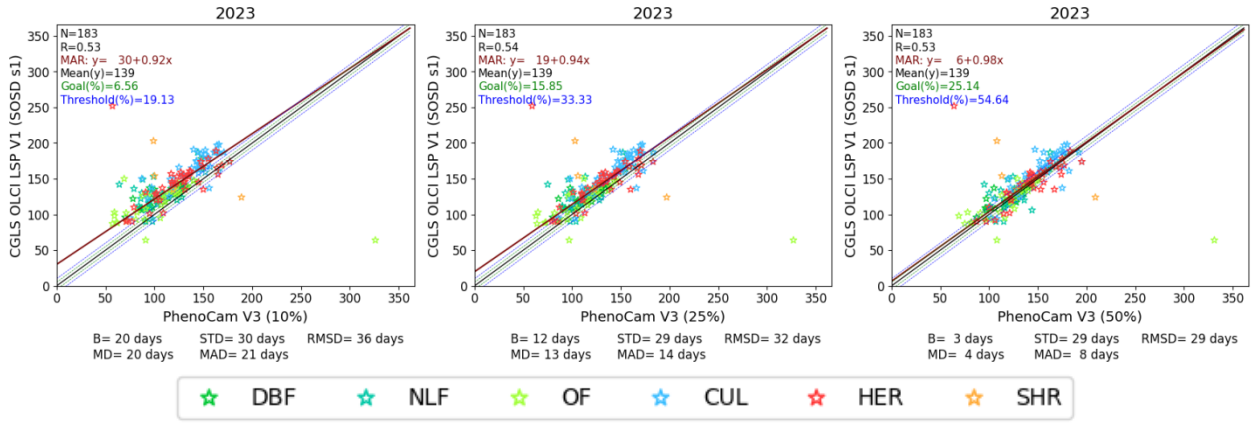



**Figure 39: CLMS OLCI LSP SOSD (good quality pixels) versus PhenoCam V3 start of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2023 (season 1).**

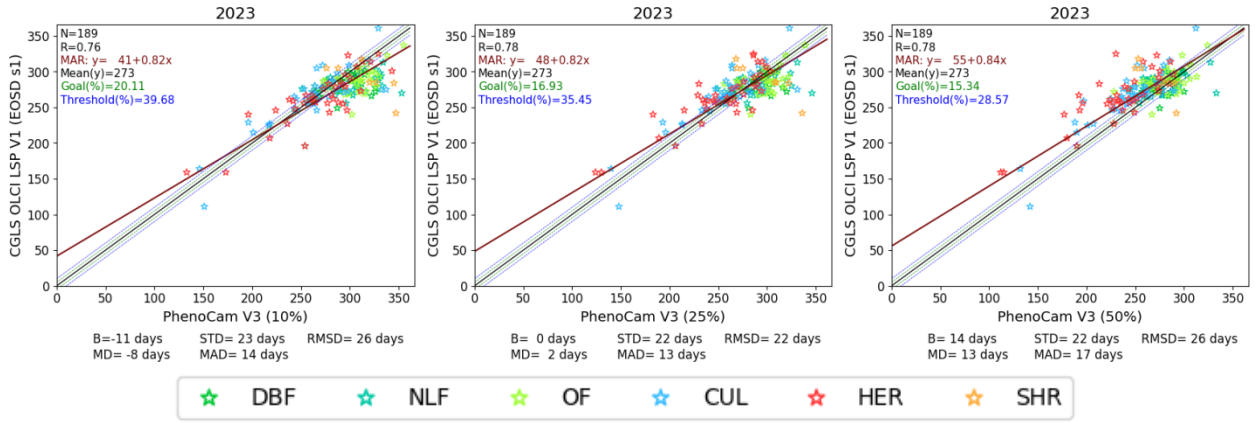



**Figure 40: CLMS OLCI LSP EOSD (good quality pixels) versus PhenoCam V3 end of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2023 (season 1).**

**Table 21: Performance statistics between CLMS OLCI LSP SOSD and EOSD with PhenoCam V3 equivalent phases defined at 10%, 25% and 50% for 2023 (season 1).**

| | CLMS OLCI LSP SOSD vs. PhenoCamV3 | | | CLMS OLCI LSP EOSD vs. PhenoCamV3 | | |
|---|---|---|---|---|---|---|
| | 10% | 25% | 50% | 10% | 25% | 50% |
| N | 183 | | | 189 | | |
| R | 0.53 | 0.54 | 0.53 | 0.76 | 0.78 | 0.78 |
| Bias (days) | 20 | 12 | 3 | -11 | 0 | 14 |
| MD (days) | 20 | 13 | 4 | -8 | 2 | 13 |
| STD (days) | 30 | 29 | 29 | 23 | 22 | 22 |
| MAD (days) | 21 | 14 | 8 | 14 | 13 | 17 |
| RMSD (days) | 36 | 32 | 29 | 26 | 22 | 26 |
| MAR | y=30+0.92x | y=19+0.94x | y=6+0.98x | y=41+0.82x | y=48+0.82x | y=55+0.84x |
| % goal | 6.6 | 15.9 | 25.1 | 20.1 | 16.9 | 15.3 |
| % threshold | 19.1 | 33.3 | 54.6 | 39.8 | 35.5 | 28.6 |

Main results of the comparison between CLMS OLCI LSP and PhenoCam V3 are:

- CLMS OLCI LSP SOSD better agrees in terms of accuracy (B, MD) and uncertainty (RMSD) with PhenoCam V3 start of season defined at 50% threshold (Bias = 3 days) than compared with Phenocam V3 defined at 10% (Bias = 20 days) and 25% (Bias = 12 days). As CLMS SOSD is defined at 25% threshold, the comparison with the same PhenoCam threshold shows that CLMS OLCI LSP SOSD is delayed as compared to PhenoCam near-ground-based estimates.

- For the comparison of CLMS OLCI LSP EOSD (defined at 15%) the best agreement in terms of bias is found with PhenoCam defined at 25% (Bias = 0 days) compared with 10% (Bias = -11 days) and 50% (Bias = 14 days). Therefore, it can be concluded that OLCI LSP EOSD is slightly anticipated as compared to PhenoCam near-ground-based estimates. Similarly, the best agreement in terms of uncertainty is found with PhenoCam defined at 25%, with RMSD of 22 days.
- The positive bias for SOSD and negative bias for EOSD is translated in shorter season of CLMS OLCI LSP as compared with this near-ground-based dataset.

### 4.4.3 Overall consistency over Europe with MR-VPP

The overall comparison between CLMS OLCI LSP SOSD and EOSD versus MR-VPP equivalent layers over the European sampling EVAL is presented in this section (Figure 41, Table 22). Main results are:

- Both products are comparable with high correlations (R > 0.81) and RMSD of 22 and 23 days for SOSD and EOSD.
- Large positive bias (B = 18 days, MD = 19 days) is found for SOSD, whilst low negative bias is found for EOSD (B = -2 days, MD = -4 days). In summary, CLMS OLCI LSP provides shorter season of around 20-23 days compared with MR-VPP. These results are in line to the tendency of the whole European area, as displayed in the difference maps of Annex I.

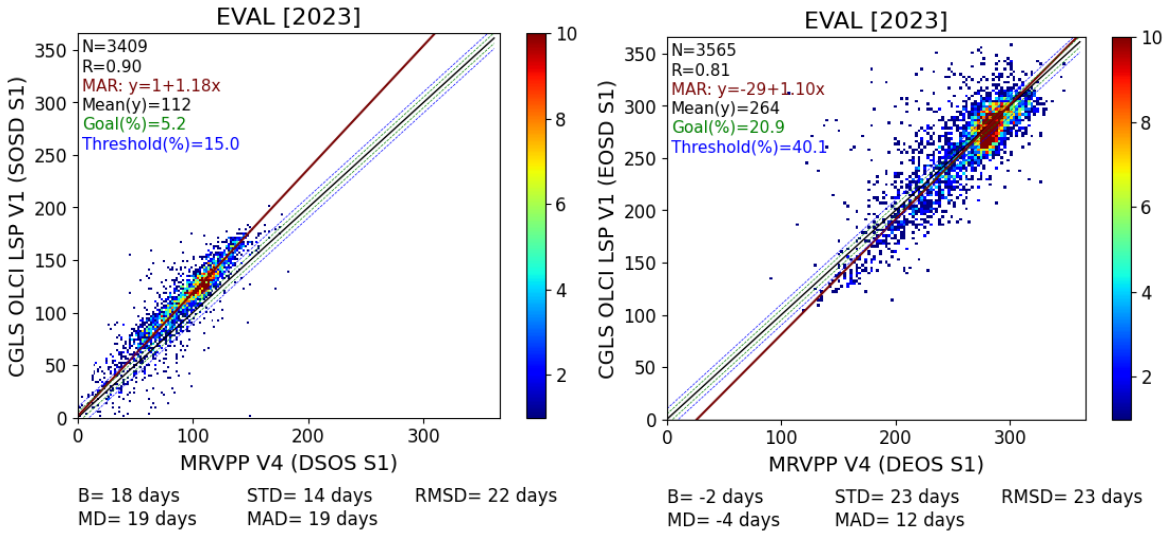



**Figure 41: CLMS OLCI LSP SOSD and EOSD versus MR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 22: Performance statistics between CLMS OLCI LSP SOSD and EOSD versus MR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP vs. MR-VPP | |
|---|---|---|
| | SOSD (days) | EOSD (days) |
| N | 3409 | 3565 |
| R | 0.90 | 0.81 |
| Bias (days) | 18 | -2 |
| MD (days) | 19 | -4 |
| STD (days) | 14 | 23 |
| MAD (days) | 19 | 12 |
| RMSD (days) | 22 | 23 |
| MAR | y=1+1.18x | y=-29+1.10x |
| % goal | 5.2 | 20.9 |
| % threshold | 15.0 | 40.1 |

The overall comparison between CLMS OLCI LSP TPROD and SPROD versus MR-VPP equivalent layers over the European sampling EVAL is presented in Figure 42, and main statistics are summarized in Table 23. Main results are:

- Both products are comparable with high correlations (R ~ 0.9).
- CLMS OLCI TPROD/SPROD show higher values (B and MD around 25%) than equivalent MR-VPP layers, despite of the fact that CLMS product shows shorter season.
- The higher productivity of CLMS than MR-VPP over EVAL sampling was also observed over almost the whole European region (see difference maps in Annex I), and is explained by the higher maximum PPI values of CLMS products.

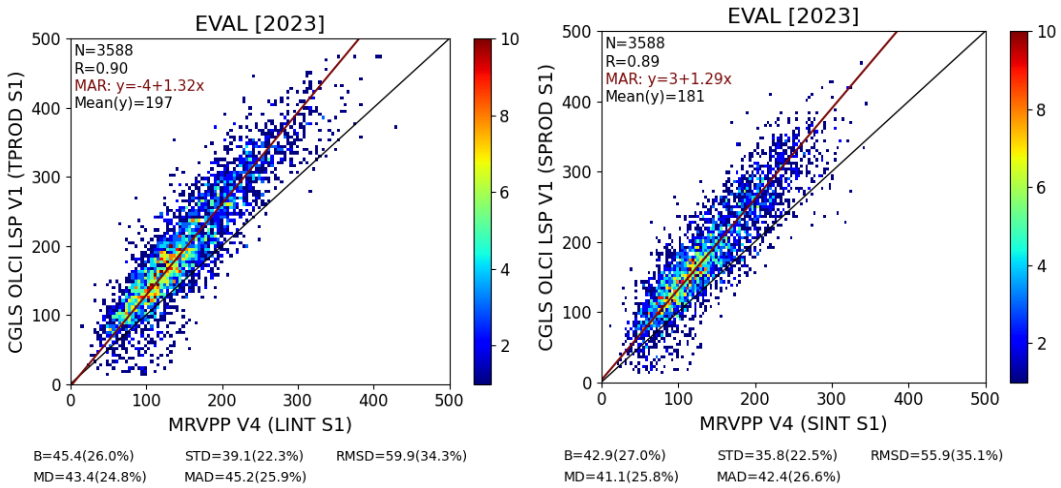



**Figure 42: CLMS OLCI LSP TPROD and SPROD versus MR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 23: Performance statistics between CLMS OLCI LSP TPROD and SPROD versus MR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP vs. MR-VPP | |
|---|---|---|
| | TPROD ($m^2 \cdot m^{-2} \cdot day$) | SPROD ($m^2 \cdot m^{-2} \cdot day$) |
| N | 3588 | 3588 |
| R | 0.90 | 0.89 |
| Bias ($m^2m^{-2}$) | 45.4 (26.0%) | 42.9 (27.0%) |
| MD ($m^2m^{-2}$) | 43.4 (24.8%) | 41.1 (25.8%) |
| STD ($m^2m^{-2}$) | 39.1 (22.3%) | 35.8 (22.5%) |
| MAD ($m^2m^{-2}$) | 45.2 (25.9%) | 42.4 (26.6%) |
| RMSD ($m^2m^{-2}$) | 59.9 (34.3%) | 55.9 (35.1%) |
| MAR | y=-4+1.32x | y=3+1.29x |

### 4.4.4 Overall consistency over Europe with HR-VPP

The overall comparison between CLMS OLCI LSP SOSD and EOSD versus HR-VPP equivalent layers over the European sampling EVAL is presented in this section (Figure 43, Table 24). Main results are:

- Both products are comparable with high correlations (R ≥ 0.82) and RMSD of 21 and 24 days for SOSD and EOSD.
- Low positive bias (B = 6 days, MD = 1 day, i.e., delay) is found for EOSD, whilst OLCI tends to provide larger positive bias (B = 14 days, MD = 12 days) for SOSD in the comparison with HR-VPP SOSD. This is translated in shorter season of OLCI products of around 8-11 days as compared to HR-VPP.

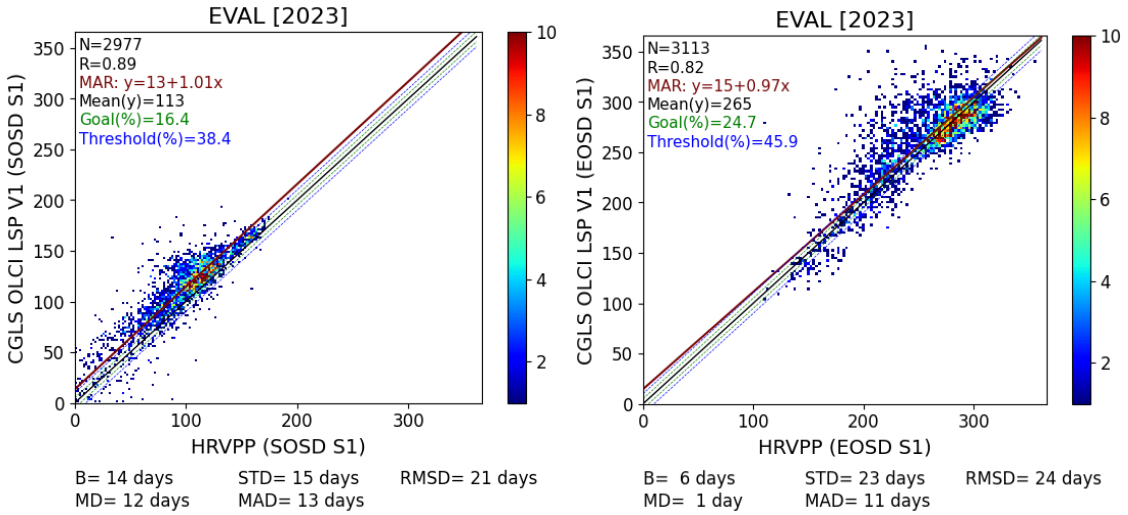



**Figure 43: CLMS OLCI LSP SOSD and EOSD versus HR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 24: Performance statistics between CLMS OLCI LSP SOSD and EOSD versus HR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP vs. HR-VPP | |
|---|---|---|
| | SOSD (days) | EOSD (days) |
| N | 2977 | 3113 |
| R | 0.89 | 0.82 |
| Bias (days) | 14 | 6 |
| MD (days) | 12 | 1 |
| STD (days) | 15 | 23 |
| MAD (days) | 13 | 11 |
| RMSD (days) | 21 | 24 |
| MAR | y=13+1.01x | y=15+0.97x |
| % goal | 16.4 | 24.7 |
| % threshold | 38.4 | 45.9 |

The overall comparison between CLMS OLCI LSP TPROD and SPROD versus HR-VPP equivalent layers over the European sampling EVAL is presented in Figure 44 and main statistics are summarized in Table 25. Main results are:

- Both products are comparable with high correlations (R > 0.88).
- CLMS OLCI TPROD/SPROD show higher values of around 30% than equivalent HR-VPP layers, despite of the fact that CLMS shows shorter season than HR-VPP.

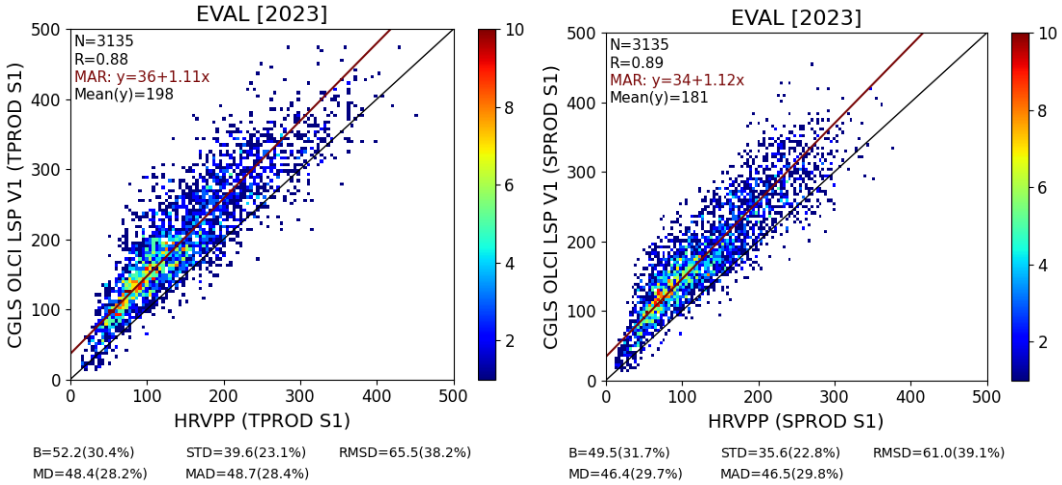



**Figure 44: CLMS OLCI LSP TPROD and SPROD versus HR-VPP equivalent layers scatter-plots over EVAL sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 25: Performance statistics between CLMS OLCI LSP TPROD and SPROD versus HR-VPP equivalent layers. Computation over EVAL sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP vs. HR-VPP | |
|---|---|---|
| | TPROD ($m^2 \cdot m^{-2} \cdot day$) | SPROD ($m^2 \cdot m^{-2} \cdot day$) |
| N | 3135 | 3135 |
| R | 0.88 | 0.89 |
| Bias ($m^2 m^{-2}$) | 52.2 (30.4%) | 49.5 (31.7%) |
| MD ($m^2 m^{-2}$) | 48.4 (28.2%) | 46.4 (29.7%) |
| STD ($m^2 m^{-2}$) | 39.6 (23.1%) | 35.6 (22.8%) |
| MAD ($m^2 m^{-2}$) | 48.7 (28.4%) | 46.5 (29.8%) |
| RMSD ($m^2 m^{-2}$) | 65.5 (38.2%) | 61.0 (39.1%) |
| MAR | y=36+1.11x | y=34+1.12x |

### 4.4.5 Overall consistency with global phenology products

This section focuses on the analysis of CLMS OLCI SOSD and EOSD phenometrics with MODIS V6.1 and VIIRS V2 LCD products for 2023. The analysis is focused on season 1, as few pixels showing high quality were found in season 2 for both reference products. Scatter-plots and validation metrics were computed for LANDVAL V2 sites for all cases and per main biome type. It should be noted that the sparse and bare area (SBA) class is not presented here as almost no concomitant pixels (less than 10 cases) between pair of products are available (MODIS and VIIRS based products do not provide estimate over deserts).

#### *4.4.5.1 CLMS OLCI LSP vs. MODIS LCD V6.1*

The scatter-plots between CLMS OLCI LSP and MODIS V6.1 for SOSD (Figure 45) and EOSD (Figure 46) are presented in this section. The main performance statistical metrics are displayed in Table 26. The three MODIS phenometrics Greenup, MidGreenup and Maturity are intercompared with SOSD while the other three MODIS phenometrics (i.e., Dormancy, MidGreendown and Senescence) are compared with EOSD. MODIS phenophases are defined at the corresponding 15%, 50% and 90% of the EVI2 amplitude.

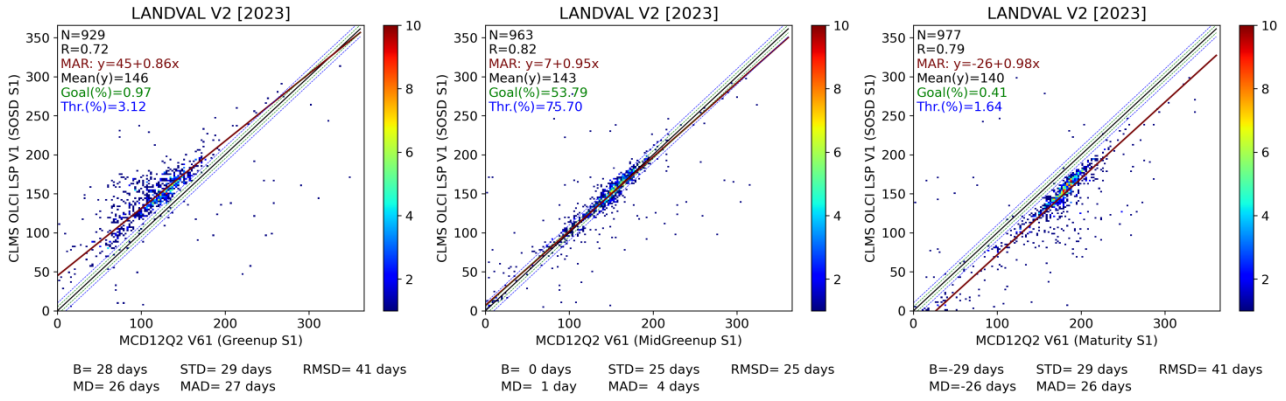



**Figure 45: CLMS OLCI LSP SOSD versus MODIS V6.1 Greenup (left), MidGreenup (middle) and Maturity (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels.**

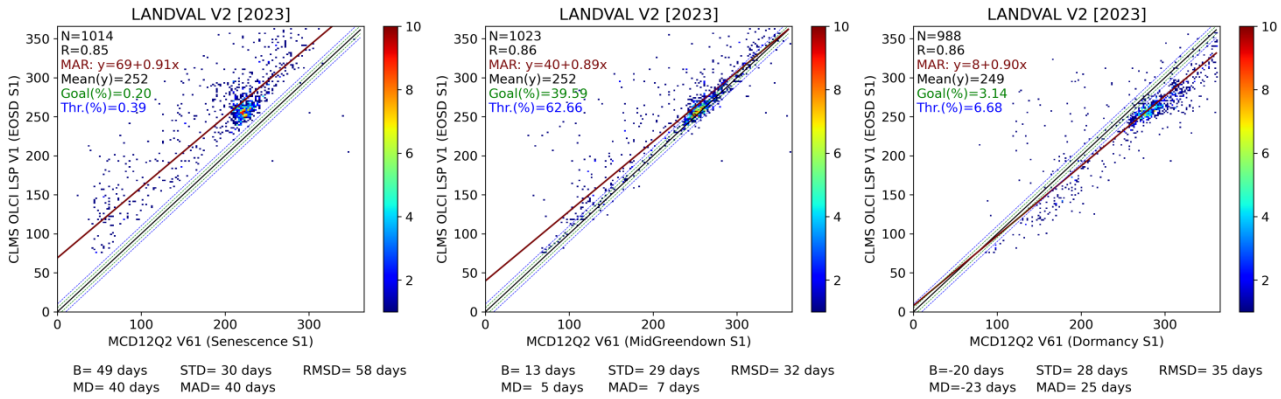



**Figure 46: CLMS OLCI LSP EOSD versus MODIS V6.1 Senescence (left), MidGreendown (middle) and Dormancy (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 26: Performance statistics between CLMS OLCI LSP SOSD and EOSD with MODIS V6.1 Greenup/MidGreenup/Maturity and Dormancy/MidGreendown/Senescence phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SOSD (25%) vs. MODIS V6.1 | | | CLMS OLCI LSP EOSD (15%) vs. MODIS V6.1 | | |
|---|---|---|---|---|---|---|
| | Greenup (15%) | MidGreenup (50%) | Maturity (90%) | Senescence (90%) | MidGreendown (50%) | Dormancy (15%) |
| N | 929 | 963 | 977 | 1014 | 1023 | 988 |
| R | 0.72 | 0.82 | 0.79 | 0.85 | 0.86 | 0.86 |
| Bias | 28 | 0 | -29 | 49 | 13 | -20 |
| MD | 26 | 1 | -26 | 40 | 5 | -23 |
| STD | 29 | 25 | 29 | 30 | 29 | 28 |
| MAD | 27 | 4 | 26 | 40 | 7 | 25 |
| RMSD | 41 | 25 | 41 | 58 | 32 | 35 |
| MAR | y=45-0.86x | y=7+0.95x | y=-26+0.98x | y=69+0.91x | y=40+0.89x | y=8+0.90x |
| % goal | 1.0 | 53.8 | 0.4 | 0.2 | 39.6 | 3.1 |
| % thr. | 3.1 | 75.7 | 1.6 | 0.4 | 62.7 | 6.7 |

Main results of the comparison of CLMS OLCI LSP versus MODIS LCD V6.1 in 2023 are:

- In case of OLCI SOSD (defined for 25% of PPI seasonal amplitude), the best correspondence was found with MODIS V6.1 MidGreenup phenometric (RMSD = 25 days, 75.7% of cases within threshold requirement) despite of the fact that the best agreement could be expected

with Greenup (RMSD = 41 days, 3.1% of cases within threshold) according to their threshold level definitions. OLCI SOSD shows no bias compared with MODIS MidGreenup (defined at 50%), whereas positive bias of 28 days is found with Greenup (defined at 15%). As expected, large negative bias of -29 days is found with MODIS Maturity (defined at 90%).

- The comparison of OLCI EOSD (defined for 15% of PPI seasonal amplitude) with equivalent MODIS phenometrics shows, again, the best overall agreement in the comparison with MidGreendown (RMSD = 32 days, 62.7% of cases within threshold requirement). OLCI EOSD, however, shows a delay compared to MODIS MidGreendown (B = 13 days) but an earlier timing compared to MODIS Dormancy (B = -20 days). Thus, OLCI EOSD falls between both MODIS MidGreendown and MODIS Dormancy dates.

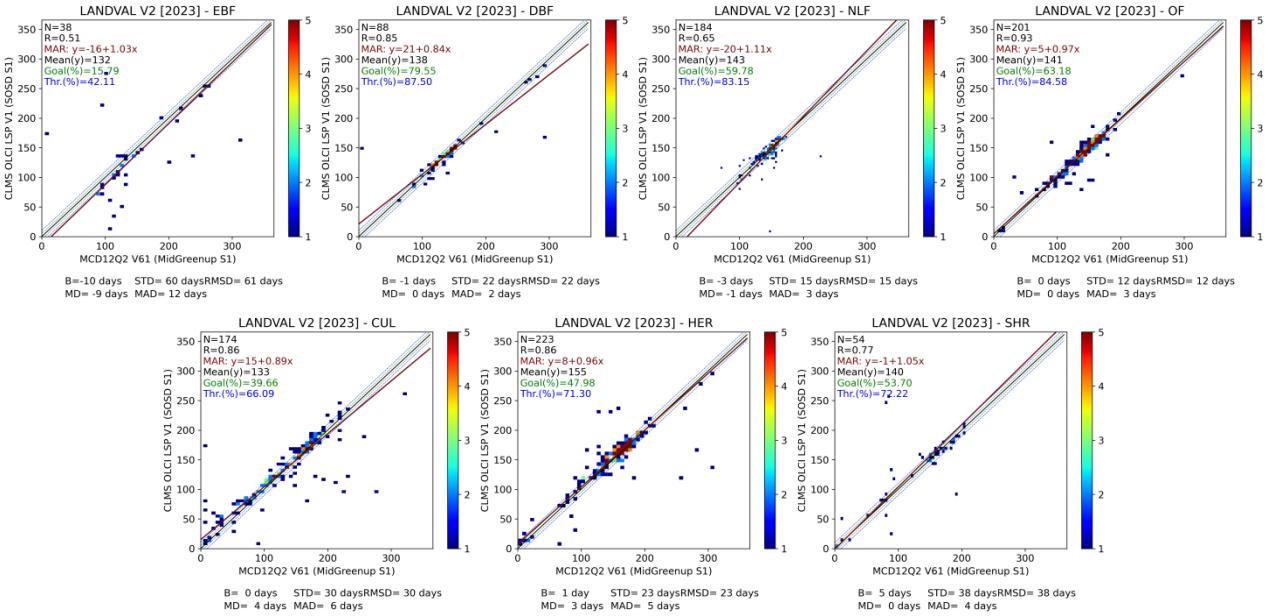



**Figure 47: CLMS OLCI LSP SOSD versus MODIS V6.1 MidGreenup products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 27: Performance statistics between CLMS OLCI LSP SOSD with MODIS V6.1 Greenup and MidGreenup phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SOSD (25%) vs. | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MODIS V6.1 Greenup (15%) | | | | | | | MODIS V6.1 MidGreenup (50%) | | | | | | |
| | EBF | DBF | NLF | OF | Cul. | Herb. | Shr. | EBF | DBF | NLF | OF | Cul. | Herb. | Shr. |
| N | 40 | 72 | 184 | 196 | 157 | 215 | 47 | 38 | 88 | 184 | 201 | 174 | 223 | 54 |
| R | 0.61 | 0.72 | 0.58 | 0.79 | 0.8 | 0.66 | 0.75 | 0.51 | 0.85 | 0.65 | 0.93 | 0.86 | 0.86 | 0.77 |
| Bias | 0 | 19 | 27 | 31 | 29 | 34 | 31 | -10 | -1 | -3 | 0 | 0 | 1 | 5 |
| MD | 19 | 24 | 24 | 26 | 29 | 28 | 29 | -9 | 0 | -1 | 0 | 4 | 3 | 0 |
| STD | 66 | 30 | 20 | 19 | 27 | 31 | 28 | 60 | 22 | 15 | 12 | 30 | 23 | 38 |
| MAD | 27 | 24 | 24 | 26 | 31 | 29 | 33 | 12 | 2 | 3 | 3 | 6 | 5 | 4 |
| RMSD | 66 | 36 | 33 | 37 | 39 | 46 | 42 | 61 | 22 | 15 | 12 | 30 | 23 | 38 |
| MAR | y=35 +0.73x | y=0.38 +0.84x | y=51 +0.80x | y=49 +0.84x | y=41 +0.90x | y=55 +0.84x | y=30 +1.01x | y=-16 +1.03x | y=21 +0.84x | y=-20 +1.11x | y=5 +0.97x | y=15 +0.89x | y=8 +0.96x | y=-1 +1.05x |
| % goal | 0.0 | 0.0 | 2.2 | 1.0 | 1.3 | 0.5 | 0.0 | 15.8 | 79.6 | 59.8 | 63.2 | 39.7 | 48.0 | 53.7 |
| %threshold | 7.5 | 2.3 | 5.4 | 1.5 | 2.6 | 2.3 | 4.3 | 42.1 | 87.5 | 83.2 | 84.6 | 66.1 | 71.3 | 72.2 |

The analysis of CLMS OLCI SOSD versus MODIS V6.1 per biome type is presented in Figure 47 using MODIS MidGreenup, which provides the best results in the comparison with OLCI SOSD. Table 27 shows the performance statistics considering both MODIS V6.1 MidGreenup and Greenup, which better fits the CLMS product definition.

Main results are:

- The comparison of OLCI SOSD with MODIS MidGreenup shows B and MD within ±5 days for all biomes (except for EBF, MD = -9 days) and the percentage of cases within threshold requirement higher than 66% (except EBF).
- The systematic positive bias of OLCI SOSD versus MODIS Greenup (Table 27, left side) is observed for most of biomes, with MD ranging between 19 and 29 days. The percentage of cases within threshold uncertainty requirement is lower than 8% for all biome cases.

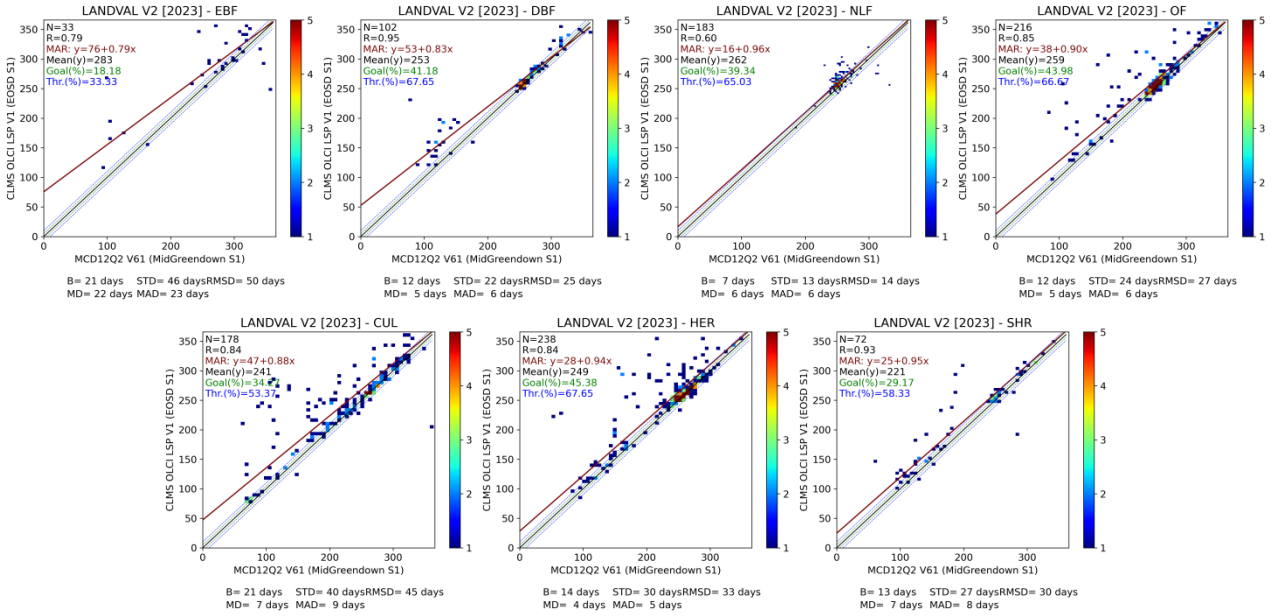



**Figure 48: CLMS OLCI LSP EOSD versus MODIS V6.1 MidGreendown products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 28: Performance statistics between CLMS OLCI LSP EOSD with MODIS V6.1 and Dormancy and MidGreendown phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP EOSD (15%) vs. | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | MODIS V6.1 Dormancy (15%) | | | | | | | MODIS V6.1 MidGreendown (50%) | | | | | | |
| | EBF | DBF | NLF | OF | Cul. | Herb. | Shr. | EBF | DBF | NLF | OF | Cul. | Herb. | Shr. |
| N | 27 | 98 | 182 | 206 | 168 | 236 | 70 | 33 | 102 | 183 | 216 | 178 | 238 | 72 |
| R | 0.84 | 0.89 | 0.73 | 0.85 | 0.82 | 0.85 | 0.93 | 0.79 | 0.95 | 0.60 | 0.85 | 0.84 | 0.84 | 0.93 |
| Bias | -12 | -22 | -23 | -21 | -11 | -20 | -26 | 21 | 12 | 7 | 12 | 21 | 14 | 13 |
| MD | -25 | -27 | -21 | -23 | -21 | -23 | -26 | 22 | 5 | 6 | 5 | 7 | 4 | 7 |
| STD | 41 | 26 | 13 | 22 | 43 | 27 | 25 | 46 | 22 | 13 | 24 | 40 | 30 | 27 |
| MAD | 28 | 28 | 21 | 24 | 27 | 24 | 27 | 23 | 6 | 6 | 6 | 9 | 5 | 8 |
| RMSD | 43 | 34 | 27 | 31 | 44 | 34 | 36 | 50 | 25 | 14 | 27 | 45 | 33 | 30 |
| MAR | y=45 +0.80x | y=-1 +0.92x | y=67 +0.68x | y=7 +0.90x | y=30 +0.83x | y=-9 +0.96x | y=-30 +1.02x | y=76 +0.79x | y=53 +0.83x | y=16 +0.96x | y=38 +0.90x | y=47 +0.88x | y=28 +0.94x | y=25 +0.95x |
| % goal | 7.4 | 1.0 | 3.3 | 3.4 | 4.2 | 2.5 | 1.4 | 18.2 | 41.2 | 39.3 | 44.0 | 34.7 | 45.4 | 29.2 |
| % threshold | 11.1 | 4.1 | 7.1 | 6.8 | 5.4 | 8.5 | 2.9 | 33.3 | 67.7 | 65.0 | 66.7 | 53.4 | 67.7 | 58.3 |

The analysis of CLMS OLCI EOSD versus MODIS V6.1 per biome type is presented in Figure 48 using MODIS MidGreendown, which provides the best results in the comparison with OLCI EOSD. Table 28 shows the performance statistics considering both MODIS V6.1 MidGreendown and Dormancy, which better fits the CLMS product definition.

Main results are:

- As observed for the overall case, the comparison of OLCI EOSD with MODIS MidGreedown shows improved overall results to that found for MODIS Dormancy, whatever the biome
- CLMS OLCI LSP EOSD shows typically slight positive differences with MODIS MidGreendown (B lower than 14 days), except for CUL and EBF (B of 21 days). Percentage of cases within threshold requirement is typically higher 53% except for EBF (33%).
- CLMS OLCI LSP shows systematic negative bias as compared with MODIS V6.1 Dormancy for all biome types, with B typically around 20 days of anticipation (except CUL and EBF with 11 and 12 days of anticipation). The percentage of cases within threshold requirements is typically lower than 9%.

#### 4.4.5.2 *CLMS OLCI LSP vs. VIIRS LCD V2*

The scatter-plots between CLMS OLCI LSP and VIIRS V2 for SOSD (Figure 49) and EOSD (Figure 50) are presented in this section. The main performance statistical metrics are displayed in Table 29. The three VIIRS V2 Onset_Greenness_Increase, Date_Mid_Greenup_Phase and Onset_Greenness_Maximum phenometrics are intercompared with SOSD while the other three VIIRS phenometrics (i.e., Onset_Greeness_Decrease, Date_Mid_Senescence_Phase and Onset_Greeness_Minimum) are compared with EOSD.

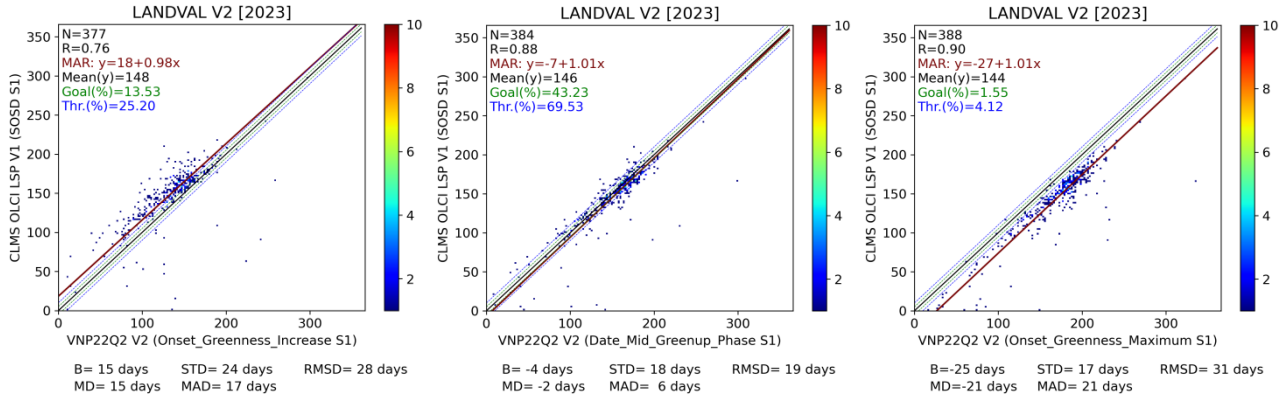



**Figure 49: CLMS OLCI LSP SOSD versus VIIRS V2 Onset_Greenness_Increase (left), Date_Mid_Greenup_Phase (middle) and Onset_Greenness_Maximum (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels.**

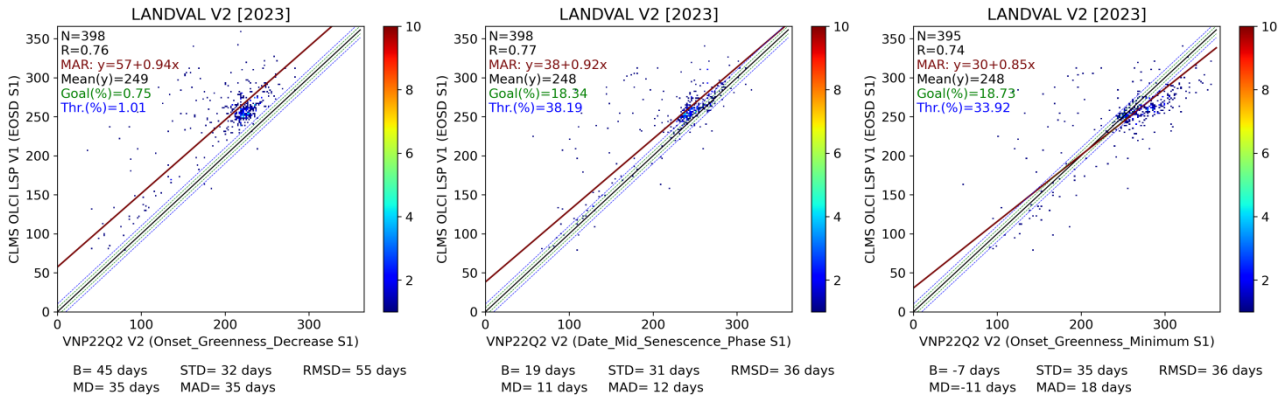



**Figure 50: CLMS OLCI LSP EOSD versus VIIRS V2 Onset_Greeness_Decrease (left), Date_Mid_Senescence_Phase (middle) and Onset_Greeness_Minimum (right) products scatter-plots over LANDVAL V2 sites during 2023 (season 1). The computation was done over good quality pixels.**

**Table 29: Performance statistics between CLMS OLCI LSP SOSD and EOSD with VIIRS V2 Onset_Greenness_Increase/Date_Mid_Greenup_Phase/Onset_Greenness_Maximum and Onset_Greeness_Decrease/Date_Mid_Senescence_Phase/Onset_Greeness_Minimum phenometrics. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SOSD (25%) vs. VIIRS V2 | | | CLMS OLCI LSP EOSD (15%) vs. VIIRS V2 | | |
|---|---|---|---|---|---|---|
| | Onset_ Greenness_ Increase | Date_Mid_ Greenup _Phase | Onset_ Greenness_ Maximum | Onset_ Greeness_ Decrease | Date_Mid_ Senescence_ Phase | Onset_ Greeness_ Minimum |
| N | 337 | 384 | 388 | 398 | 398 | 395 |
| R | 0.76 | 0.88 | 0.90 | 0.76 | 0.77 | 0.74 |
| Bias | 15 | -4 | -25 | 45 | 19 | -7 |
| MD | 15 | -2 | -21 | 35 | 11 | -11 |
| STD | 24 | 18 | 17 | 32 | 31 | 35 |
| MAD | 17 | 6 | 21 | 35 | 12 | 18 |
| RMSD | 28 | 19 | 31 | 55 | 36 | 36 |
| MAR | y=18-0.98x | y=-7+1.01x | y=-27+1.01x | y=57+0.94x | y=38+0.92x | y=30+0.85x |
| % goal | 13.5 | 43.2 | 1.6 | 0.8 | 18.3 | 18.7 |
| % thr. | 25.2 | 69.5 | 4.1 | 1.0 | 38.2 | 33.9 |

The comparison of CLMS OLCI LSP and VIIRS V2 (best-quality pixels) during the year 2023 shows:

- In case of OLCI SOSD (defined for 25% of PPI seasonal amplitude), the best correspondence was found with VIIRS V2 Mid_Greenup_Phase (RMSD = 19 days, 69.5% of cases within

threshold requirement) despite of the fact that the best agreement could be expected with Onset_Greeness_Increase (RMSD = 28 days, 25.2% of cases within threshold). OLCI SOSD shows almost no bias (MD = -2 days) with VIIRS Mid_Greenup_Phase, whereas positive bias (MD = 15 days) is found with Onset_Greeness_Increase. Large negative bias (MD = -21 days) is found with Onset_Greenness_Maximum as expected.

- The comparison of OLCI EOSD (defined for 15% of PPI seasonal amplitude) with VIIRS V2 phenometrics shows that OLCI EOSD falls between VIIRS V2 Date_Mid_Senescence_Phase (MD = 11 days) and Onset_Greeness_Minimum (MD = -11 days). Similar overall uncertainty (RMSD = 36 days) compared with both VIIRS V2 phases, with slightly better results in the comparison with Date_Mid_Senescence_Phase in terms of correlation, precision (STD, MAD) and cases within threshold uncertainty requirement. Worse correspondence is found with VIIRS V2 Onset_Greeness_Decrease (MD = 35 days) as expected.

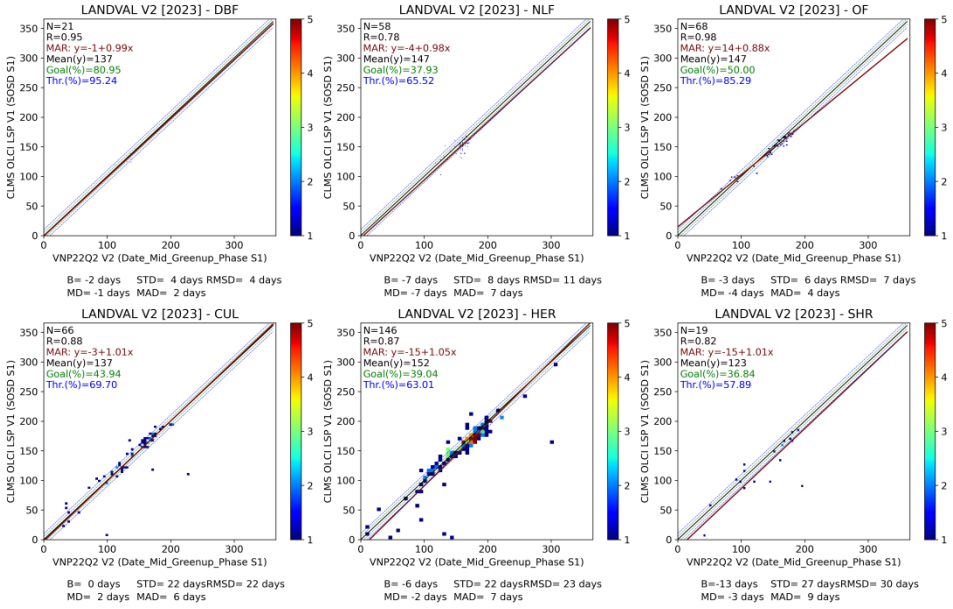



**Figure 51: CLMS OLCI LSP SOSD versus VIIRS V2 Date_Mid_Greenup_Phase products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

**Table 30: Performance statistics between CLMS OLCI LSP SOSD with VIIRS V2 Onset_Greenness_Increase and Date_Mid_Greenup_Phase. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SOSD vs. VIIRS V2 | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Onset_Greenness_Increase | | | | | | Date_Mid_Greenup_Phase | | | | | |
| | DBF | NLF | OF | Cul. | Herb. | Shr. | DBF | NLF | OF | Cul. | Herb. | Shr. |
| N | 21 | 58 | 68 | 65 | 142 | 17 | 21 | 58 | 68 | 66 | 146 | 19 |
| R | 0.90 | 0.75 | 0.96 | 0.69 | 0.78 | 0.36 | 0.95 | 0.78 | 0.98 | 0.88 | 0.87 | 0.82 |
| Bias | 15 | 14 | 16 | 17 | 13 | 6 | -2 | -7 | -3 | 0 | -6 | -13 |
| MD | 15 | 14 | 13 | 22 | 15 | 20 | -1 | -7 | -4 | 2 | -2 | -3 |
| STD | 6 | 11 | 10 | 33 | 25 | 46 | 4 | 8 | 6 | 22 | 22 | 27 |
| MAD | 15 | 14 | 13 | 24 | 17 | 22 | 2 | 7 | 4 | 6 | 7 | 9 |
| RMSD | 17 | 18 | 19 | 37 | 29 | 46 | 4 | 11 | 7 | 22 | 23 | 30 |
| MAR | y=37 +0.83x | y=45 +0.77x | y=50 +0.74x | y=12 +1.05x | y=7 +1.05x | y=25 +0.85x | y=-1 +0.99x | y=-4 +0.98x | y=14 +0.88x | y=-3 +1.01x | y=-15 +1.05x | y=-15 +1.01x |
| % goal | 4.7 | 20.7 | 8.8 | 3.1 | 20.4 | 0.0 | 81.0 | 37.9 | 50.0 | 43.9 | 39.0 | 36.8 |
| % threshold | 14.3 | 36.2 | 33.8 | 6.2 | 29.6 | 5.9 | 95.2 | 65.5 | 85.3 | 69.7 | 63.0 | 57.9 |

The analysis of CLMS OLCI SOSD versus VIIRS V2 per biome type is presented in Figure 51 using VIIRS V2 Date_Mid_Greenup_Phase, which provides the best results in the comparison with OLCI SOSD. Table 30 shows the performance statistics considering both VIIRS V2 Date_Mid_Greenup_Phase and Onset_Greeness_Increase phases. Note that SBA class is not displayed as very few (<10) concomitant samples between both products are found. EBF class is not displayed as VIIRS V2 does not provide valid best-quality retrievals. Main conclusions are:

- The comparison of OLCI SOSD with VIIRS Date_Mid_Greenup_Phase shows slight early start of season (MD within -7 and -2 days) except for cultivated (MD = 2 days). The percentage of cases within threshold requirement is typically higher than 60%.
- The systematic positive bias of OLCI SOSD versus VIIRS Onset_Greenness_Increase (Table 30, left side) is observed for all biomes, with MD ranging between 13 and 22 days.

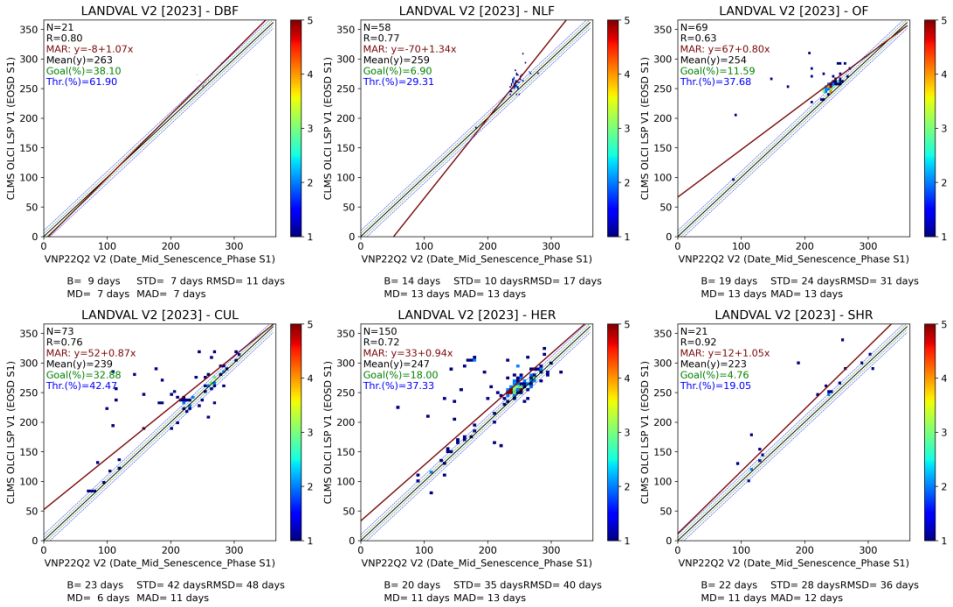



**Figure 52: CLMS OLCI LSP V1 EOSD versus VIIRS V2 Date_Senescence_Greenup_Phase products scatter-plots per aggregated biome type. Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

**Table 31: Performance statistics between CLMS OLCI LSP EOSD with VIIRS V2 Date_Mid_ Senescence _Phase (left) and Onset_Greeness_Minimum (right). Computation over LANDVAL V2 sites for best-quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP EOSD vs. VIIRS V2 | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Date_Mid_Senescence_Phase | | | | | | Onset_Greeness_Minimum | | | | | |
| | DBF | NLF | OF | Cul. | Herb. | Shr. | DBF | NLF | OF | Cul. | Herb. | Shr. |
| N | 21 | 58 | 69 | 73 | 150 | 21 | 21 | 58 | 68 | 72 | 148 | 20 |
| R | 0.80 | 0.77 | 0.63 | 0.76 | 0.72 | 0.92 | 0.86 | 0.78 | 0.73 | 0.72 | 0.70 | 0.87 |
| Bias | 9 | 14 | 19 | 23 | 20 | 22 | -27 | -10 | -5 | -4 | -5 | -4 |
| MD | 7 | 13 | 13 | 6 | 11 | 11 | -25 | -9 | -6 | -17 | -10 | -13 |
| STD | 7 | 10 | 24 | 42 | 35 | 28 | 10 | 11 | 21 | 50 | 39 | 36 |
| MAD | 7 | 13 | 13 | 11 | 13 | 12 | 25 | 10 | 10 | 28 | 18 | 19 |
| RMSD | 11 | 17 | 31 | 48 | 40 | 36 | 28 | 15 | 22 | 50 | 39 | 36 |
| MAR | y=-8 +1.07x | y=-70 +1.34x | y=67 +0.80x | y=52 +0.87x | y=33 +0.94x | y=12 +1.05x | y=76 +0.65x | y=4 +0.94x | y=52 +0.78x | y=51 +0.78x | y=23 +0.89x | y=-25 +1.09x |
| % goal | 38.1 | 6.9 | 11.6 | 32.9 | 18.0 | 4.8 | 4.8 | 19.0 | 33.8 | 8.1 | 18.9 | 25.0 |
| % threshold | 61.9 | 29.3 | 37.7 | 42.5 | 37.3 | 19.1 | 4.8 | 48.3 | 50.0 | 16.2 | 35.1 | 25.0 |

The analysis of CLMS OLCI EOSD versus VIIRS V2 per biome type is presented in Figure 52 using VIIRS V2 Date_Mid_Senescence_Phase. Table 31 shows the performance statistics considering both VIIRS V2 Date_Mid_Senescence_Phase and Onset_Greeness_Minimum phases. Note that SBA class is not displayed as very few (<10) concomitant samples between both products are found. EBF class is not displayed as VIIRS V2 does not provide valid best-quality retrievals. Main conclusions are:

- As observed for the overall case, OLCI EOSD falls between VIIRS V2 Date_Mid_Senescence_Phase (MD within 6 and 13 days) and Onset_Greeness_Minimum (MD within -25 and -6 days) for all biome cases.
- Similar values are found in comparison with both VIIRS V2 phases (i.e., similar MD with opposite sign and similar RMSD), but the comparison with Onset_Greeness_Minimum shows considerably worse results for DBF.

### 4.4.6 Overall consistency with global productivity products

#### *4.4.6.1 CLMS OLCI LSP vs. CLMS OLCI GPP*

The CLMS OLCI LSP SPROD and TPROD productivity parameters are compared with CLMS OLCI GPP product. CLMS GPP are expressed in $kgC \cdot m^{-2} \cdot day$, whereas CLMS OLCI LSP SPROD and TPROD are expressed in $m^2 \cdot m^{-2} \cdot day$. Due to the different definition (units) of both products related to productivity, only correlation and regression fit statistics between them are provided. As CLMS LSP SPROD/TPROD products are defined as the sum of all PPI daily values between SOSD and EOSD, the equivalent CLMS OLCI GPP values were also computed between the same period (i.e., season between SOSD and EOSD). The comparison was made at 1 km resolution (i.e. 3x3 CLMS OLCI 300m pixels).

The scatter-plots of CLMS OLCI LSP SPROD and TPROD versus the aggregated values of CLMS GPP are displayed in Figure 53 and the main performance statistics are shown in Table 32. The scatter-plots per main biome type and their main performance statistics can be found in Figure 54 and Table 33 for SPROD and in Figure 55 and Table 34 for TPROD. It should be noted that SBA case is not included due to the limited sampling of this biome (N=22).

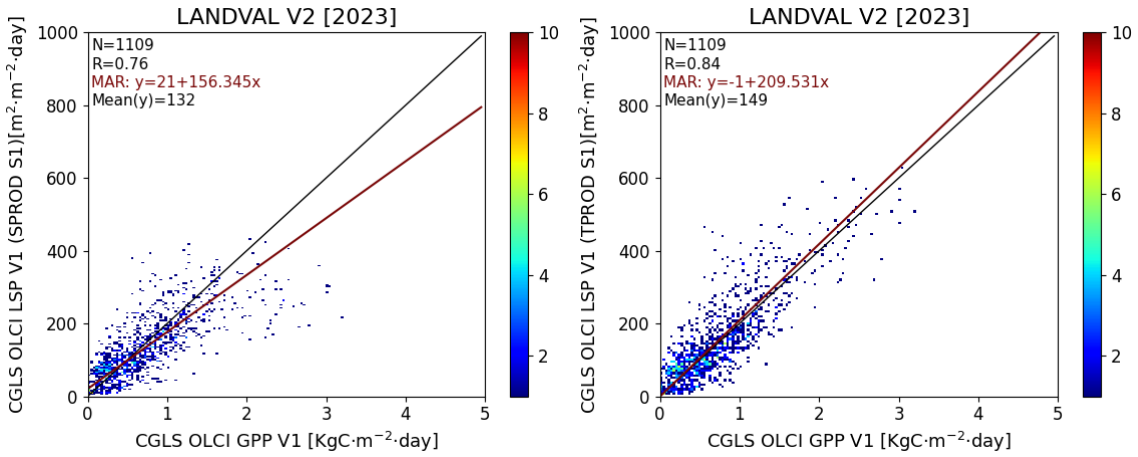



**Figure 53: Scatter-plots between CLMS OLCI LSP SPROD (left) and TPROD (right) ($m^2 \cdot m^{-2} \cdot day$) and the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1).**

**Table 32: Performance statistics between CLMS OLCI LSP SPROD and TPROD ($m^2 \cdot m^{-2} \cdot day$) versus CLMS OLCI GPP aggregated values ($kgC \cdot m^{-2} \cdot day$) for the same season length. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SPROD/TPROD vs. CLMS OLCI GPP (between SOSD and EOSD) | |
|---|---|---|
| | SPROD vs. GPP | TPROD vs. GPP |
| N | 1109 | 1109 |
| R | 0.76 | 0.84 |
| MAR | y=21+156x | y=-1+210x |

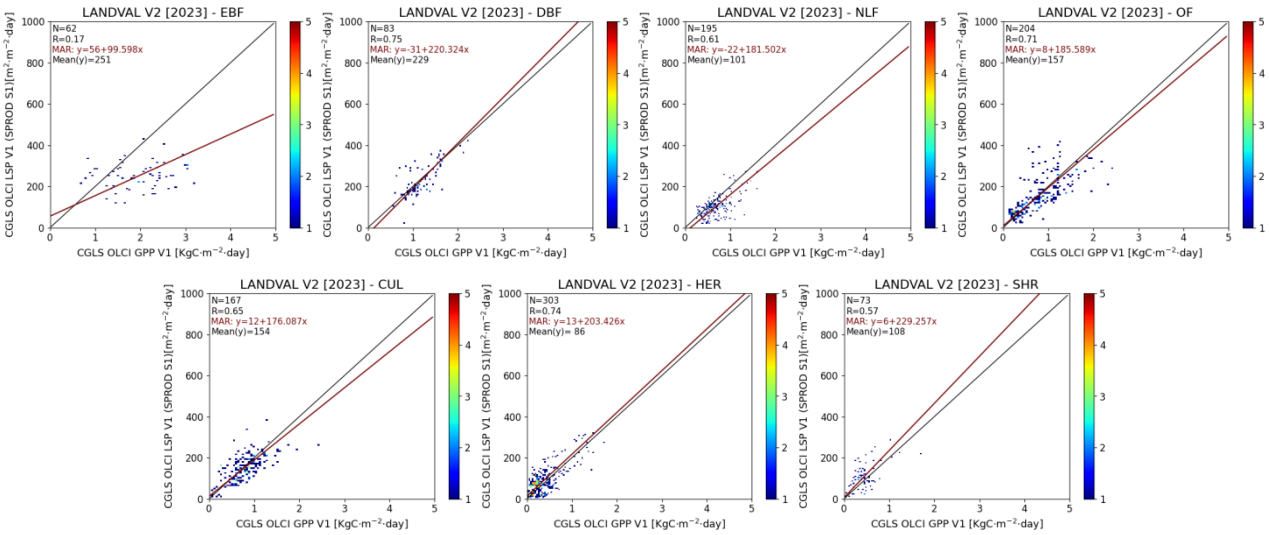



**Figure 54: Scatter-plots CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1).**

**Table 33: Performance statistics between CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SPROD vs. CLMS OLCI GPP (between SOSD and EOSD) | | | | | | |
|---|---|---|---|---|---|---|---|
| | EBF | DBF | NLF | OF | CUL | HER | SHR |
| N | 62 | 83 | 195 | 204 | 167 | 303 | 73 |
| R | 0.17 | 0.75 | 0.61 | 0.71 | 0.65 | 0.74 | 0.57 |
| MAR | y=56+100x | y=-31+222x | y=-22+182x | y=8+186x | y=12+176x | y=13+203x | y=6+229x |

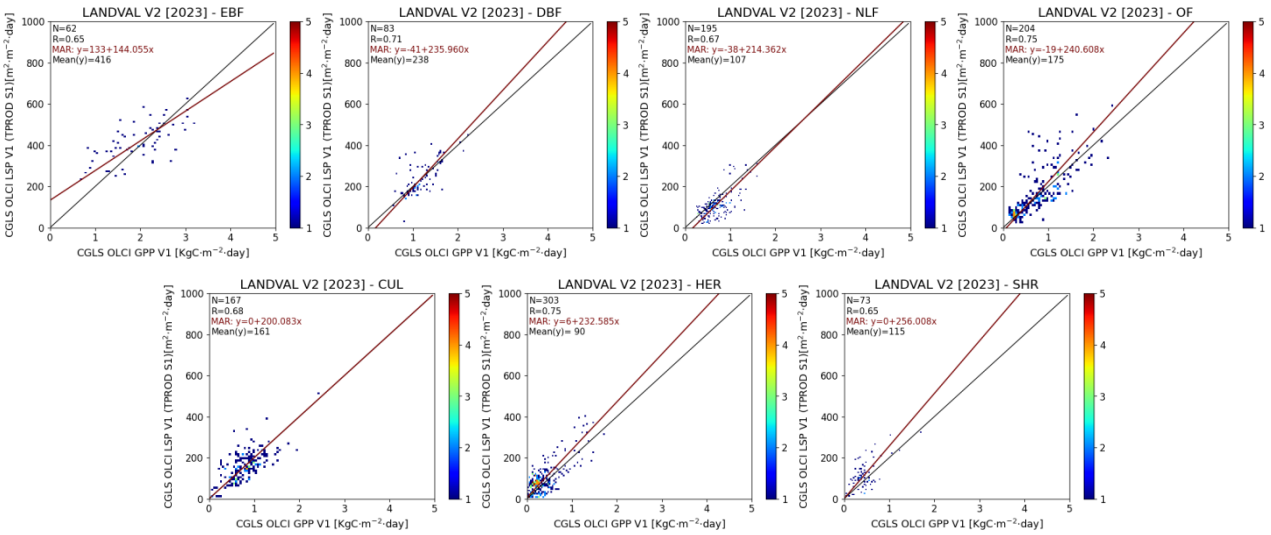



**Figure 55: Scatter-plots CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1).**

**Table 34: Performance statistics between CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS OLCI GPP values ($kgC \cdot m^{-2} \cdot day$) between OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP TPROD vs. CLMS OLCI GPP (between SOSD and EOSD) | | | | | | |
|---|---|---|---|---|---|---|---|
| | EBF | DBF | NLF | OF | CUL | HER | SHR |
| N | 62 | 83 | 195 | 204 | 167 | 303 | 73 |
| R | 0.65 | 0.71 | 0.67 | 0.75 | 0.68 | 0.75 | 0.65 |
| MAR | y=133+144x | y=-41+236x | y=-38+214x | y=-19+241x | y=200x | y=6+233x | y=256x |

Main conclusions from the comparison of CLMS OLCI LSP SPROD and TPROD productivity parameters with CLMS OLCI GPP product are:

- A strong relationship between SPROD and TPROD as compared to the temporal aggregated CLMS OLCI GPP products is found, with correlations of 0.76 and 0.84, respectively.
- The analysis per biome type shows correlations greater than 0.6 for all biomes, except for EBF (R=0.17) and SHR (R=0.57) in case of SPROD versus GPP.

#### *4.4.6.2 CLMS OLCI LSP vs. MODIS GPP*

The CLMS OLCI LSP SPROD and TPROD productivity parameters are compared with MODIS GPP (MOD17A2HGF V6.1, which consists in cumulative 8-day composites of values). As CLMS products are defined as the sum of all PPI daily values between SOSD and EOSD, the sum of MODIS GPP cumulative values was also computed between CLMS SOSD and EOSD dates. MODIS GPP are expressed in kgC·m$^{-2}$·day, whereas CLMS OLCI LSP SPROD and TPROD are expressed in m$^2$·m$^{-2}$·day. Due to the difference in definition and units, the intercomparison of this section is relative and only correlation and regression fit statistics are provided. The comparison was made at 1 km resolution (i.e. 3x3 CLMS OLCI pixels, 2x2 MODIS pixels).

The scatter-plots between CLMS OLCI LSP SPROD and TPROD as compared with the sum of MODIS GPP values between the OLCI LSP SOSD and EOSD are displayed in Figure 56 and the main performance statistics can be found in Table 35. The analyses per main biome type are showed in Figure 57 and Table 36 for SPROD and in Figure 58 and Table 37 for TPROD. It should be noted that SBA case is not included due to the limited sampling of this biome (N=5).

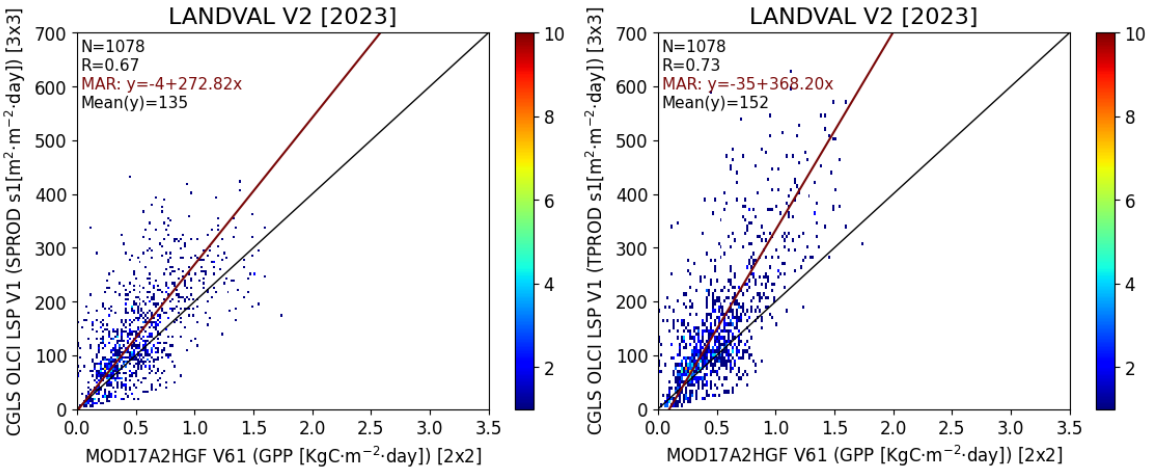



**Figure 56: Scatter-plots between CLMS OLCI LSP SPROD and TPROD (m$^2$·m$^{-2}$·day) and the sum of MODIS GPP values (kgC·m$^{-2}$·day) between CLMS OLCI LSP SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1).**

**Table 35: Performance statistics between CLMS OLCI LSP SPROD and TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD an EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SPROD and TPROD vs. sum of MOD17A2HGF V6.1 GPP between SOSD and EOSD | |
|---|---|---|
| | SPROD | TPROD |
| N | 1078 | 1078 |
| R | 0.67 | 0.73 |
| MAR | y=-4+272.8x | y=-35+368.2x |

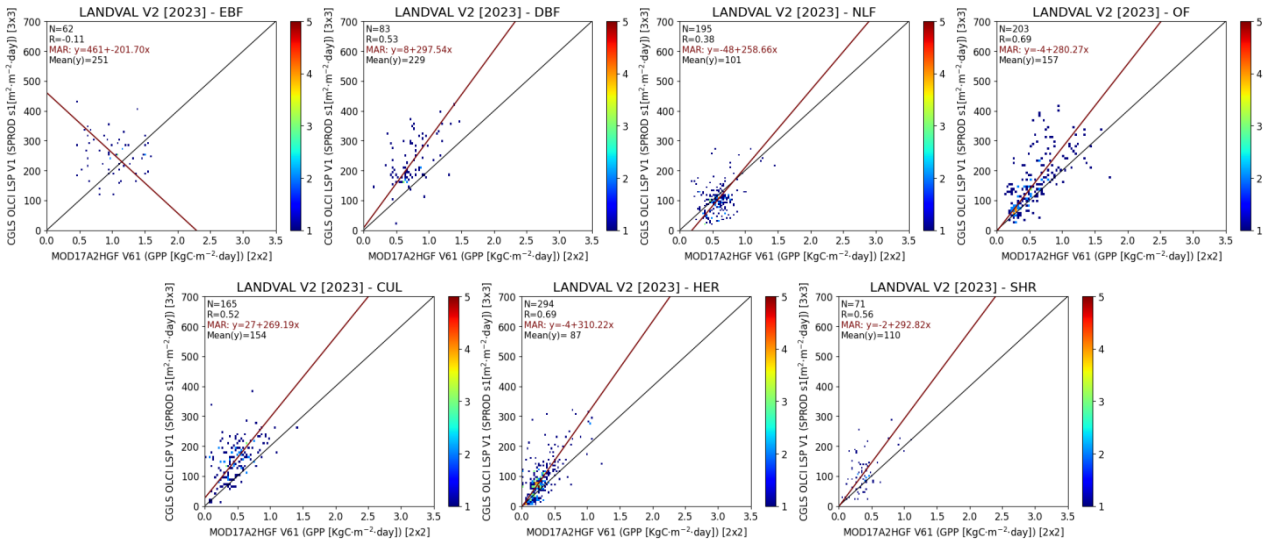



**Figure 57: Scatter-plots CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD and EOS. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1).**

**Table 36: Performance statistics between CLMS OLCI LSP SPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP SPROD vs. sum of MOD17A2HGF V6.1 GPP between SOSD and EOSD | | | | | | |
|---|---|---|---|---|---|---|---|
| | EBF | DBF | NLF | OF | CUL | HER. | SHR |
| N | 62 | 83 | 195 | 203 | 165 | 294 | 71 |
| R | -0.11 | 0.53 | 0.38 | 0.69 | 0.52 | 0.69 | 0.56 |
| MAR | y=461-202x | y=8+298x | y=-48+259x | y=-4+280x | y=27+269x | y=-4+310x | y=-2+293x |

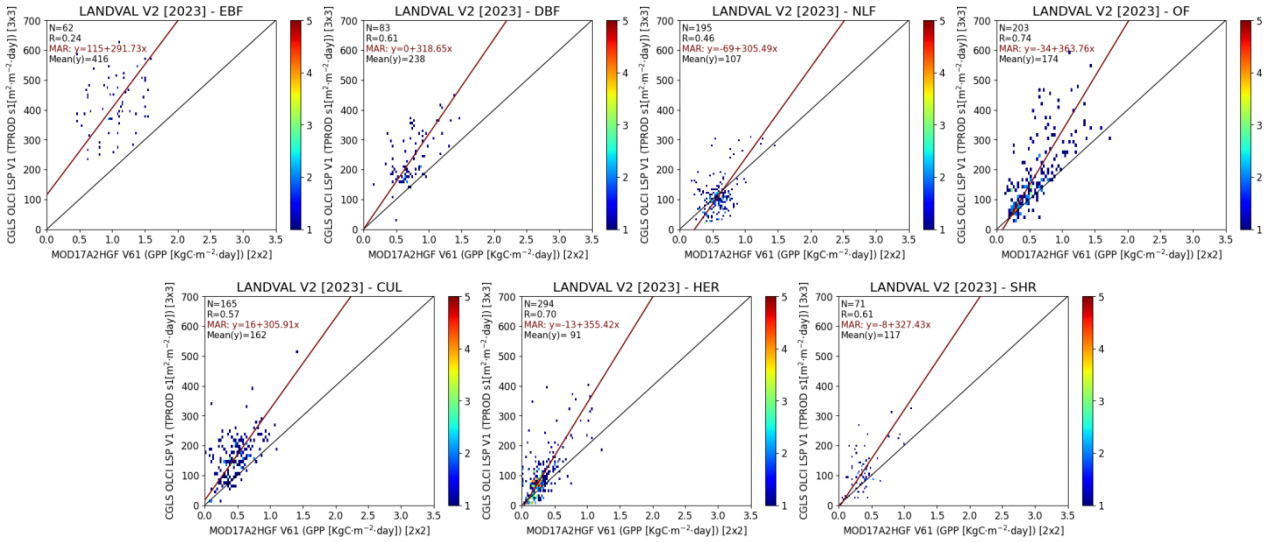



**Figure 58: Scatter-plots CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI SOSD and EOSD. Computation over good quality pixels of LANDVAL V2 sites during 2023 (season 1).**

**Table 37: Performance statistics between CLMS OLCI LSP TPROD ($m^2 \cdot m^{-2} \cdot day$) versus the sum of MODIS GPP values ($kgC \cdot m^{-2} \cdot day$) between CLMS OLCI LSP SOSD and EOSD. Computation over LANDVAL V2 sites for good quality pixels during 2023 (season 1).**

| | CLMS OLCI LSP TPROD vs. sum of MOD17A2HGF V6.1 GPP between SOSD and EOSD | | | | | | |
|---|---|---|---|---|---|---|---|
| | EBF | DBF | NLF | OF | CUL | HER | SHR |
| N | 62 | 83 | 195 | 203 | 165 | 294 | 71 |
| R | 0.23 | 0.61 | 0.46 | 0.74 | 0.57 | 0.70 | 0.61 |
| MAR | y=115+292x | y=319x | y=-69+306x | y=-34+364x | y=16+306x | y=-13+355x | y=-8+327x |

The comparison between CLMS OLCI LSP SPROD and TPROD and the productivity of MODIS GPP over OLCI LSP season shows:

- Correlations of 0.67 and 0.73 are found in the comparison of SPROD and TPROD, respectively, with the sum of MODIS GPP values between SOSD and EOSD. As expected, lower correlations are found than in the comparison with CLMS OLCI GPP, because CLMS OLCI products are retrieved from the same input data with the same TOA2TOC pre-processing.

- The analyses per biome type show correlations typically greater than 0.5 except for EBF (R=-0.11/0.23 for SPROD/TPROD) and NLF (R=0.38/0.46).


© C-GLOPS Lot1 consortium

# 5 SUMMARY AND CONCLUSIONS

The quality assessment of CLMS OLCI LSP products was conducted for year 2023. The methodology is described in the CLMS Product Quality Assurance Document [CGLOPS1_PQAD_I2.30]. Two main validation approaches are defined: direct validation (i.e., comparison of satellite products with in situ observations) and indirect validation (or product intercomparison). Several criteria of performance are analyzed, including the visualization of the product content, and the evaluation of the product completeness, spatial consistency and error (accuracy, precision and uncertainty).

The direct validation of SOSD and EOSD is performed with PEP725, TEMPO and NEON datasets, which provide phenological observations. PhenoCam V3 dataset, which infers phenometrics from digital cameras, is also used for direct comparison. The product intercomparison with global satellite-based products includes the quantitative and qualitative comparison of SOSD/EOSD with MODIS LCD V6.1 and VIIRS LCD V2 phenometrics and quantitative comparisons of TPROD/SPROD with CLMS OLCI and MODIS (MOD17A2HGF V6.1) GPP products over LANDVAL V2 network of sites. Finally, the comparison over Europe with equivalent MR-VPP and HR-VPP parameters is also performed over different sampling areas.

Main results are:

**Product content**

- Reliable values are found for all CLMS OLCI LSP layers. However, OLCI products display unexpected seasons over few desert areas (e.g., Sahara and Arabia deserts).

**Product completeness**

- OLCI LSP shows similar overall completeness (75% of valid data for 2023) than VIIRS V2. OLCI LSP provides improved completeness over tropical areas but unexpected values over few desert areas. MODIS V6.1 shows the worse completeness (60% of valid data).
- Considering the quality flag (Table 7), less than 5% and 2% of OLCI and MODIS data is filtered out. VIIRS quality flag (‘overall good quality’) is more restrictive regarding best-quality pixels, filtering out more than 50% of data.
- All products provide very few retrievals for season 2 (around 2%-5%).

**Spatial consistency**

- The analysis of OLCI LSP maps at full resolution over subcontinental regions and AOIs shows consistent values in most cases (except over few desert areas, with unrealistic seasons).
- The maps of residuals of CLMS OLCI LSP versus MR-VPP over Europe show around 60% of pixels within threshold for SOSD and 35% for EOSD. Maps of differences show delayed CLMS SOSD in most cases and large negative bias for EOSD mainly over Eastern Europe, which translates in shorter season in case of CLMS OLCI LSP.
- CLMS OLCI LSP and MR-VPP productivity layers show residuals typically within ±40 $m^2.m^{-2}$.days, with main spatial discrepancies over southern sparsely vegetated regions. CLMS OLCI LSP tends to provide higher productivity than MR-VPP, even when the season is shorter, as the OLCI PPI reaches higher maximum values than MR-VPP.
- CLMS OLCI LSP products (SOSD, EOSD, SPROD and TPROD layers) are also spatially consistent with HR-VPP over selected AOIs in Europe, showing similar spatial patterns for most cases.
- The spatial consistency of OLCI LSP with global reference products show:
    - Map of differences of OLCI SOSD with MODIS V6.1 MidGreenup shows differences typically within ±15 days, showing large differences over tropical areas (OLCI<MODIS) and southern hemisphere (OLCI>MODIS). Map of differences of OLCI EOSD with MODIS V6.1 MidGreendown shows, as expected, systematically positive differences (OLCI > MODIS). Histograms of differences are centred of around 3 and 5 days for SOSD and EOSD.
    - Map of differences of OLCI SOSD with VIIRS V2 Date_Mid_Greenup_Phase show, as expected, typically negative sign (OLCI < VIRS) except over temperate latitudes. The comparison of OLCI EOSD with VIIRS V2 Date_Mid_Senescence_Phase shows as expected systematically positive differences (OLCI > VIIRS). Histograms of differences are centred of around -3 and 12 days for SOSD and EOSD.
- The analysis of SOSD and EOSD variations per latitude (latitudinal transect from Norway to South Africa) shows the expected trend and consistent values than MODIS and VIIRS layers, except for desert areas and sparsely vegetated areas (around -25º latitude) and tropical forest (around equator) where larger discrepancies are found in case of VIIRS.
- The spatial variability over homogeneous areas shows good precision for SOSD (STD typically ≤ 4 days, MAD ≤ 3 days) and for EOSD (STD and MAD typically ≤ 7 days) over most of the selected ROIs, except for the EBF area where larger variability is found.
- The spatial variability of TPROD/SPROD is ranges typically between 6% and 25% in terms of STD and MAD.

**Error evaluation (direct validation)**

- The comparison of CLMS OLCI LSP SOSD and EOSD versus ground measurements of PEP725, TEMPO and NEON datasets for 2023 shows:
  - For SOSD, bias close to zero or slight delay (up to 5 days) is found in the comparison with ground phenophases related to the first stages of the growing season (e.g., 1st leaves separated (PEP725), germination (TEMPO) and breaking leaves (NEON)). As compared to phenophases related to more developed leaf growth (i.e., leaf development (TEMPO) and leaf unfolding ≥50% (PEP725)) negative bias (i.e., anticipated SOSD) between -7 and -12 days is found. These results suggest a good accuracy of the OLCI LSP SOSD, which represents a date between the first emerging buds/leaves and a more advance stage of leaf development. The exception is for emerging leaves (NEON) where large negative bias is found, which can be partly due to the low representativity of this class (only 11 samples). Better accuracy that MR-VPP products is found for comparisons with both PEP725 and TEMPO.
  - For EOSD, the bias is typically negative (i.e., anticipated end of season), and CLMS EOSD is typically more related to colouration stages (PEP725 and NEON) or beginning of dormancy (TEMPO) (bias ranging from -14 days to 4 days) than to the fall of leaves (up to -33 days). Similar accuracy than MR-VPP with PEP725 and TEMPO observations is found.
  - RMSD is typically ranging between 20 and 30 days for SOSD and from 20 to 40 days in case of EOSD.
  - The percentage of cases CLMS OLCI LSP within goal requirements typically ranges between 10% and 20% in case of SOSD (25%-40% for threshold) and between 15% and 20% in case of EOSD (30%-45% for threshold).
- The results of the comparison with near-ground-based PhenoCam V3 for 2023 are:
  - OLCI LSP SOSD better agrees in terms of accuracy and uncertainty with PhenoCam start of season defined at 50% threshold (Bias = 3 days, RMSD = 29 days) than compared with PhenoCam at 25% (Bias = 12 days, RMSD = 23 days), which indicates a delay of the SOSD in OLCI LSP as compared to PhenoCam GCC phenology estimates
  - For EOSD, the best agreement in terms of bias and uncertainty is found with PhenoCam defined at 25% (Bias = 0, MD = 2, RMSD = 22 days). Negative bias (Bias = -11 days) is found compared with PhenoCam defined at 10%.
  - GCLS OLCI LSP tends to provide shorter seasons as compared with near-ground-based datasets

**Error evaluation (product intercomparison)**

- The comparison of CLMS OLCI LSP with MR-VPP equivalent layers over Europe (EVAL sampling) shows:
  - Shorter season of CLMS, with delayed SOSD (B = 18 days) respect to MR-VPP and similar EOSD (B = -2 days). RMSD of 22 and 23 days for SOSD and EOSD.
  - CLMS TPROD/SPROD overestimate equivalent MR-VPP similar layers (B and MD around 25%) despite the shorter seasons, which is explained by the larger maximum values of PPI.
- The comparison of CLMS OLCI LSP with HR-VPP equivalent layers over Europe (EVAL sampling) shows:
  - Shorter season of CLMS as a result of delayed SOSD (B = 14 days) of CLMS respect to HR-VPP and slightly delayed EOSD (B = 6 days). RMSD of 21 and 24 days for SOSD and EOSD.
  - CLMS TPROD and SPROD overestimate (B around 30%) equivalent HR-VPP productivity layers, explained by the larger maximum CLMS PPI values.
- The results of OLCI SOSD and EOSD with MODIS V6.1 phenological metrics shows:
  - OLCI SOSD shows the best correspondence with MODIS MidGreenup (MD = 1 day, RMSD = 25 days) despite of the fact that the best agreement could be expected, due to the product definition, with Greenup (MD = 26 days, RMSD = 41 days).
  - For EOSD, OLCI falls between MODIS MidGreendown (B = 5 days, RMSD = 32 days) and MODIS Dormancy (B = -23 days, RMSD = 35 days).
- The comparison of OLCI LSP with VIIRS V2 phenological metrics shows:
  - the best agreement of OLCI SOSD with VIIRS V2 Mid_Greenup_Phase (MD =-2 days, RMSD = 19 days) compared with VIIRS V2 Onset_Greeness_Increase (MD = 15 days, RMSD=28 days).
  - OLCI EOSD falls between VIIRS V2 Date_Mid_Senescence_Phase (MD = 11 days) and Onset_Greeness_Minimum (MD = -11 days). Similar overall uncertainty (RMSD = 36 days) id obtained when OLCI EOSD is compared with both VIIRS V2 phases.
- Comparisons of CLMS OLCI TPROD and SPROD with other productivity GPP products show consistent values: $R \geq 0.76$ with CLMS OLCI GPP and $R \geq 0.67$ with MODIS GPP.

**Summary and conclusions**

In summary, CLMS OLCI LSP products show overall good quality, reliable values at global scale and consistent with ground references and equivalent global (MODIS V6.1, VIIRS V2) and regional (Europe) medium-resolution and high-resolution satellite sensors estimates, except for the non-expected values over few desert areas. The spatial precision over homogeneous areas covering different biomes is good, with STD and MAD typically lower than 4 and 7 days for SOSD and EOSD. Equatorial regions displayed larger spatial variability which could be either due to natural variability in the area or an impact of the noise in the input data due to the frequent cloud cover over these regions. Large spatial variability is found for productivity SPROD and TPROD layers (STD and MAD typically between 10% and 25%). The variations of SOSD and EOSD per latitude (latitudinal transect from Norway to South Africa) show the expected trend and consistent values with estimates from VIIRS (from a previous year) except over the Sahara and scrubs areas in the south.

CLMS OLCI LSP provides good accuracy for SOSD with bias typically ranging between one week with the phenophases related to start of the season in PEP725, TEMPO and NEON, and larger negative bias with phenophases related to more developed vegetation (e.g., >50% leaf developed). For EOSD, the bias is typically negative, and more similar to colouration stages or beginning of dormancy that to the fall of leaves. The percentages of CLMS OLCI LSP cases within goal requirements are typically ranging between 10% and 20% for SOSD (25%-40% for threshold) and between 15% and 20% for EOSD (30%-45% for threshold).

OLCI LSP tends to provide shorter seasons as compared with near-ground-based Phenocam and satellite-based datasets over Europe. OLCI LSP shows typically delayed SOSD than MR-VPP (B=18 days) but better accuracy with ground observations, HR-VPP (B=14 days) and PhenoCam (B=12 days). In case of EOSD, OLCI shows similar values than MR-VPP and HR-VPP (B of -2 and 4 days) and anticipated EOSD than PhenoCam (defined at 10% GCC amplitude). The comparison with global products (MODIS and VIIRS) shows that OLCI SOSD is similar to MODIS and VIIRS mid-green-up dates, whilst EOSD falls between MODIS and VIIRS mid-green-down and dormancy (or minimum) dates.

The comparison of productivity layers (SPROD and TPROD) shows high correlations and higher productivity (positive bias around 25%-30%) than MR-VPP and HR-VPP. Comparisons of CLMS TPROD and SPROD with other productivity GPP global products show consistent values: $R \geq 0.76$ with CLMS OLCI GPP and $R \geq 0.67$ with MOD17A2HGF V6.1.

The validation of SOSD and EOSD for 2024 compared with PEP725, TEMPO and NEON ground references and near-ground-based Phenocam data showed similar results than in 2023, with some exceptions mainly explained with the different samplings of different years.

# 6 REFERENCES

Bolton, D.K., Gray, J.M., Melaas, E.K., Moon, M., Eklundh, L., Friedl, M.A., 2020. Continental-scale land surface phenology from harmonized Landsat 8 and Sentinel-2 imagery. Remote Sens. Environ. 240, 111685. https://doi.org/10.1016/J.RSE.2020.111685

Buchhorn, M., Lesiv, M., Tsendbazar, N.-E., Herold, M., Bertels, L., Smets, B., 2020. Copernicus Global Land Cover Layers—Collection 2. Remote Sens. 12, 1044. https://doi.org/10.3390/rs12061044

Cai, Z., Jönsson, P., Jin, H., Eklundh, L., 2017. Performance of Smoothing Methods for Reconstructing NDVI Time-Series and Estimating Vegetation Phenology from MODIS Data. Remote Sens. 2017, Vol. 9, Page 1271 9, 1271. https://doi.org/10.3390/RS9121271

Camacho, F., Sánchez-Zapero, J., Martínez-Sánchez, E., Roses, C., 2024. European eVALuation sites (EVAL): A European sampling for regional satellite product intercomparison. [Dataset]. https://doi.org/10.5281/ZENODO.11125601

Fernandes, R., Plummer, S., Nightingale, J., Baret, F., Camacho, F., Fang, H., Garrigues, S., Gobron, N., Lang, M., Lacaze, R., Leblanc, S., Meroni, M., Martinez, B., Nilson, T., Pinty, B., Pisek, J., Sonnentag, O., Verger, A., Welles, J., Weiss, M., Widlowski, J.-L., Schaepman-Strub, G., Roman, M., Nickeson, J., 2014. Global Leaf Area Index Product Validation Good Practices, in: Fernandes, R., Plummer, S., Nightingale, J. (Eds.), Best Practice for Satellite-Derived Land Product Validation. Land Product Validation Subgroup (Committee on Earth Observation Satellites Working Group on Calibration and Validation). https://doi.org/10.5067/doc/ceoswgcv/lpv/lai.002

Fuster, B., Sánchez-Zapero, J., Camacho, F., García-Santos, V., Verger, A., Lacaze, R., Weiss, M., Baret, F., Smets, B., 2020. Quality Assessment of PROBA-V LAI, fAPAR and fCOVER Collection 300 m Products of Copernicus Global Land Service. Remote Sens. 12, 1017. https://doi.org/10.3390/rs12061017

Harper, W. V., 2014. Reduced Major Axis regression: teaching alternatives to Least Squares. Proc. Ninth Int. Conf. Teach. Stat. 1–4. https://doi.org/10.1016/B978-0-12-420228-3.00013-0

Jin, H., Eklundh, L., 2014. A physically based vegetation index for improved monitoring of plant phenology. Remote Sens. Environ. 152, 512–525. https://doi.org/10.1016/J.RSE.2014.07.010

Jönsson, P., Eklundh, L., 2004. TIMESAT—a program for analyzing time-series of satellite sensor data. Comput. Geosci. 30, 833–845. https://doi.org/10.1016/J.CAGEO.2004.05.006

Martínez-Sánchez, E., Sánchez-Zapero, J., Camacho, F., 2024. LAND VALidation (LANDVAL) V2: Representative global sampling for satellite product intercomparison and calibration. [Dataset]. https://doi.org/10.5281/ZENODO.10559901

Maury, O., Quidoz, M.-C., Garcia de Cortazar Atauri, I., Chuine, I., El Hasnaoui, M., Tromel, L., 2023. Portail de données phénologiques du réseau TEMPO / TEMPO Data Portal - TEMPO [Dataset]. https://doi.org/https://doi.org/10.57745/NQ9HRV

Melaas, E.K., Sulla-Menashe, D., Gray, J.M., Black, T.A., Morin, T.H., Richardson, A.D., Friedl, M.A., 2016. Multisite analysis of land surface phenology in North American temperate and

boreal deciduous forests from Landsat. Remote Sens. Environ. 186, 452–464. https://doi.org/10.1016/J.RSE.2016.09.014

Mira, M., Weiss, M., Baret, F., Courault, D., Hagolle, O., Gallego-Elvira, B., Olioso, A., 2015. The MODIS (collection V006) BRDF/albedo product MCD43D: Temporal course evaluated over agricultural landscape. Remote Sens. Environ. 170, 216–228. https://doi.org/10.1016/j.rse.2015.09.021

Monteith, J.L., 1972. Solar Radiation and Productivity in Tropical Ecosystems. J. Appl. Ecol. 9, 747. https://doi.org/10.2307/2401901

Moon, M., Richardson, A.D., Friedl, M.A., 2021. Multiscale assessment of land surface phenology from harmonized Landsat 8 and Sentinel-2, PlanetScope, and PhenoCam imagery. Remote Sens. Environ. 266, 112716. https://doi.org/10.1016/J.RSE.2021.112716

Nestola, E., Sánchez-Zapero, J., Latorre, C., Mazzenga, F., Matteucci, G., Calfapietra, C., Camacho, F., 2017. Validation of PROBA-V GEOV1 and MODIS C5 & C6 fAPAR Products in a Deciduous Beech Forest Site in Italy. Remote Sens. 9, 126. https://doi.org/10.3390/rs9020126

Nunes, M.H., Camargo, J.L.C., Vincent, G., Calders, K., Oliveira, R.S., Huete, A., Mendes de Moura, Y., Nelson, B., Smith, M.N., Stark, S.C., Maeda, E.E., 2022. Forest fragmentation impacts the seasonality of Amazonian evergreen canopies. Nat. Commun. 13. https://doi.org/10.1038/S41467-022-28490-7

Olson, D.M., Dinerstein, E., Wikramanayake, E.D., Burgess, N.D., Powell, G.V.N., Underwood, E.C., D'Amico, J.A., Itoua, I., Strand, H.E., Morrison, J.C., Loucks, C.J., Allnutt, T.F., Ricketts, T.H., Kura, Y., Lamoreux, J.F., Wettengel, W.W., Hedao, P., Kassem, K.R., 2001. Terrestrial ecoregions of the world: A new map of life on Earth. Bioscience 51, 933–938. https://doi.org/10.1641/0006-3568(2001)051[0933:TEOTWA]2.0.CO;2

Peng, D., Zhang, X., Wu, C., Huang, W., Gonsamo, A., Huete, A.R., Didan, K., Tan, B., Liu, X., Zhang, B., 2017. Intercomparison and evaluation of spring phenology products using National Phenology Network and AmeriFlux observations in the contiguous United States. Agric. For. Meteorol. 242, 33–46. https://doi.org/10.1016/J.AGRFORMET.2017.04.009

Rodriguez-Galiano, V.F., Dash, J., Atkinson, P.M., 2015. Intercomparison of satellite sensor land surface phenology and ground phenology in Europe. Geophys. Res. Lett. 42, 2253–2260. https://doi.org/10.1002/2015GL063586

Román, M.O., Justice, C., Paynter, I., Boucher, P.B., Devadiga, S., Endsley, A., Erb, A., Friedl, M., Gao, H., Giglio, L., Gray, J.M., Hall, D., Hulley, G., Kimball, J., Knyazikhin, Y., Lyapustin, A., Myneni, R.B., Noojipady, P., Pu, J., Riggs, G., Sarkar, S., Schaaf, C., Shah, D., Tran, K.H., Vermote, E., Wang, D., Wang, Z., Wu, A., Ye, Y., Shen, Y., Zhang, S., Zhang, X., Zhao, M., Davidson, C., Wolfe, R., 2024. Continuity between NASA MODIS Collection 6.1 and VIIRS Collection 2 land products. Remote Sens. Environ. 302, 113963. https://doi.org/10.1016/J.RSE.2023.113963

Sánchez-Zapero, J., Camacho, F., Martínez-Sánchez, E., Lacaze, R., Carrer, D., Pinault, F., Benhadj, I., Muñoz-Sabater, J., 2020. Quality Assessment of PROBA-V Surface Albedo V1 for the Continuity of the Copernicus Climate Change Service. Remote Sens. 2020, Vol. 12, Page 2596 12, 2596. https://doi.org/10.3390/rs12162596

Sánchez-Zapero, J., Martínez-Sánchez, E., Camacho, F., Wang, Z., Carrer, D., Schaaf, C., García-Haro, F.J., Nickeson, J., Cosh, M., 2023. Surface ALbedo VALidation (SALVAL) Platform: Towards CEOS LPV Validation Stage — Application to Three Global Albedo Climate Data Records. Remote Sens. 2023, Vol. 15, Page 1081 15, 1081. https://doi.org/10.3390/RS15041081

Tian, F., Cai, Z., Jin, H., Hufkens, K., Scheifinger, H., Tagesson, T., Smets, B., Van Hoolst, R., Bonte, K., Ivits, E., Tong, X., Ardö, J., Eklundh, L., 2021. Calibrating vegetation phenology from Sentinel-2 using eddy covariance, PhenoCam, and PEP725 networks across Europe. Remote Sens. Environ. 260, 112456. https://doi.org/10.1016/J.RSE.2021.112456

Wang, L., Zhu, H., Lin, A., Zou, L., Qin, W., Du, Q., 2017. Evaluation of the Latest MODIS GPP Products across Multiple Biomes Using Global Eddy Covariance Flux Data. Remote Sens. 2017, Vol. 9, Page 418 9, 418. https://doi.org/10.3390/RS9050418

Zhang, X., Friedl, M.A., Schaaf, C.B., 2006. Global vegetation phenology from Moderate Resolution Imaging Spectroradiometer (MODIS): Evaluation of global patterns and comparison with in situ measurements. J. Geophys. Res. Biogeosciences 111, 4017. https://doi.org/10.1029/2006JG000217

Zhang, X., Jayavelu, S., Liu, L., Friedl, M.A., Henebry, G.M., Liu, Y., Schaaf, C.B., Richardson, A.D., Gray, J., 2018a. Evaluation of land surface phenology from VIIRS data using time series of PhenoCam imagery. Agric. For. Meteorol. 256–257, 137–149. https://doi.org/10.1016/J.AGRFORMET.2018.03.003

Zhang, X., Liu, L., Liu, Y., Jayavelu, S., Wang, J., Moon, M., Henebry, G.M., Friedl, M.A., Schaaf, C.B., 2018b. Generation and evaluation of the VIIRS land surface phenology product. Remote Sens. Environ. 216, 212–229. https://doi.org/10.1016/J.RSE.2018.06.047

Zhao, M., Heinsch, F.A., Nemani, R.R., Running, S.W., 2005. Improvements of the MODIS terrestrial gross and net primary production global data set. Remote Sens. Environ. 95, 164–176. https://doi.org/10.1016/j.rse.2004.12.011

Zhu, X., Pei, Y., Zheng, Z., Dong, J., Zhang, Y., Wang, J., Chen, L., Doughty, R.B., Zhang, G., Xiao, X., 2018. Underestimates of Grassland Gross Primary Production in MODIS Standard Products. Remote Sens. 2018, Vol. 10, Page 1771 10, 1771. https://doi.org/10.3390/RS10111771

# ANNEX I: DIFFERENCE MAPS BETWEEN CLMS OLCI LSP AND MR-VPP

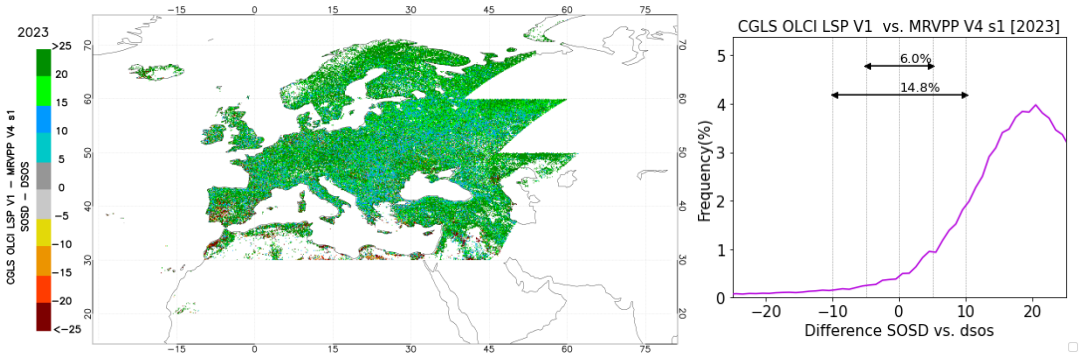



**Figure 59: Top: Maps and histograms of differences between CLMS OLCI LSP SOSD and MR-VPP DSOS for 2023 (season 1). Unit: days.**

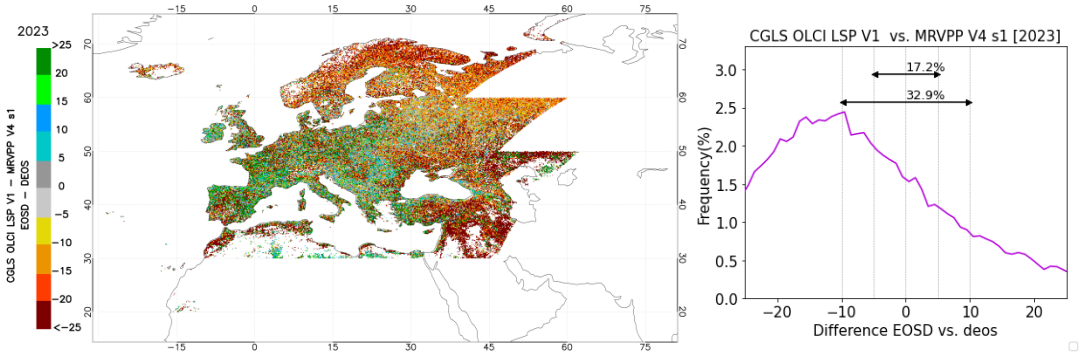



**Figure 60: Top: Maps and histograms of differences between CLMS OLCI LSP EOSD and MR-VPP DEOS for 2023 (season 1). Unit: days.**

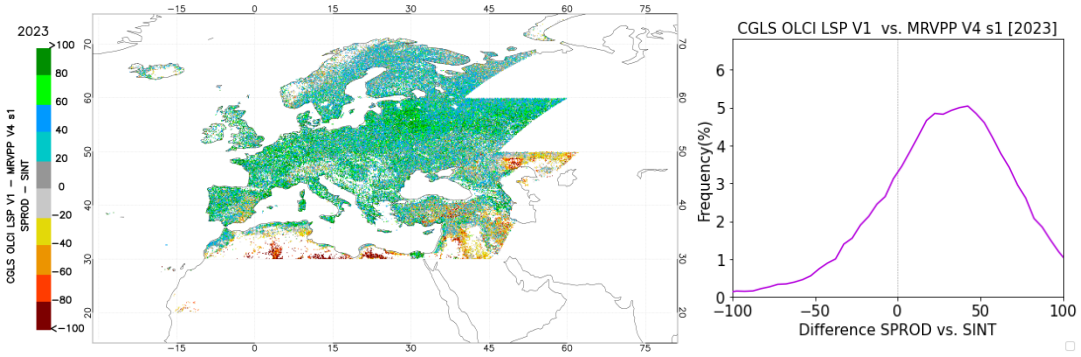



**Figure 61: Top: Maps and histograms of differences between CLMS OLCI LSP SPROD and MR-VPP SINT for 2023 (season 1). Unit: $m^2 \cdot m^{-2} \cdot day$.**

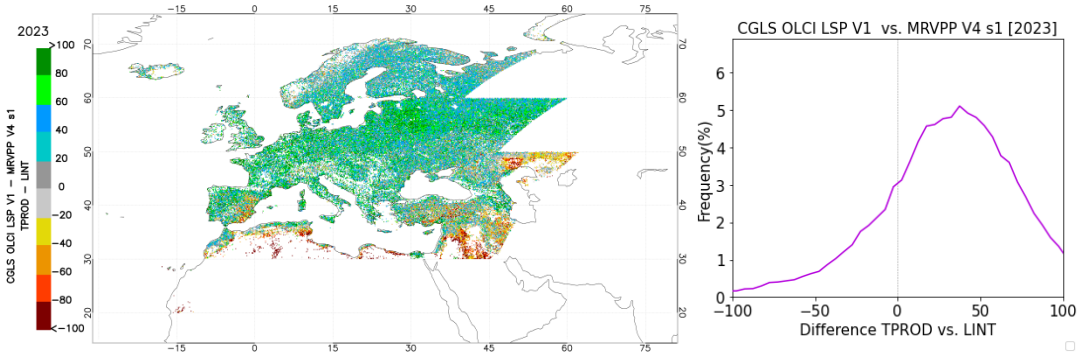



**Figure 62: Top: Maps and histograms of differences between CLMS OLCI LSP TPROD and MR-VPP LINT for 2023 (season 1). Unit: $m^2 \cdot m^{-2} \cdot day$.**

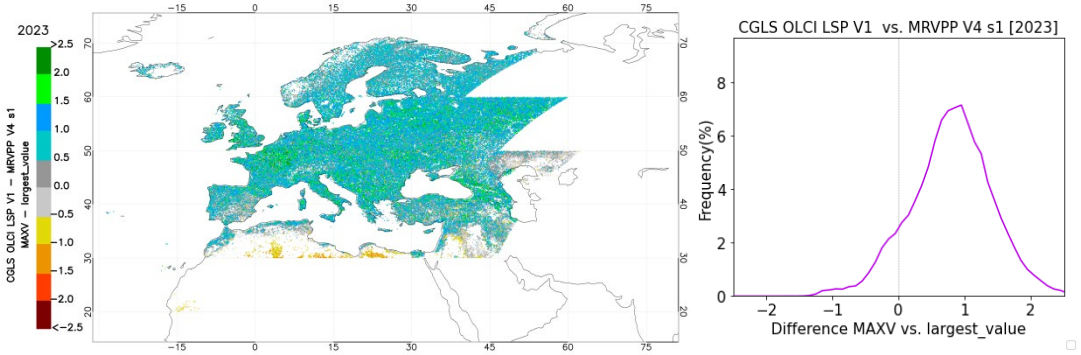



**Figure 63: Top: Maps and histograms of differences between CLMS OLCI LSP MAXV and MR-VPP 'largest values' for 2023 (season 1). Unit: $m^2 \cdot m^{-2}$.**

# ANNEX II: SPATIAL CONSISTENCY BETWEEN CLMS OLCI LSP AND HR-VPP OVER AOIS

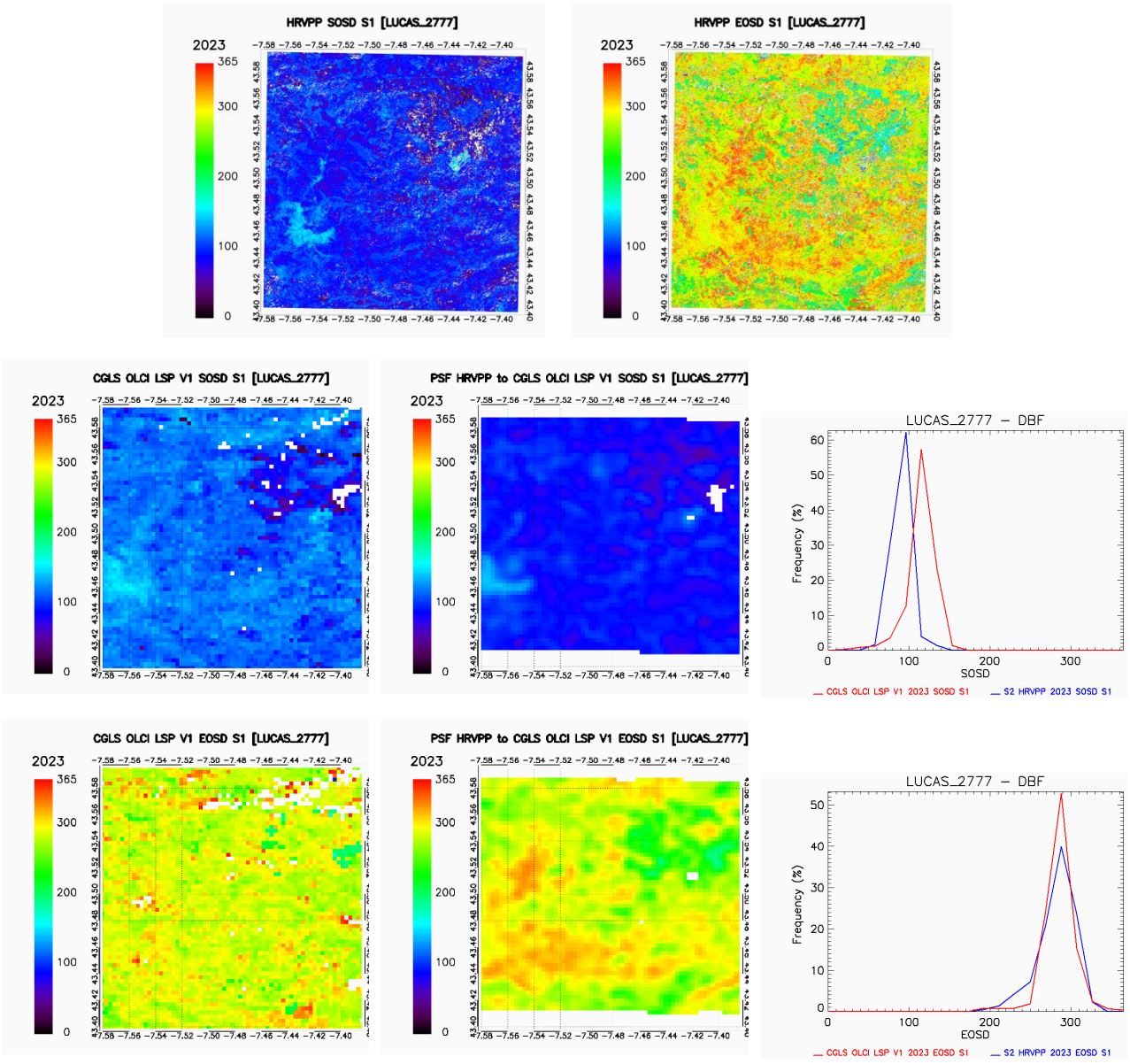



**Figure 64: Spatial consistency over LUCAS 2777 DBF site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SOSD (left) and EOSD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SOSD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for EOSD.**

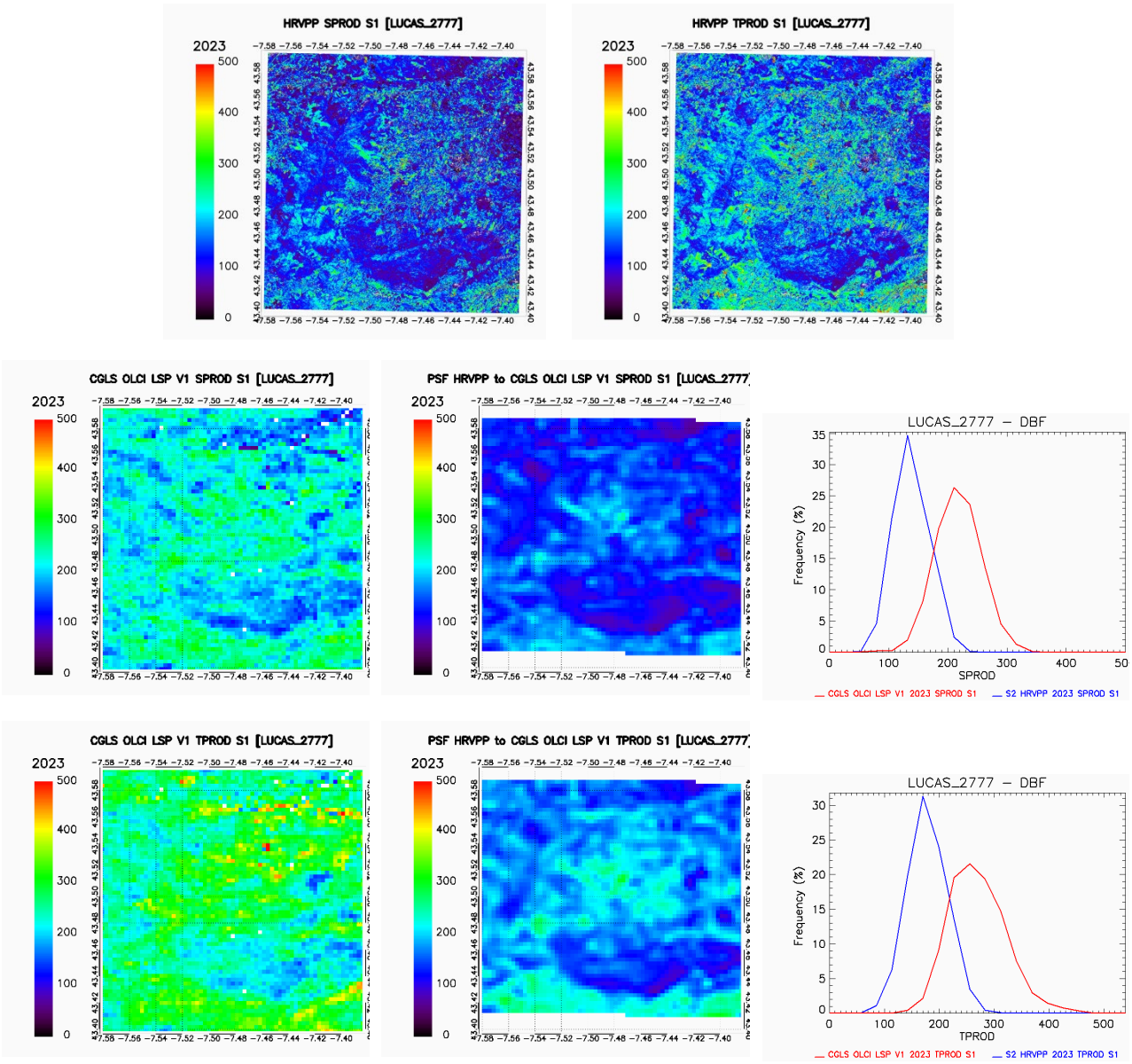



**Figure 65: Spatial consistency over LUCAS 2777 DBF site (20 km x 20 km area) between CLMS OLCI LSP and HR-VPP in 2023 season 1. Line 1: HR-VPP SPROD (left) and TPROD (right) at 10 m resolution. Line 2: CLMS OLCI LSP SPROD (left), aggregated S2 HR-VPP to 300 m using PSF (center) and their associated histograms (right). Line 3: same as in Line 2, but for TPROD.**

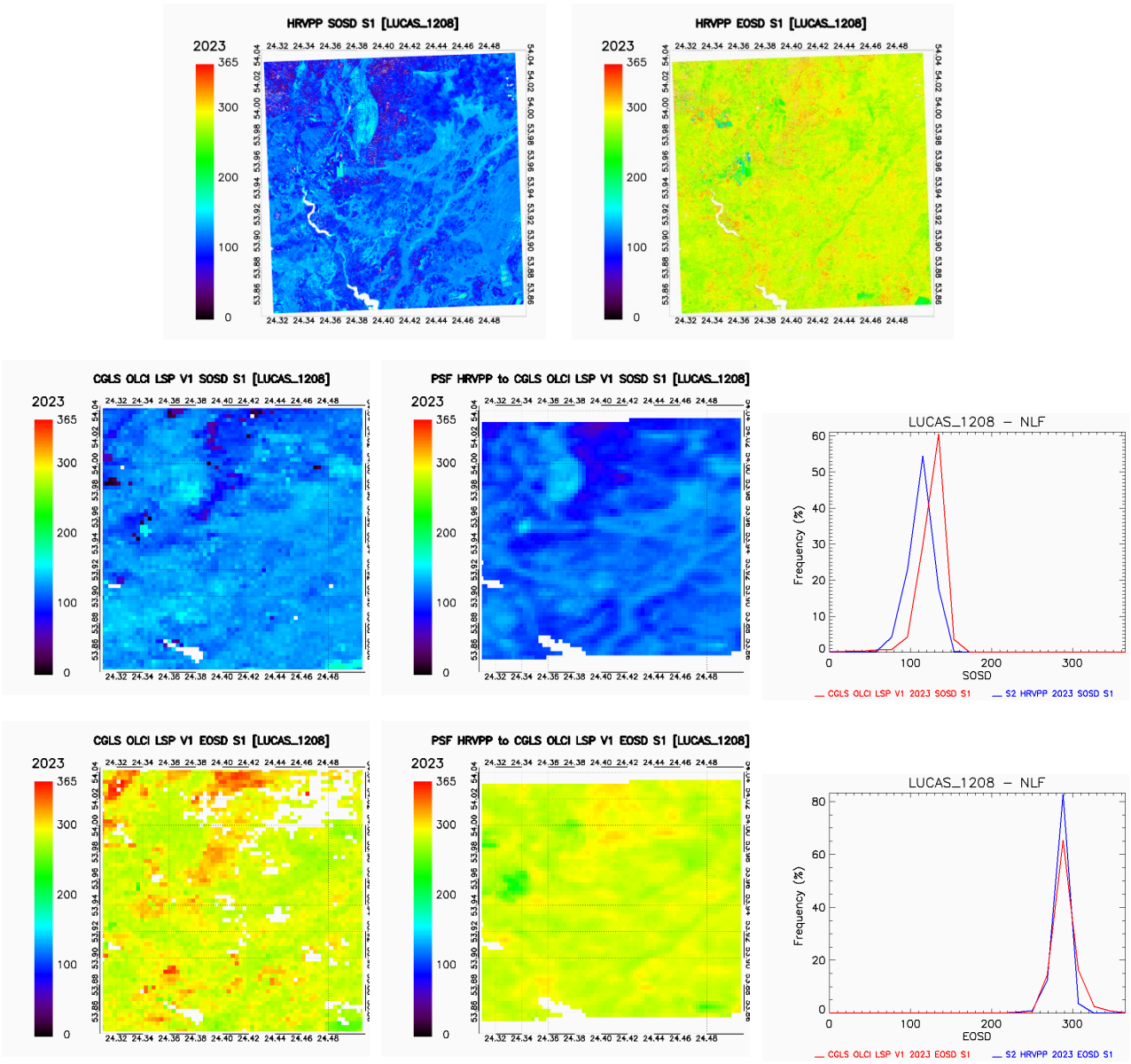



**Figure 66: Same as in Figure 64, but for LUCAS 1208 NLF site.**

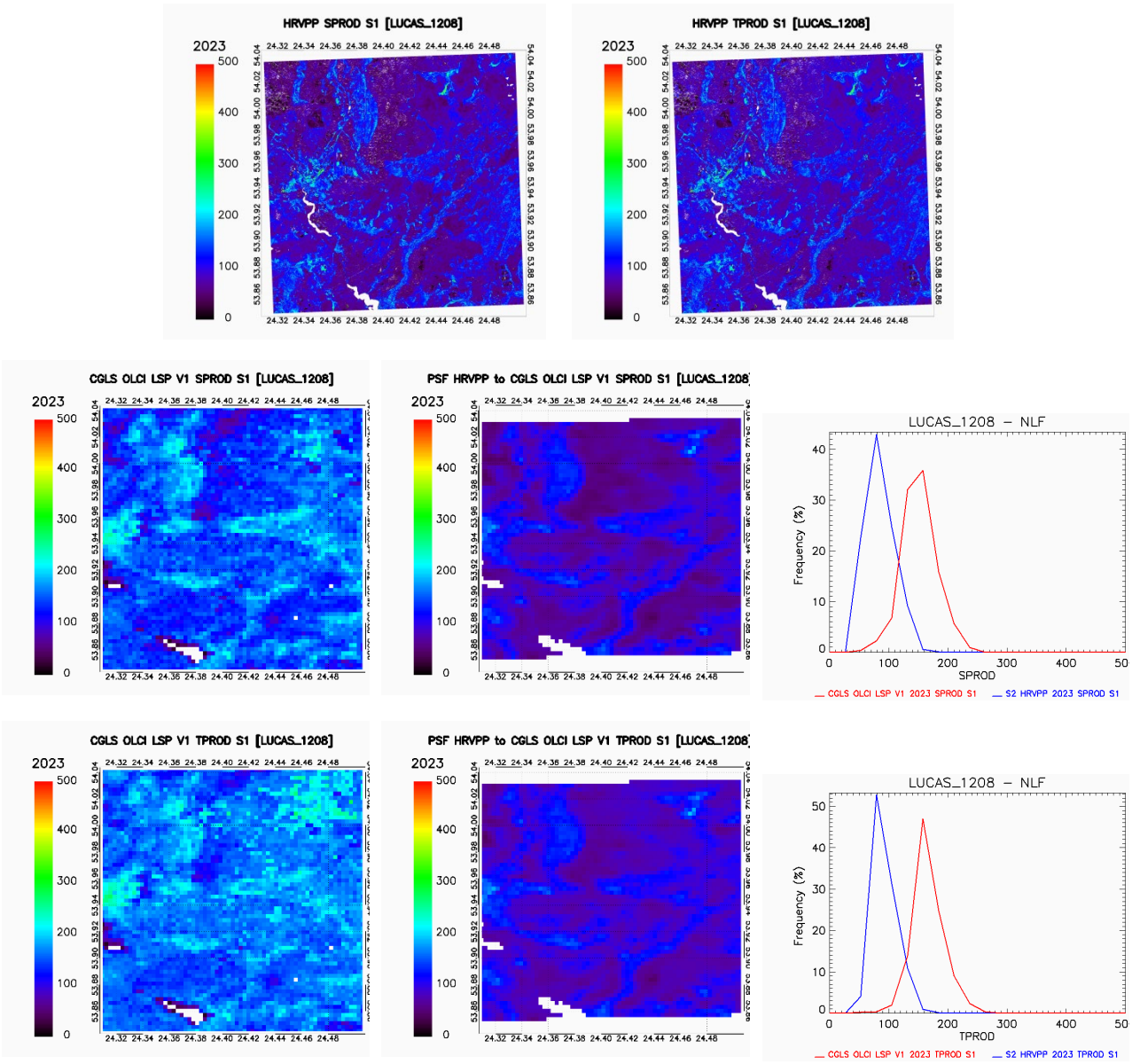



**Figure 67: Same as in Figure 65, but for LUCAS 1208 NLF site.**

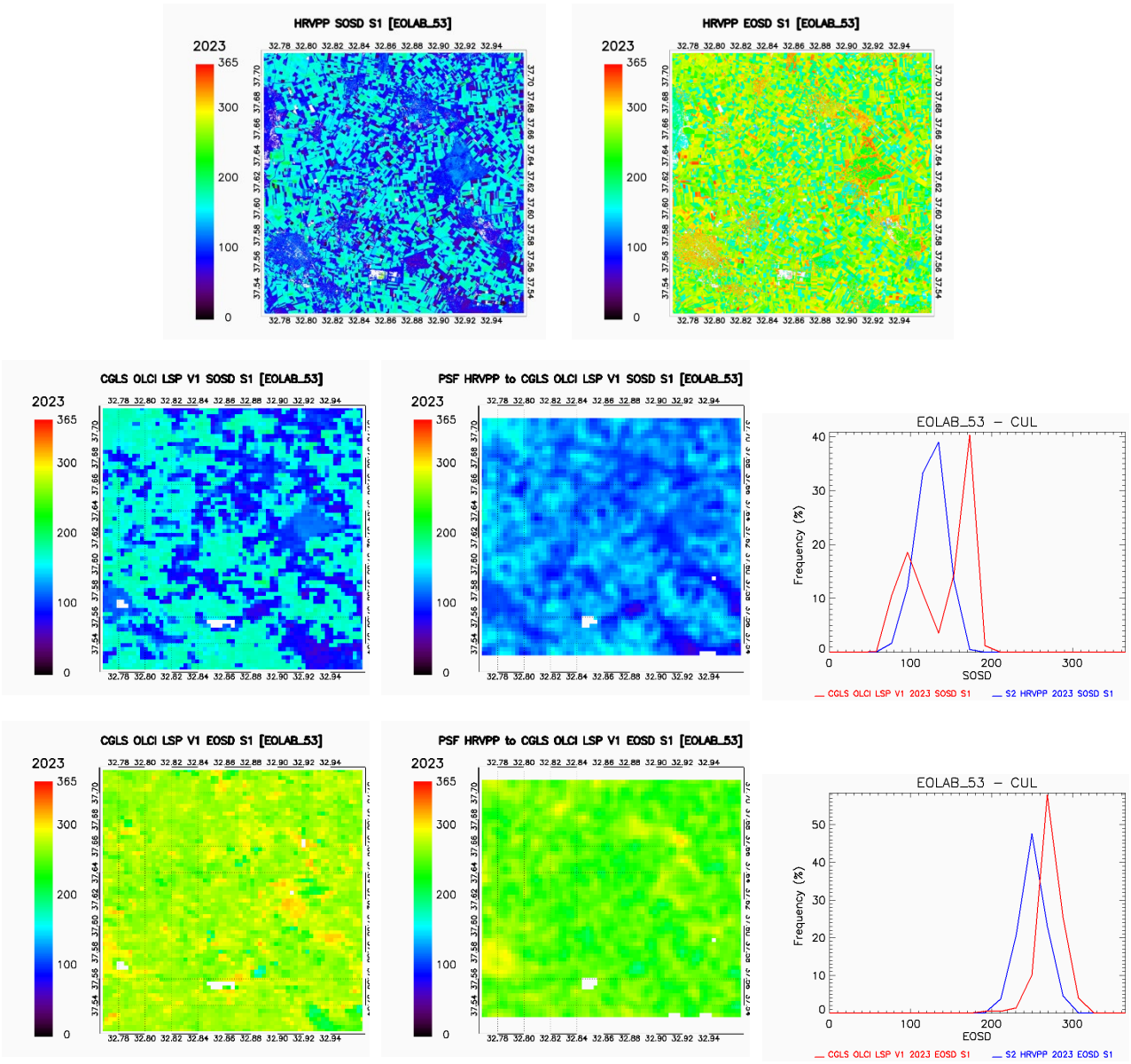



**Figure 68: Same as in Figure 64, but for EOLAB 53 cultivated site.**

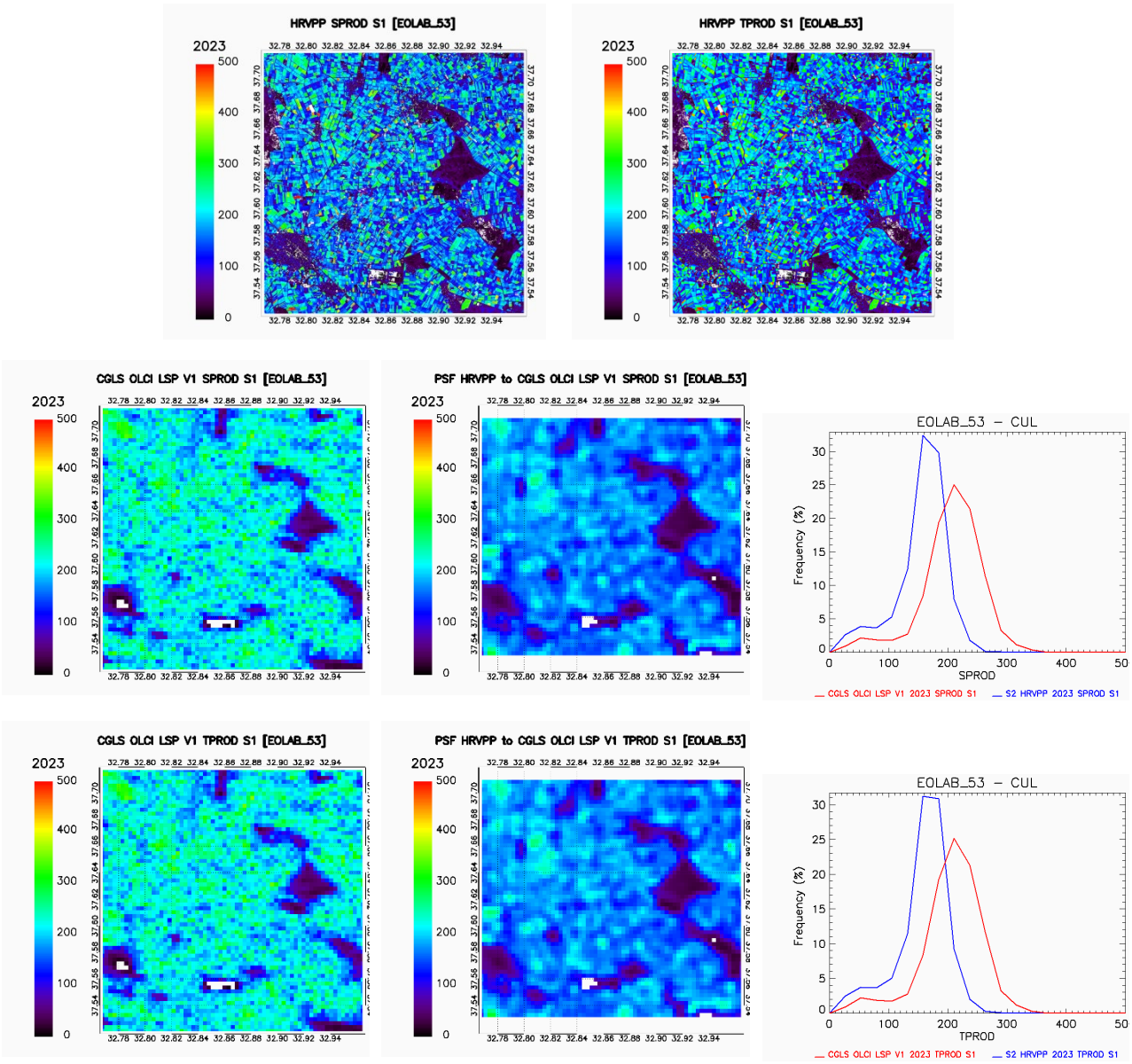



**Figure 69: Same as in Figure 65, but for EOLAB 53 cultivated site.**

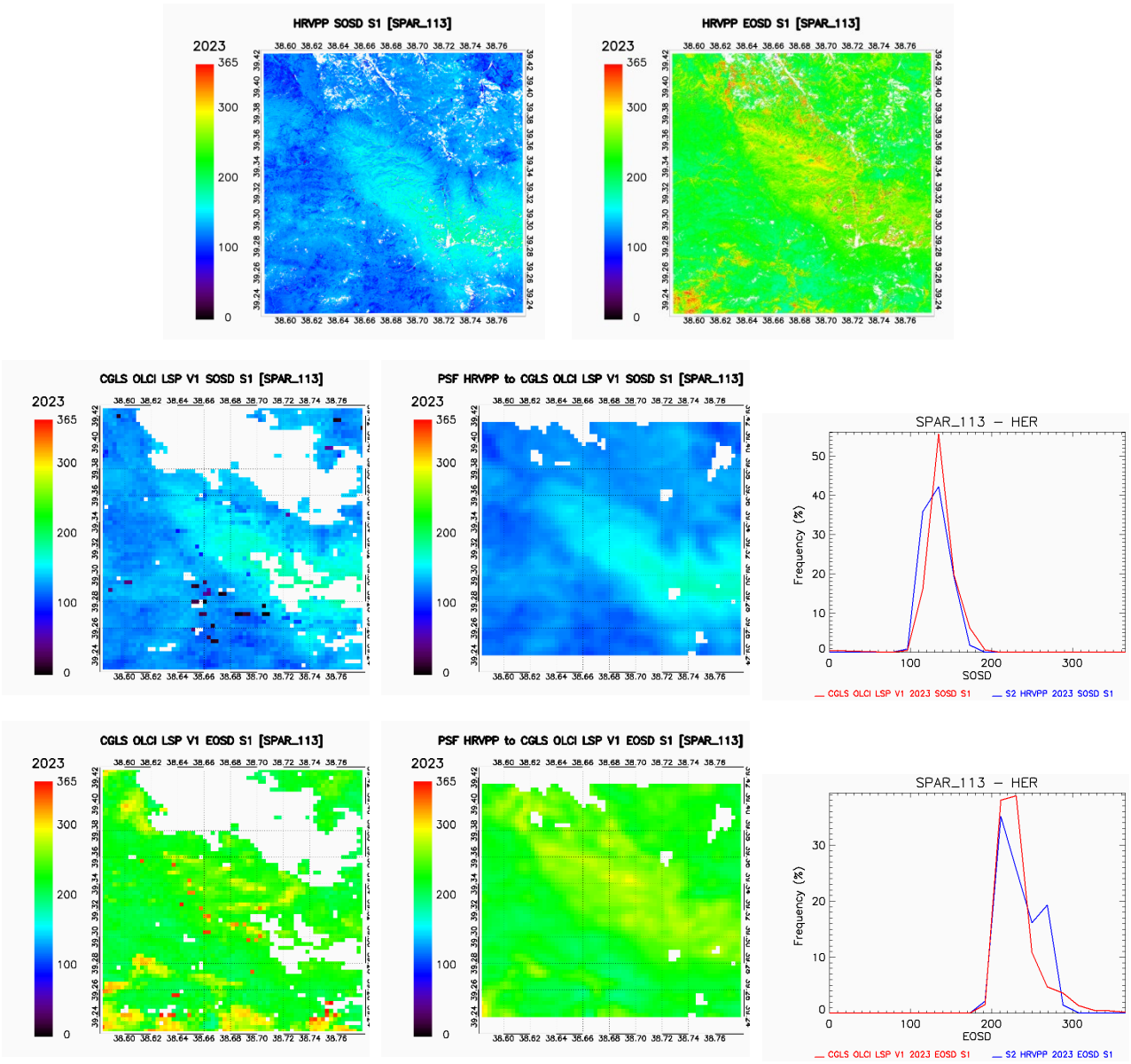



**Figure 70: Same as in Figure 64, but for SPAR 113 shrublands site.**


© C-GLOPS Lot1 consortium

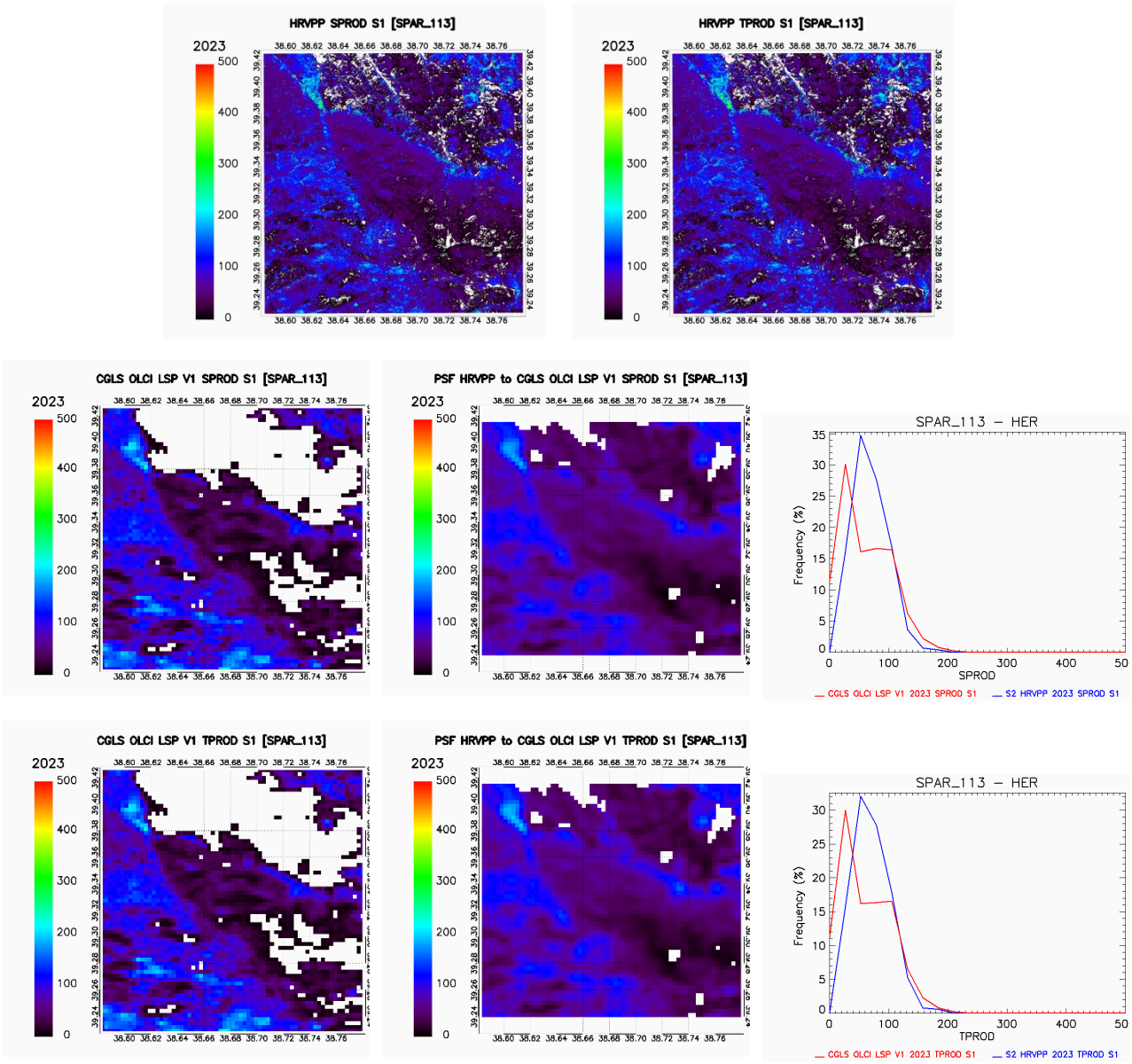



**Figure 71: Same as in Figure 65, but for SPAR 113 shrublands site.**


© C-GLOPS Lot1 consortium

# ANNEX III: MAPS AND SPATIAL PRECISION OF CLMS OLCI LSP OVER SELECTED ROIS

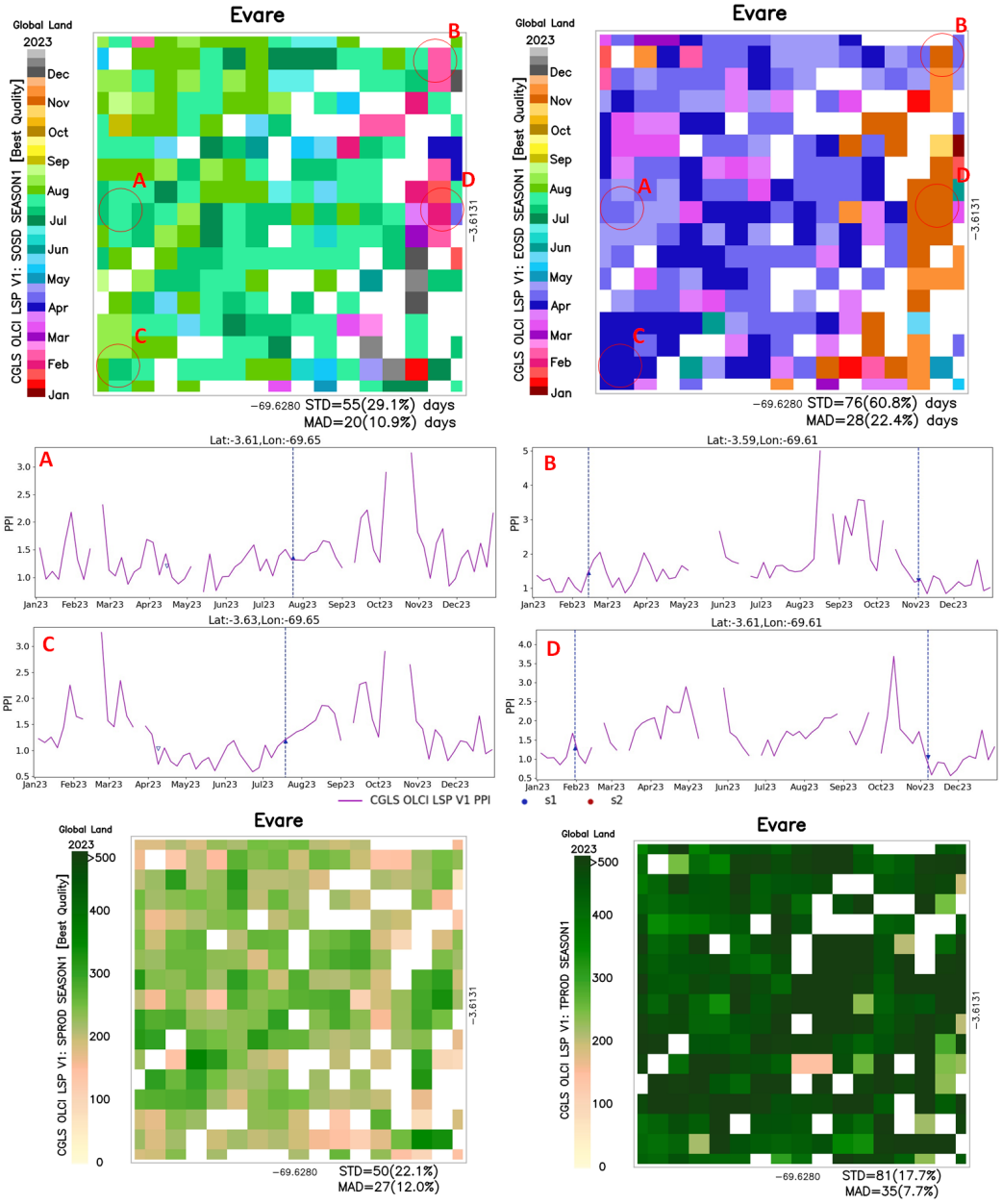



**Figure 72: Spatial precision over 5 km x 5 km area of Evare EBF site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). In the temporal profiles, unfilled inverted triangles of EOSD represent the value corresponding to next year (EOSD > 365).**

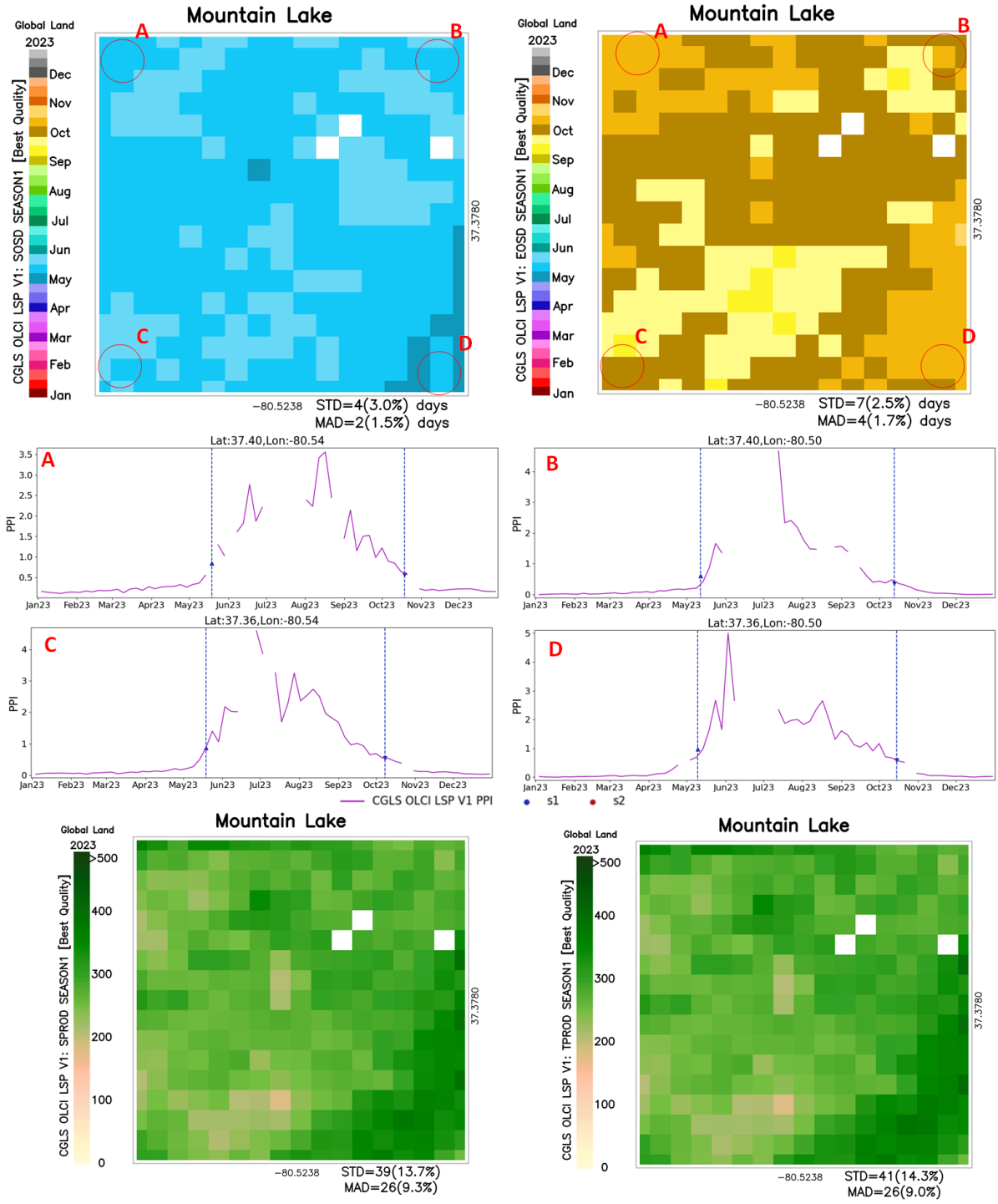



**Figure 73: Spatial precision over 5 km x 5 km area of Mountain Lake DBF site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right).**

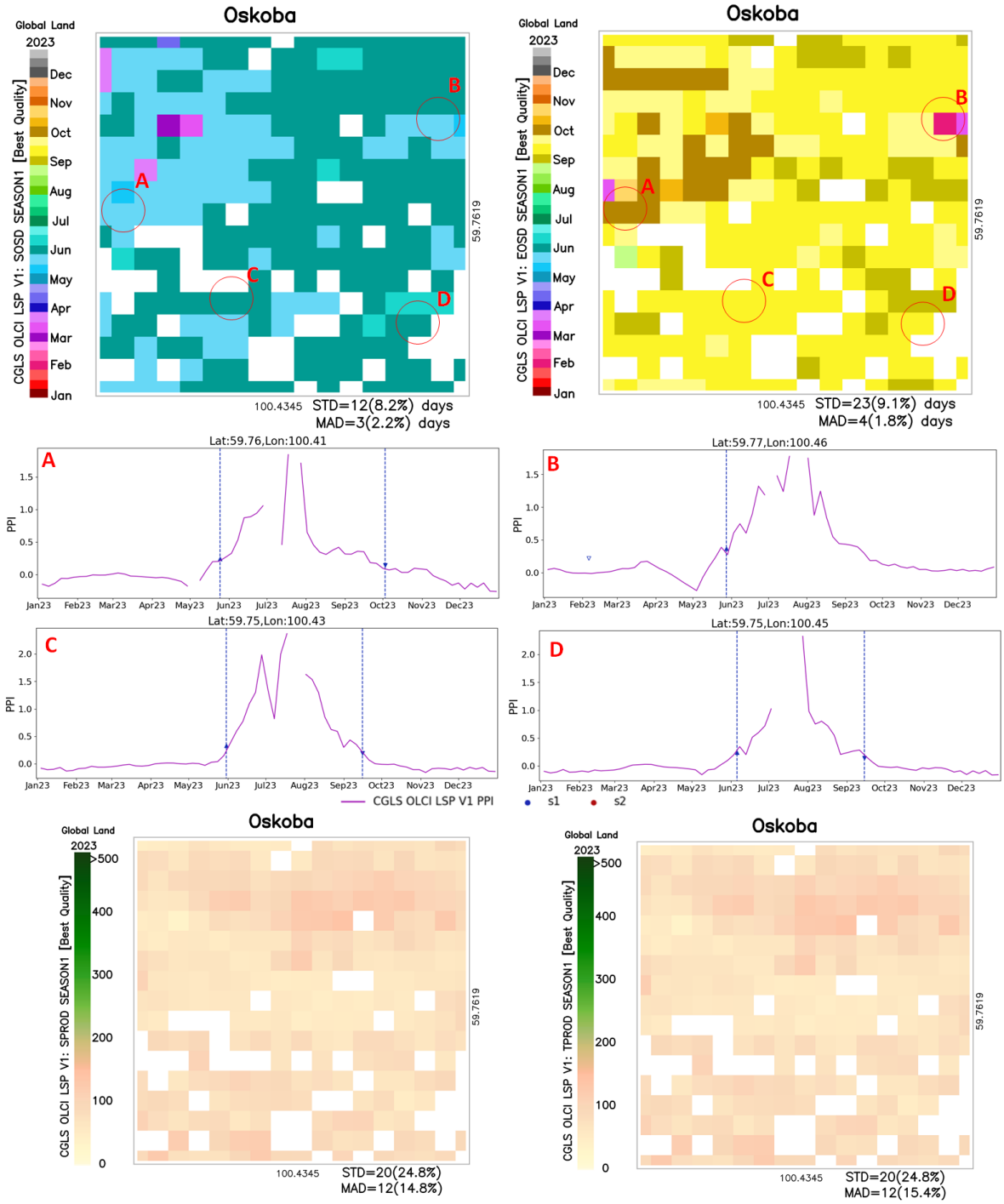



**Figure 74: Spatial precision over 5 km x 5 km area of Oskoba NLF site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right). In the temporal profiles, unfilled inverted triangles of EOSD represent the value corresponding to next year (EOSD > 365).**

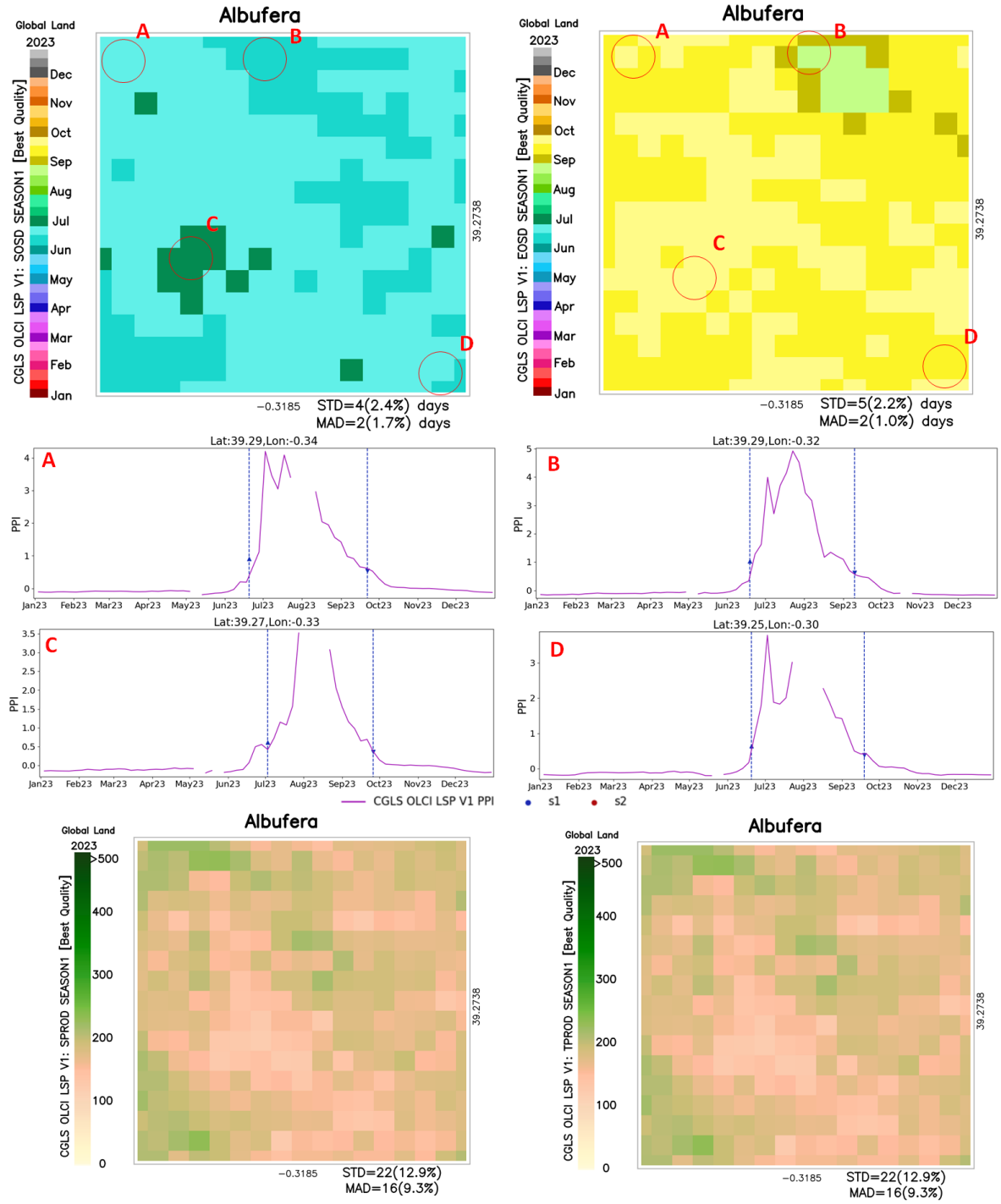



**Figure 75: Spatial precision over 5 km x 5 km area of Albufera CUL site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right).**

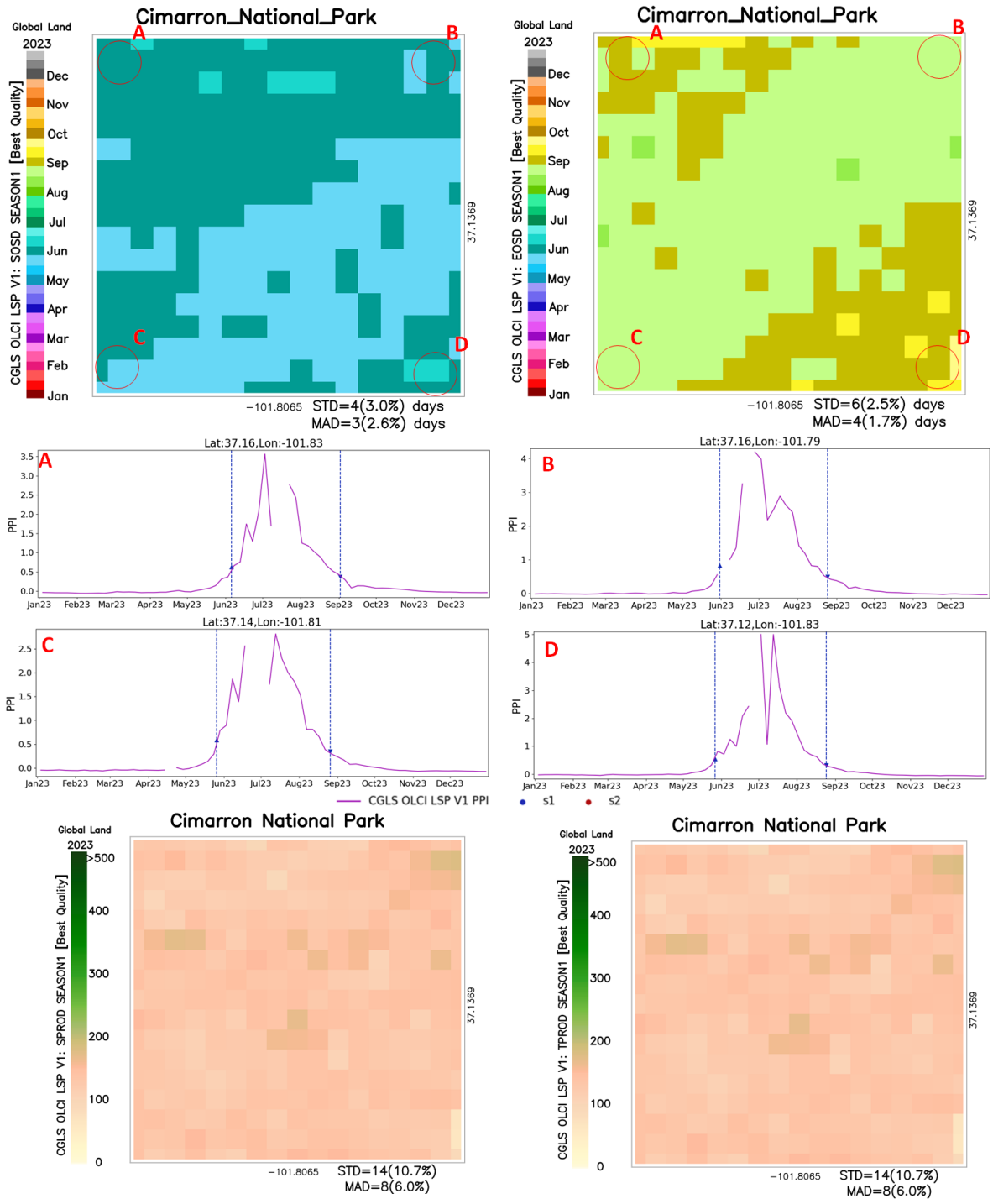



**Figure 76: Spatial precision over 5 km x 5 km area of Cimarron National Grassland Park HER site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right).**

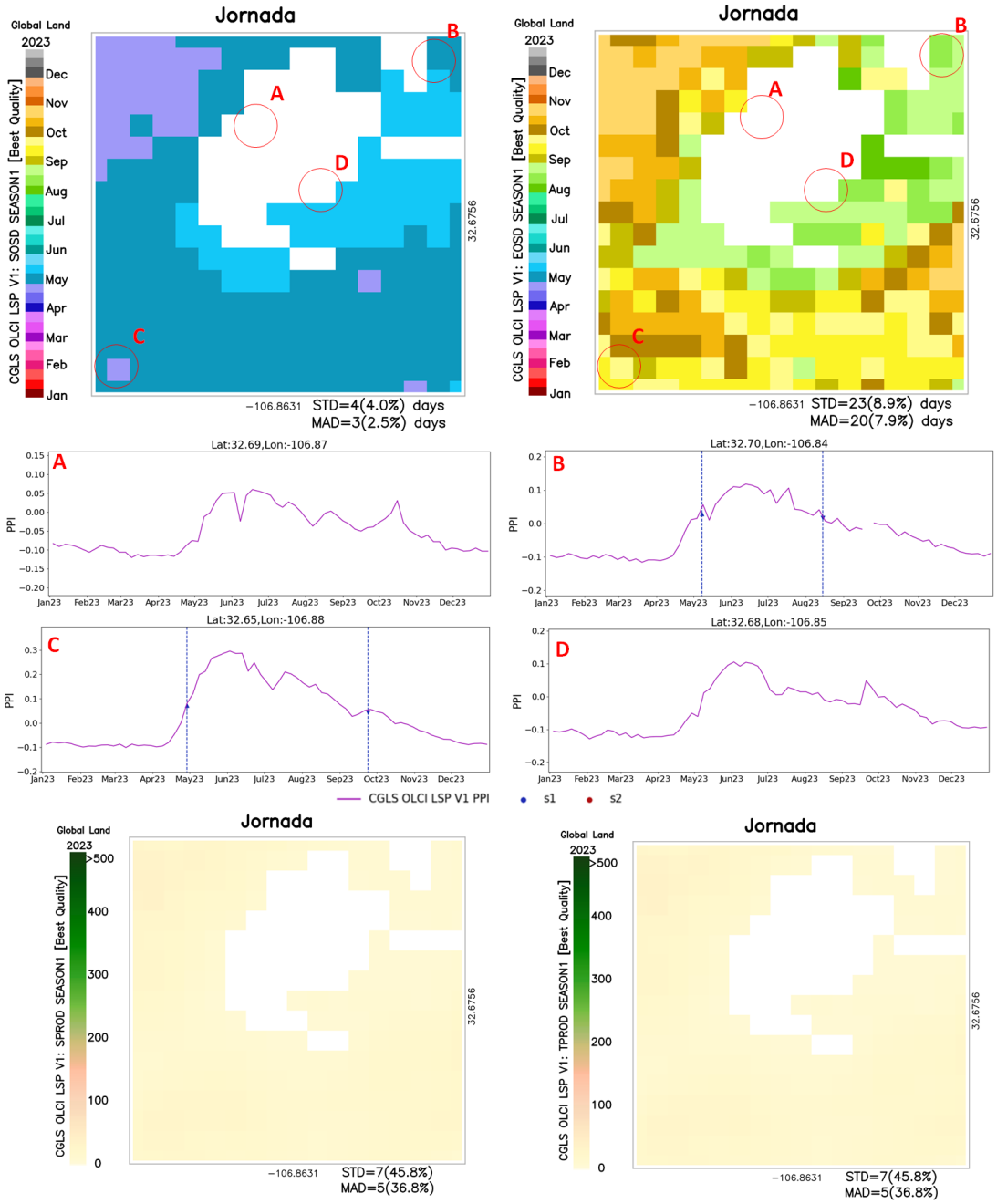



**Figure 77: Spatial precision over 5 km x 5 km area of Jornada SHR site in 2023 (season 1). From Top to Bottom: CLMS OLCI LSP SOSD (line 1, left), EOSD (line 1, right), temporal profiles of CLMS OLCI PPI where SOSD and EOSD are represented with triangles and inverted triangles for four selected pixels (A and B in line 2, C and D in line 3), SPROD (line 4, left) and TPROD (line 4, right).**

# ANNEX IV: DIRECT VALIDATION RESULTS FOR 2024 YEAR

The direct validation results of CLMS OLCI LSP for the year 2024 are presented in this annex. The methodology for processing in-situ reference products is the same as described in section 3.3, but using different sampling depending on the availability of ground data.

## PEP725

A total of 321 out of 1078 available sites providing data in 2024 are used in the validation after performing the vegetation group selection with availability of selected phenophases.

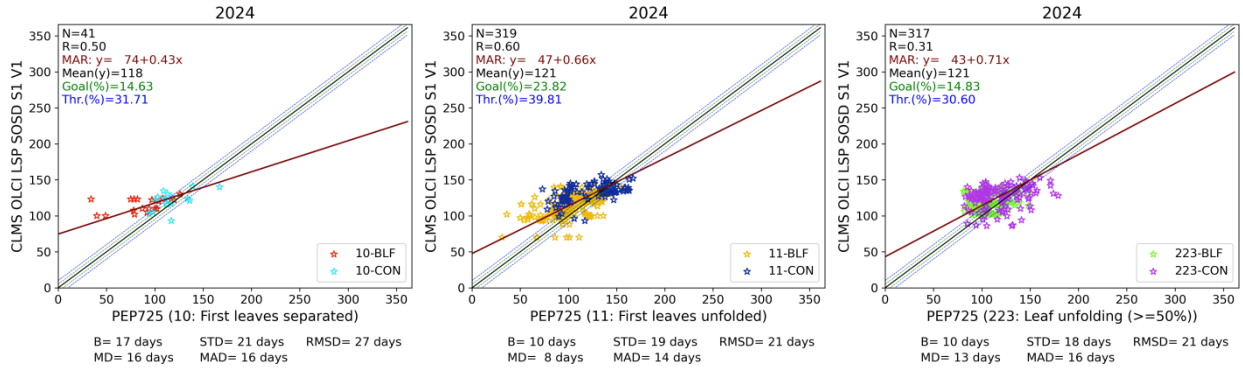



**Figure 78: CLMS OLCI LSP SOSD (good quality pixels) versus PEP725 phases 10 ('first leaves separated', left), 11 ('first leaves unfolded', middle) and 223 ('leaf unfolding ≥ 50%', right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

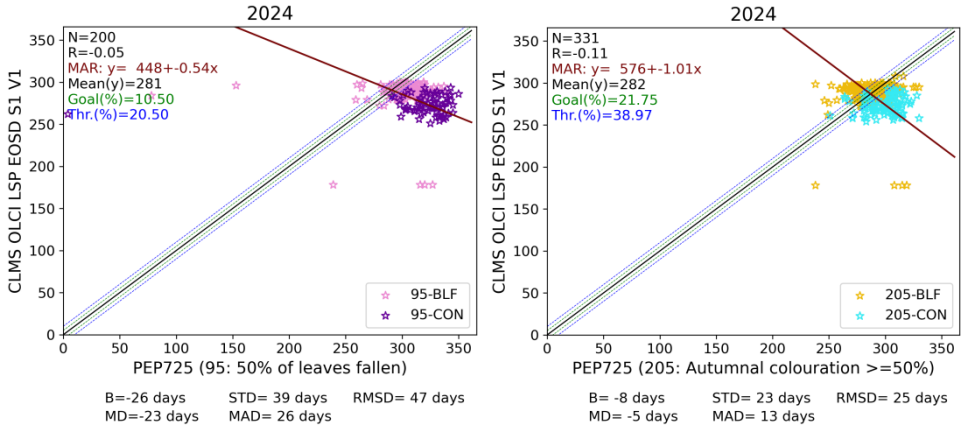



**Figure 79: CLMS OLCI LSP EOSD (good quality pixels) versus PEP725 phases 95 ('50% of leaves fallen', left), and 205 ('autumnal colouration ≥ 50%', right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

**Table 38: Performance statistics between CLMS OLCI LSP versus PEP725 database: SOSD versus PEP725 Phases 10, 11 and 223; and EOSD versus PEP725 Phases 95 and 205. Analysis for 2024 (season 1).**

| | CLMS OLCI LSP SOSD vs. PEP725 | | | CLMS OLCI LSP EOSD vs. PEP725 | |
|---|---|---|---|---|---|
| | **Phase 10** 'first leaves separated' | **Phase 11** 'first leaves unfolded' | **Phase 223** 'leaf unfolding ≥ 50%' | **Phase 95** '50% of leaves fallen' | **Phase 205** 'autumnal colouration ≥ 50%' |
| N | 41 | 319 | 317 | 200 | 331 |
| R | 0.50 | 0.60 | 0.31 | -0.05 | -0.11 |
| Bias (days) | 17 | 10 | 10 | -26 | -8 |
| MD (days) | 16 | 8 | 13 | -23 | -5 |
| STD (days) | 21 | 19 | 18 | 39 | 23 |
| MAD (days) | 16 | 14 | 16 | 26 | 13 |
| RMSD (days) | 27 | 21 | 21 | 47 | 25 |
| MAR | y=74+0.43x | y=47+0.66x | y=43+0.71x | y=448-0.54x | y=576-1.01x |
| % goal | 14.6 | 23.8 | 14.8 | 10.5 | 21.8 |
| % threshold | 31.7 | 39.8 | 30.6 | 20.5 | 39.0 |

The results in 2024 indicate:

- CLMS OLCI SOSD shows positive bias (i.e., delay of start of season) in comparison with PEP725 phases 10 and 11, corresponding to first leaves separated (B=17 days) and first leaves unfolded (B=10 days). Similar sign of bias is found compared with the year 2023, but showing higher differences. However, different sign of bias in 2024 (B=10 days) compared with 2023 (B=-12 days) is found in the comparison of SOSD with PEP725 223 leaf unfolding ≥ 50%), which could be explained by the different samplings.
- OLCI EOSD shows negative bias (i.e., anticipated end of season) of -26 and -8 days in comparison with PEP725 phases 95 ('50% of leaves fallen') and 205 ('autumnal colouration ≥50%'), showing that the OLCI LSP product represents better the 'colouration' phenophase than the 'leaf fallen'. Similar results were found in 2023.
- OLCI SOSD and EOSD show similar RMSD values compared with PEP725 in 2024 to that found 2023.

### TEMPO

The current dataset counts with more than 80 species and 990 sites in 2024. In total, 652 sites are available after performing the vegetation group selection with availability of selected phenophases.

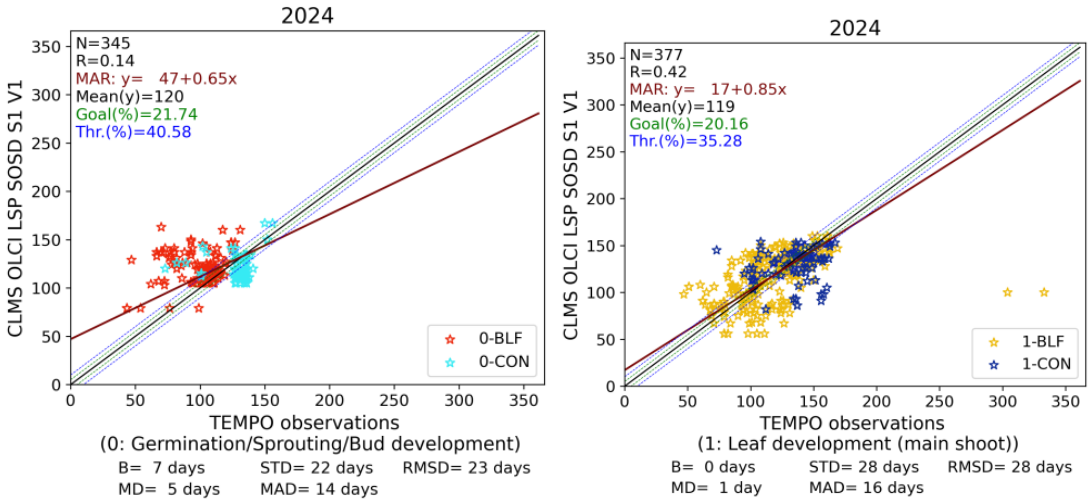



**Figure 80: CLMS OLCI LSP SOSD (good quality pixels) versus TEMPO phases 0 ('germination/sprouting/bud development', left) and 1 ('leaf development/main shoot', right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

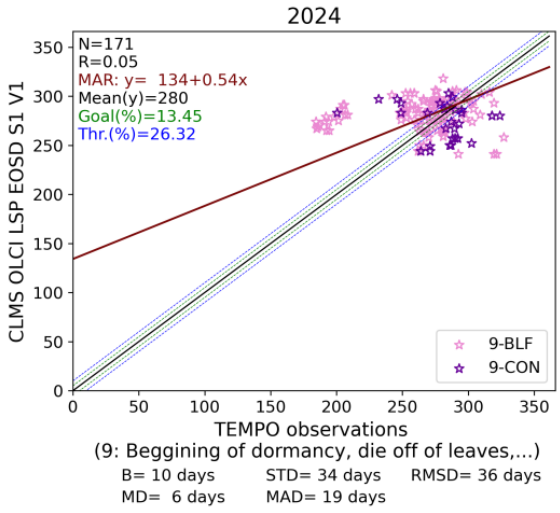



**Figure 81: CLMS OLCI LSP EOSD (good quality pixels) versus TEMPO phase 9 ('beginning of dormancy/die off of leaves') scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

**Table 39: Performance statistics between CLMS OLCI LSP versus TEMPO database: SOSD versus TEMPO Phases 0 and 1; and EOSD versus TEMPO Phase 9. Analysis for 2024 (season 1).**

| | CLMS OLCI LSP SOSD vs. TEMPO | | CLMS OLCI LSP EOSD vs. TEMPO |
|---|---|---|---|
| | **Phase 0** 'germination / sprouting / bud development' | **Phase 1** 'leaf development / main shoot' | **Phase 9** 'beginning of dormancy / die off of leaves' |
| N | 345 | 377 | 171 |
| R | 0.14 | 0.42 | 0.05 |
| Bias (days) | 7 | 0 | 10 |
| MD (days) | 5 | 1 | 6 |
| STD (days) | 22 | 28 | 34 |
| MAD (days) | 14 | 16 | 19 |
| RMSD (days) | 23 | 28 | 36 |
| MAR | y=47+0.65x | y=17+0.85x | y=134+0.54x |
| % goal | 21.7 | 20.2 | 13.5 |
| % threshold | 40.6 | 35.3 | 26.3 |

The results in 2024 indicate:

- OLCI SOSD shows a slight positive bias (B = 7 days) with TEMPO phase 0 (i.e., germination) and no mean bias with TEMPO phase 1 (leaf development), showing slightly worse agreement to that found in 2023 with TEMPO phase 0 (B = 1 day) and best agreement with TEMPO phase 1 (B = -7 days).
- OLCI EOSD shows a positive bias (i.e., delayed end of season) of 10 days (MD=6 days) in comparison with TEMPO phase 9 ('beginning of dormancy'), even though most of the points are clustered around the 1:1 line. This represents a delay of around 7 days (in terms of MD) as compared to 2023.

### NEON

The whole dataset counts with more than 600 species and has 46 sites and 2 different plots per site. A total of 48 plots, available in 2024, provide data for the growth forms and phenophases selected for validation.

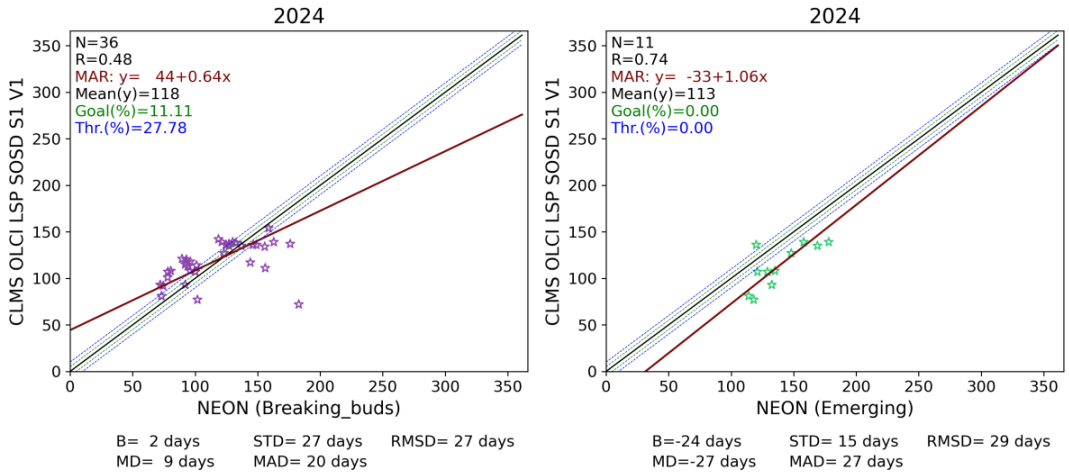



**Figure 82: CLMS OLCI LSP SOSD (good quality pixels) versus NEON phases breaking leaves/buds (left) and emerging leaves/buds (right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

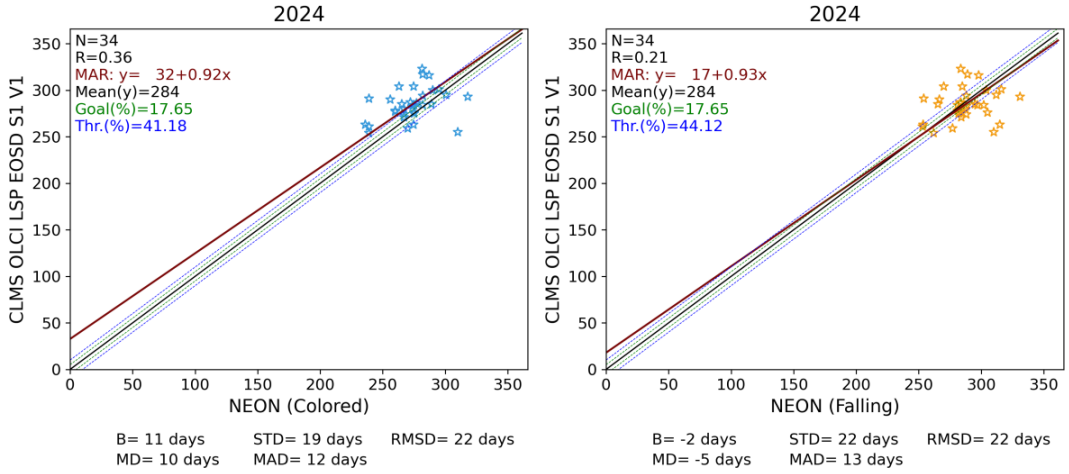



**Figure 83: CLMS OLCI LSP EOSD (good quality pixels) versus NEON phases colored leaves/buds (left), and falling leaves/buds (right) scatter-plots during 2024 (season 1). Green and blue lines around the 1:1 line correspond to goal and threshold requirements.**

**Table 40: Performance statistics between CLMS OLCI LSP SOSD and EOSD with NEON (2024) breaking&emerging, colored and falling leaves/buds.**

| | CLMS OLCI LSP SOSD vs. NEON | | CLMS OLCI LSP EOSD vs. NEON | |
|---|---|---|---|---|
| | breaking leaves/buds | emerging leaves/buds | colored leaves/buds | falling leaves/buds |
| N | 36 | 11 | 34 | 34 |
| R | 0.48 | 0.74 | 0.36 | 0.21 |
| Bias (days) | 2 | -24 | 11 | -2 |
| MD (days) | 9 | -27 | 10 | -5 |
| STD (days) | 27 | 15 | 19 | 22 |
| MAD (days) | 20 | 27 | 12 | 13 |
| RMSD (days) | 27 | 29 | 22 | 22 |
| MAR | y=44+0.64x | y=-33+1.06x | y=32+0.92x | y=17+0.93x |
| % goal | 11.1 | 0.0 | 17.7 | 17.7 |
| % threshold | 27.8 | 0.0 | 41.2 | 44.1 |

The results in 2024 indicate:

- SOSD shows slight positive bias of 2 days (i.e., delay) compared with NEON ‘breaking leaves/buds’ phase and large negative bias (i.e., anticipated start of season) compared with NEON ‘emerging leaves/buds’, in line to that found in 2023. Note that only 11 samples are available for ‘emerging leaves/buds’, which makes this class less representative.
- EOSD shows positive bias of 11 days compared with ‘colored leaves/buds’ and negative bias of -2 days in comparison with NEON ‘falling leaves/buds’. That means that OLCI LSP EOSD is between the ‘colored leaves’ and ‘falling leaves’, which seem to be quite realistic and are in line with what was found in 2023.

### PhenoCam V3

PhenoCam V3 counts with a total of 1320 ROI's masks and data from 2000 to 2025. A total of 203 ROIs are used in 2024 after applying the visual homogeneity filter.

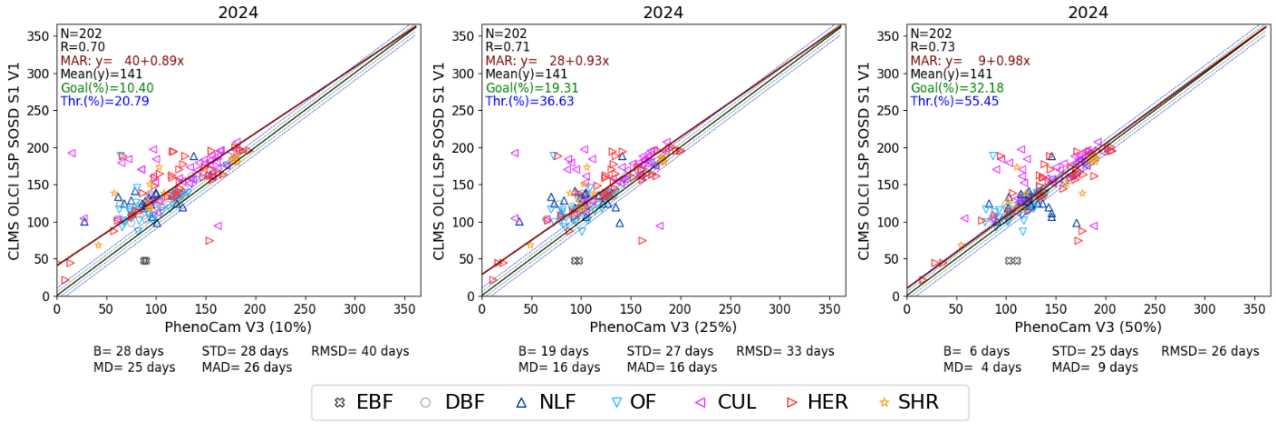



**Figure 84: CLMS OLCI LSP SOSD (good quality pixels) versus PhenoCam V3 start of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2024 (season 1).**

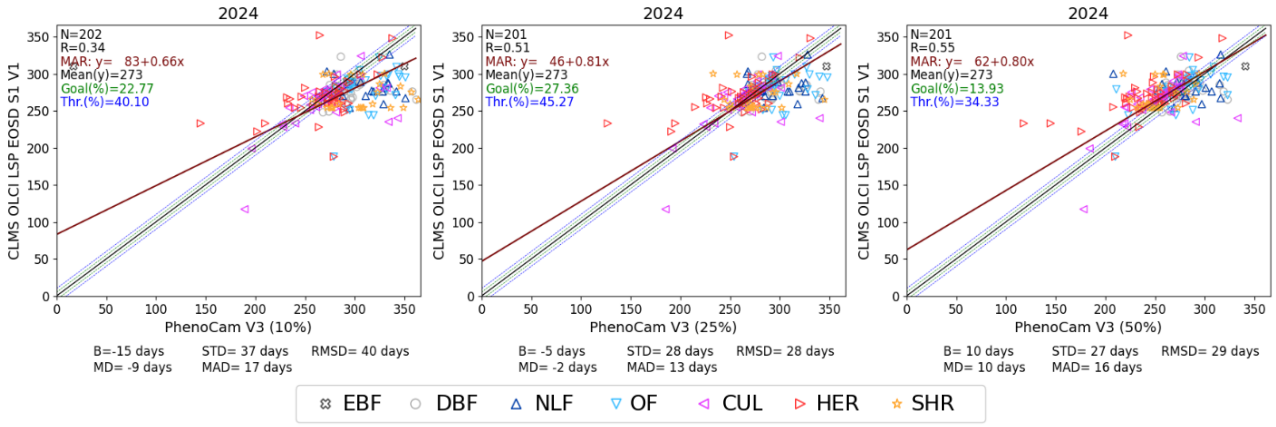



**Figure 85: CLMS OLCI LSP EOSD (good quality pixels) versus PhenoCam V3 end of season defined at 10% (left), 25% (middle) and 50% (right) scatter-plots during 2024 (season 1).**

**Table 41: Performance statistics between CLMS OLCI LSP SOSD and EOSD with PhenoCam V3 equivalent phases defined at 10%, 25% and 50% for 2024 (season 1).**

| | CLMS OLCI LSP SOSD vs. PhenoCamV3 | | | CLMS OLCI LSP EOSD vs. PhenoCamV3 | | |
|---|---|---|---|---|---|---|
| | 10% | 25% | 50% | 10% | 25% | 50% |
| N | 202 | 202 | 202 | 202 | 201 | 201 |
| R | 0.70 | 0.71 | 0.73 | 0.34 | 0.51 | 0.55 |
| Bias (days) | 28 | 19 | 6 | -15 | -5 | 10 |
| MD (days) | 25 | 16 | 4 | -9 | -2 | 10 |
| STD (days) | 28 | 27 | 25 | 37 | 28 | 27 |
| MAD (days) | 26 | 16 | 9 | 17 | 13 | 16 |
| RMSD (days) | 40 | 33 | 26 | 40 | 28 | 29 |
| MAR | y=40+0.89x | y=28+0.93x | y=9+0.98x | y=83+0.66x | y=46+0.81x | y=62+0.80x |
| % goal | 10.4 | 19.3 | 32.2 | 22.8 | 27.4 | 13.9 |
| % threshold | 20.8 | 36.6 | 55.5 | 40.1 | 45.3 | 34.3 |

The results for 2024 indicate:

- As observed in 2023, CLMS OLCI LSP SOSD better agrees in terms of accuracy (B, MD) and uncertainty (RMSD) with PhenoCam V3 start of season defined at 50% threshold (Bias = 6 days) than compared with Phenocam V3 defined at 10% (Bias = 28 days) and 25% (Bias = 19 days). As CLMS SOSD is defined at 25% threshold, the comparison with the same PhenoCam threshold shows that OLCI SOSD is delayed as compared to PhenoCam near-ground-based estimates based on GCC index.
- Similar conclusions are found for EOSD than in 2023. OLCI EOSD (defined at 15%) better agrees in terms of bias is found with PhenoCam defined at 25% (Bias = -5 days) compared with 10% (Bias = -15 days) and 50% (Bias = 10 days). Therefore, it can be concluded that OLCI LSP EOSD is slightly anticipated as compared to PhenoCam near-ground-based estimates.
- The positive bias for SOSD and negative bias for EOSD is translated in shorter season of CLMS OLCI LSP as compared with this near-ground-based dataset.